

Coefficients: Predicting density and assessing factors driving marbled murrelet distribution in Puget Sound using hierarchical distance sampling

Code by S.J. Gillman

Required Packages

```
library(tidyverse)

library(MCMCvis)
library(nimble)

library(ggplot2)
library(cowplot)
library(patchwork)
library(sf)
library(viridis)
library(scico)
library(ggtext)
```

Required Files Previously Made or Provided

- Survey_Grid_Info_NEW.RData
- chain_1.rds
- chain_2.rds
- chain_3.rds
- nimConstants.rds
- nimData.rds
- grid2km_hex.gpkg
- predicted_month_years.rds
- predicted_season_years.rds
- predicted_months.rds
- predicted_seasonsB.rds
- predicted_seasonsNB.rds
- WA_State_Boundary.shp downloaded from the [Washington State Department of Natural Resources GIS Open Data Portal](#)
- WSDOT_-_Major_Shorelines.shp also downloaded from the [Washington State Department of Natural Resources GIS Open Data Portal](#)

```

load("../DataAnalysis/DataFiles/Survey_Grid_Info_NEW.RData")
samples_1 <- readRDS("RESULTS/COMPRESSED/chain_1.rds")
samples_2 <- readRDS("RESULTS/COMPRESSED/chain_2.rds")
samples_3 <- readRDS("RESULTS/COMPRESSED/chain_3.rds")

nimConstants <- readRDS("RESULTS/COMPRESSED/nimConstants.rds")
nimData <- readRDS("RESULTS/COMPRESSED/nimData.rds")

chains <- coda::mcmc.list(samples_1,samples_2,samples_3)

result_summary <- MCMCsummary(chains, probs = c(0.025,0.25,0.5,0.75, 0.975)) %>%
  rownames_to_column("parameter") %>%
  mutate(
    variable = sub("\\[.*", "", parameter),
    model_id = gsub("\\]", "", sub(".*\\]", "", parameter)))

round(max(result_summary$Rhat, na.rm =T),2)

## [1] 1

# posterior direction
pd_dir <- bayestestR::p_direction(chains)

```

Result Outputs

```

stat_vals <- as.data.frame(t(t(colnames(nimData[["STAT_COVS"]])))) %>%
  rownames_to_column("model_id") %>%
  mutate(V1 = toupper(gsub("_scale", "", V1)))

sea_vals <- as.data.frame(t(t(colnames(nimData[["DYNAMIC"]])))) %>%
  rownames_to_column("model_id") %>%
  mutate(V1 = toupper(gsub("_scale", "", V1)),
    model_id = paste0("1, ", model_id)) %>%
  rbind(as.data.frame(t(t(colnames(nimData[["DYNAMIC"]])))) %>%
    rownames_to_column("model_id") %>%
    mutate(V1 = toupper(gsub("_scale", "", V1)),
      model_id = paste0("2, ", model_id)))

result_coef <- result_summary %>%
  full_join(pd_dir %>% rename(parameter = Parameter),
    by = "parameter") %>%
  filter(variable == "eps_mon") %>%
  rename(mNum = model_id) %>%
  full_join(survey_info %>%
    mutate(mNum = as.character(mNum)) %>%
    distinct(month, mNum), by = "mNum") %>%
  mutate(parameter = paste0("eps-", month)) %>%
  dplyr::select(-month) %>%
  rename(model_id = mNum) %>%

```

```

# simple
rbind(result_summary %>%
  full_join(pd_dir %>% rename(parameter = Parameter),
    by = "parameter") %>%
  filter(variable == "alpha0" |
    variable == "mu_gt" |
    variable == "mu_tide" |
    variable == "mu_delta" |
    variable == "rate_r" |
    variable == "shape_r" |
    variable == "sig_delta" |
    variable == "sig_gt" |
    variable == "sig_pair" |
    variable == "sig_tide" |
    variable == "sig_mon" |
    variable == "sigma_mean" |
    startsWith(variable, "beta"))) %>%

# static
rbind(result_summary %>%
  full_join(pd_dir %>% rename(parameter = Parameter),
    by = "parameter") %>%
  filter(variable == "alpha_stat") %>%
  full_join(stat_vals, by = "model_id") %>%
  mutate(parameter= V1) %>%
  dplyr::select(-V1)) %>%

# seasonal
rbind(result_summary %>%
  full_join(pd_dir %>% rename(parameter = Parameter),
    by = "parameter") %>%
  filter(variable == "alpha_sea") %>%
  full_join(sea_vals, by = "model_id") %>%
  mutate(parameter= ifelse(startsWith(model_id, "1,"),
    paste0(V1, " B"),
    paste0(V1, " NB"))) %>%

  dplyr::select(-V1)) %>%
rbind(result_summary %>%
  full_join(pd_dir %>% rename(parameter = Parameter),
    by = "parameter") %>%
  filter(variable == "eps_pair")) %>%

# sigma
mutate(parameter = ifelse(parameter == "beta", "TIDE",
  ifelse(parameter == "alpha0[1]", "ALPHA0 B",
  ifelse(parameter == "alpha0[2]", "ALPHA0 NB",
  ifelse(variable == "eps_pair",
    paste0("EPS_PAIR-", model_id),
    toupper(parameter)))))) %>%
mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
rename(ESS = `n.eff`)

result_coef2 <- result_coef %>%

```

```
dplyr::select(-c(model_id,variable))
```

```
write_csv(result_coef2, file = "RESULTS/parameter_coefs.csv")
```

Figure 2: Habitat

```
alphas_hab <- result_coef %>%
  filter(startsWith(variable, "alpha_")) %>%
  filter(!startsWith(parameter, "PAR") & !startsWith(parameter, "DO") &
    !startsWith(parameter, "GT")) %>%
  mutate(Comparison = ifelse(startsWith(parameter, "EI-") |
    parameter == "MI" |
    parameter == "OCN",
    "Shoreline Type",
    ifelse(parameter == "DEPTH" |
      parameter == "BPI" |
      parameter == "RIE", "Static",
      ifelse(startsWith(parameter, "SHORE") |
        startsWith(parameter, "SUIT"),
        "Dynamic", "Seafloor Substrate"))),
    season = ifelse(endsWith(parameter, " NB"), "Non Breeding",
      ifelse(endsWith(parameter, " B"),
        "Breeding", "Constant")) %>%
  mutate(parameter = case_when(parameter == "EI-MIX_FINE" ~
    "Estuarian-intertidal habitat, Mixed Fine Substrate",
    parameter == "EI-BOULDER" ~ "Estuarian-intertidal habitat, Bedrock",
    parameter == "EI-GRAVEL" ~ "Estuarian-intertidal habitat, Gravel Substrate",
    parameter == "EI-SAND" ~ "Estuarian-intertidal habitat, Sand Substrate",
    parameter == "EI-ARTIFICIAL" ~ "Estuarian-intertidal habitat, Artificial Substrate",
    parameter == "OCN" ~ "Open Ocean",
    parameter == "MI" ~ "Marine-intertidal habitat, All Substrates",
    #seafloor
    parameter == "MUD" ~ "Muddy Seafloor",
    parameter == "COBBLE" ~ "Cobble Seafloor",
    parameter == "GRAVEL" ~ "Gravel Seafloor",
    parameter == "FUS" ~ "Fine Unconsolidated Substrate Seafloor",
    parameter == "UMS" ~ "Unconsolidated Mineral Substrate Seafloor",
    parameter == "CUS" ~ "Coarse Unconsolidated Substrate Seafloor",
    parameter == "SAND" ~ "Sandy Substrate Seafloor",
    parameter == "SANDYMUD" ~ "Mixed Sandy-Mud Substrate Seafloor",

    parameter == "SHORE B" ~ "SHORE",
    parameter == "SHORE NB" ~ "SHORE",
    parameter == "SHORESQ B" ~ "SHORESQ",
    parameter == "SHORESQ NB" ~ "SHORESQ",
    parameter == "SUIT NB" ~ "SUIT",
    parameter == "SUIT B" ~ "SUIT",
    parameter == "DEPTH" ~ "DEPTH",
    parameter == "BPI" ~ "BPI",
    parameter == "RIE" ~ "RIE")) %>%
  mutate(parameter = factor(parameter,
```

```

        levels =c("Mixed Sandy-Mud Substrate Seafloor",
                  "Muddy Seafloor",
                  "Unconsolidated Mineral Substrate Seafloor",
                  "Coarse Unconsolidated Substrate Seafloor",
                  "Fine Unconsolidated Substrate Seafloor",
                  "Sandy Substrate Seafloor",
                  "Gravel Seafloor",
                  "Cobble Seafloor",

                  #shoreline
                  "Marine-intertidal habitat, All Substrates",
                  "Estuarian-intertidal habitat, Mixed Fine Substrate",
                  "Open Ocean",
                  "Estuarian-intertidal habitat, Bedrock",
                  "Estuarian-intertidal habitat, Sand Substrate",
                  "Estuarian-intertidal habitat, Artificial Substrate",
                  "Estuarian-intertidal habitat, Gravel Substrate",

                  # Continuous static
                  "DEPTH",
                  "RIE",
                  "BPI",

                  # Continuous dynamic
                  "SHORESQ",
                  "SHORE",
                  "SUIT")),
season = factor(season, levels = c("Breeding", "Non Breeding", "Constant"))

my_labs4 <- c("Mixed Sandy-Mud Substrate Seafloor" =
              "Mixed Sandy-Mud<br>Substrate Seafloor",
              "Muddy Seafloor" = "Muddy Seafloor",
              "Unconsolidated Mineral Substrate Seafloor" =
              "Unconsolidated<br>Mineral Seafloor",
              "Coarse Unconsolidated Substrate Seafloor" =
              "Coarse Unconsolidated<br>Substrate Seafloor",
              "Fine Unconsolidated Substrate Seafloor" =
              "Fine Unconsolidated<br>Seafloor",
              "Sandy Seafloor" = "Sandy Seafloor",
              "Gravel Seafloor" = "Gravel Seafloor",
              "Cobble Seafloor" = "Cobble Seafloor",

              #shoreline
              "Marine-intertidal habitat, All Substrates" =
              "Marine-intertidal<br>All Substrates",
              "Estuarian-intertidal habitat, Mixed Fine Substrate" =
              "Estuarian-intertidal<br>Mixed Fine Substrate",
              "Open Ocean" = "Open Ocean",
              "Estuarian-intertidal habitat, Bedrock" =
              "Estuarian-intertidal<br>Bedrock Substrate",
              "Estuarian-intertidal habitat, Sand Substrate" =
              "Estuarian-intertidal habitat<br>Sand Substrate",

```

```

"Estuarian-intertidal habitat, Artificial Substrate" =
  "Estuarian-intertidal<br>Artificial Substrate",
"Estuarian-intertidal habitat, Mud Substrate" =
  "Estuarian-intertidal<br>Mud Substrate",
"Estuarian-intertidal habitat, Gravel Substrate" =
  "Estuarian-intertidal<br>Gravel Substrate",

DEPTH = "Depth (m)",
RIE = "Roughness<br>Index-Elevation",
BPI = "Bathymetric<br>Position Index",
SHORESQ = "Distance to<br>Shore Squared (m)",
SHORE = "Distance to<br>Shore (m)",
SUIT = "Proportion of High<br>Quality Habitat")

sep_lines <- data.frame(y = c(8.5, 15.5, 18.5))
sep_labs <- data.frame(y = c(8.1, 15.1, 18.1),
  ylabels = c("Seafloor Type",
    "Shoreline Type", "Constant"))

ggplot(data= alphas_hab,
  aes(y = parameter, x = `50%`, fill=season))+
  geom_errorbar(aes(xmin = `2.5%`, xmax = `97.5%`,
    width = 0, linewidth = 2, color = "lightgray",
    position = position_dodge(width = 0.55))+
  geom_errorbar(aes(xmin = `25%`, xmax = `75%`,
    width = 0, linewidth = 3, color = "gray50",
    alpha=1, position = position_dodge(width = 0.55))+
  geom_point(data = alphas_hab,
    aes(y = parameter, x = `50%`), size = 3,
    shape = 21, position= position_dodge(width = 0.55)) +
  geom_text(data = sep_labs, aes(x = -1.15, y = y,
    label = ylabels),
    inherit.aes = FALSE, hjust = 0, vjust = 0, size = 3.3) +
  geom_segment(data = sep_lines,
    aes(x = -Inf, xend = Inf, y = y, yend = y),
    inherit.aes = FALSE,
    linewidth = 0.4, color = "black") +
  theme_minimal() +
  scale_fill_manual(values = c("#DF9D56", "#6EB5E1", "gray4"),
    breaks = c("Breeding", "Non Breeding")) +
  scale_x_continuous(breaks = c(-1.15, -1, -0.75, -0.5, -0.25,
    0, 0.25, 0.5, 0.75, 1, 1.25),
    limits = c(-1.15, 1.25)) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "black") +

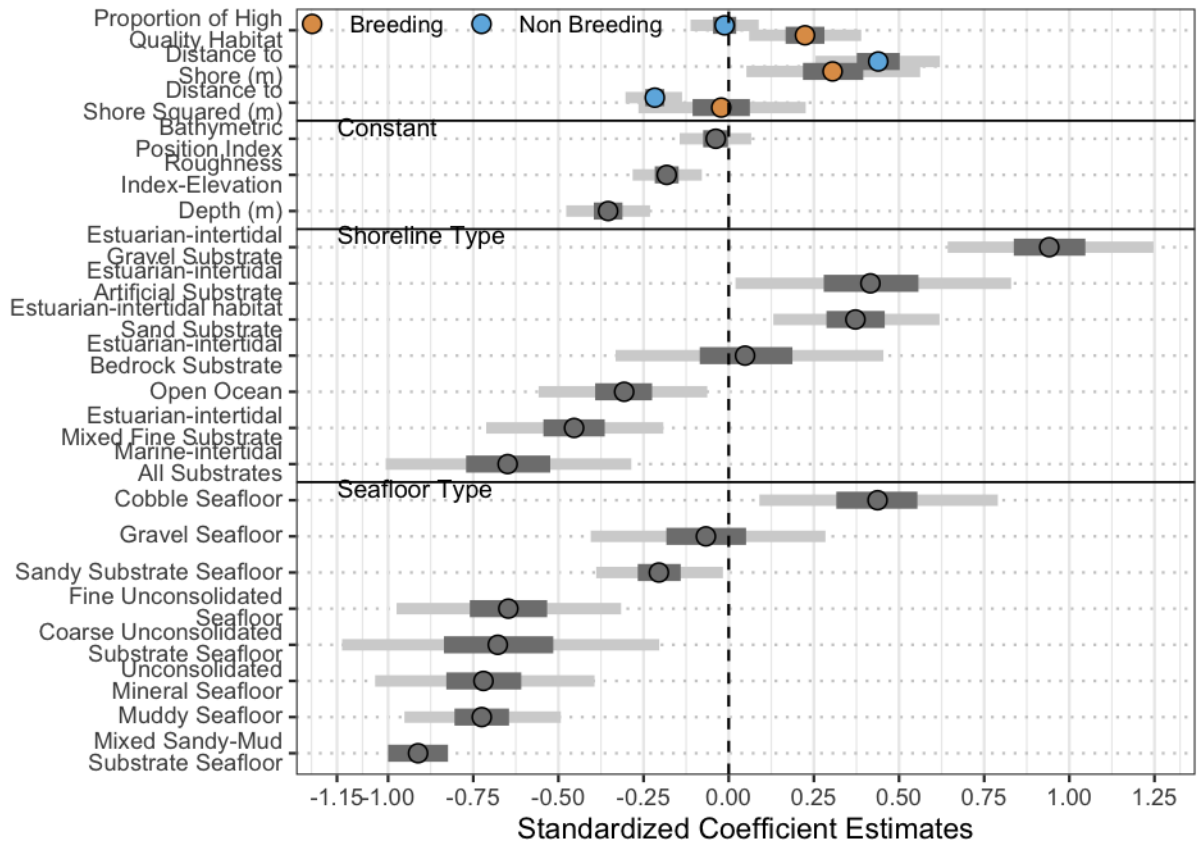
  xlab("Standardized Coefficient Estimates") +
  ylab("")+
  theme_bw() +
  scale_y_discrete(labels = my_labs4) +
  theme(axis.text.y = element_markdown(),
    panel.grid.major.y = element_line(color = "grey80",
    linetype = "dotted"),

```

```

legend.direction = "horizontal",
legend.background = element_blank(),
legend.key = element_blank(),
legend.position = c(0.2, 0.98),
legend.title=element_blank())

```



```

#ggsave("RESULTS/Figure2.png")
#ggsave("RESULTS/Figure2.pdf")

```

Figure 3: Ocean Conditions

```

alphas <- result_coef %>%
  filter(startsWith(variable, "alpha_sea") &
    !startsWith(parameter, "SHORE") &
    !startsWith(parameter, "SUIT")) %>%
  mutate(season = ifelse(endsWith(parameter, " NB"), "Non Breeding",
    ifelse(endsWith(parameter, " B"),
      "Breeding", "No Variation"))) %>%
  mutate(season = factor(season, levels=c("Breeding",
    "Non Breeding", "No Variation")),
    parameter2 = gsub(" NB", "", parameter),
    parameter2 = gsub(" B", "", parameter2),
    parameter = gsub(" ", "", parameter),

```

```

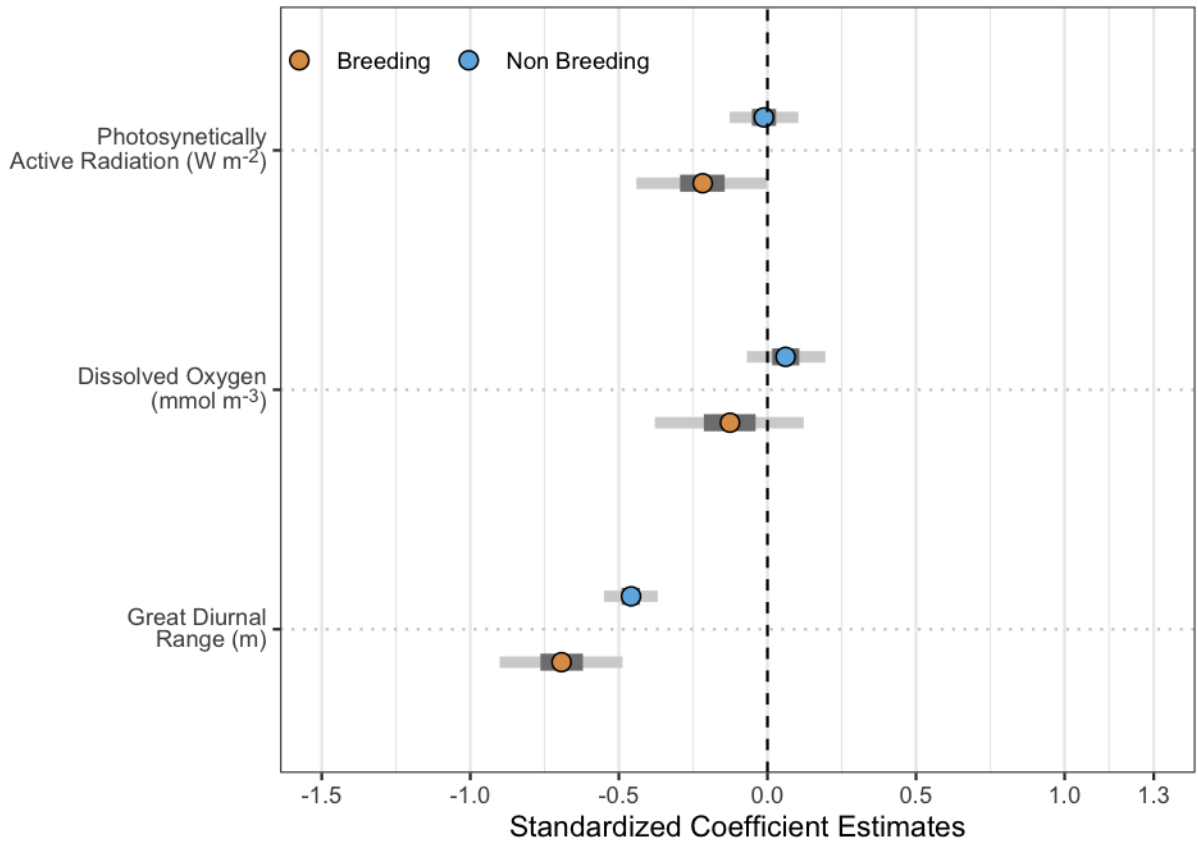
parameter = factor(parameter, levels = c("GTB",
      "GTNB",
      "DOB",
      "DONB",
      "PARB",
      "PARNB")),
parameter2 = factor(parameter2, levels = c(
      "GT",
      "DO",
      "PAR"))))

my_labs5 <- c(GT = "Great Diurnal<br> Range (m)",
      PAR = "Photosynthetically<br>Active Radiation (W m<sup>-2</sup>)",
      DO = "Dissolved Oxygen<br>(mmol m<sup>-3</sup>)")

ggplot(data= alphas,
      aes(y = parameter2, x = `50%`, fill=season))+
geom_errorbar(aes(xmin = `2.5%`, xmax = `97.5%`),
      width = 0, linewidth = 2,color = "lightgray",
      position = position_dodge(width =0.55))+
geom_errorbar(aes(xmin = `25%`, xmax = `75%`),
      width = 0, linewidth = 3, color = "gray50",
      alpha=1, position = position_dodge(width =0.55))+
geom_point(data = alphas,
      aes(y = parameter2, x = `50%`), size = 3,
      shape = 21, position= position_dodge(width =0.55)) +
theme_minimal() +
scale_fill_manual(values = c("#DF9D56", "#6EB5E1", "gray4"),
      breaks = c("Breeding", "Non Breeding")) +
scale_x_continuous(breaks = c(-1.5,-1,-0.5,0,0.5,1,1.3),
      limits = c(-1.5, 1.3)) +
geom_vline(xintercept = 0, linetype = "dashed", color = "black") +
xlab("Standardized Coefficient Estimates") +
ylab("")+
theme_bw() +
scale_y_discrete(labels = my_labs5) +
theme(axis.text.y = element_markdown(),
      panel.grid.major.y = element_line(color = "grey80",
      linetype = "dotted"),

      legend.background = element_blank(),
      legend.key = element_blank(),
      legend.direction = "horizontal",
      legend.position = c(0.2, .93),
      legend.title=element_blank())

```



```
#ggsave("RESULTS/Figure3.png", width = 7, height = 3.5, units="in")
#ggsave("RESULTS/Figure3.pdf", width = 7, height = 3.5, units="in")
```

Figure 4: Heat Maps

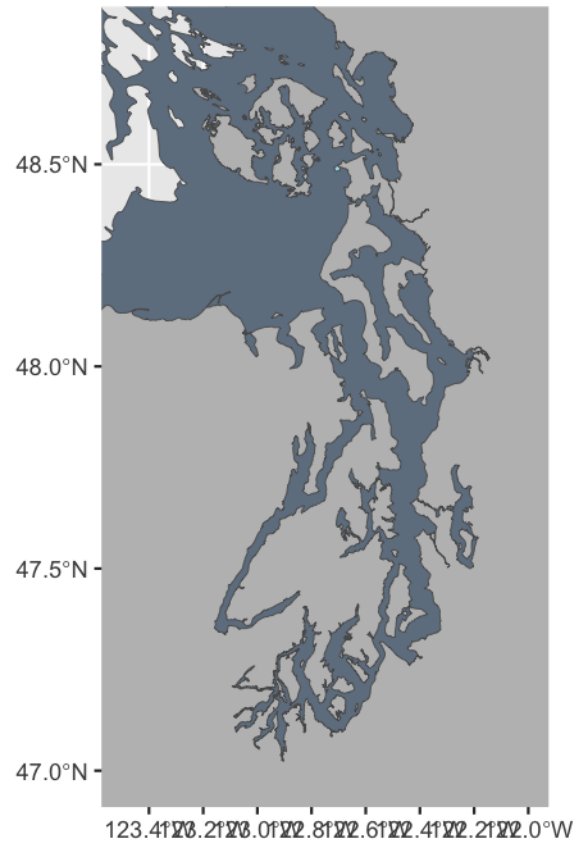
After creating pdfs of individual months/seasons, they were combined with inkscape.

import files

```
grids <- st_read("../CovariateProcessing/Outputs/grid2km_hex.gpkg")

WA_State <- st_read("WA_State_Boundary/WA_State_Boundary.shp")
puget <- st_read("WSDOTMajor_Shorelines/WSDOT_-_Major_Shorelines.shp")
WA_State <- st_transform(WA_State, crs = st_crs(grids))
puget <- st_transform(puget, crs = st_crs(grids))

ggplot()+
  geom_sf(data = WA_State, fill = "gray") +
  geom_sf(data = puget, fill = "slategray") +
  geom_sf(data = grids %>% filter(ID == "G317"), fill = "lightblue") +
  coord_sf(xlim = c(-122, -123.5), ylim = c(47, 48.8), crs = st_crs(4326))
```



import predictions

```
month_year_den <- readRDS("RESULTS/predicted_month_years.rds")
season_year_den <- readRDS("RESULTS/predicted_season_years.rds")
month_ave_den <- readRDS("RESULTS/predicted_months.rds")
B_ave_den <- readRDS("RESULTS/predicted_seasonsB.rds")
NB_ave_den <- readRDS("RESULTS/predicted_seasonsNB.rds")
```

```
dynamic_ranges <- survey_info %>%
  group_by(month) %>%
  summarise(par_min = min(par, na.rm=T),
            par_max = max(par, na.rm = T),
            do_min = min(do, na.rm=T),
            do_max = max(do, na.rm=T),
            gt_min = min(grand_range, na.rm=T),
            gt_max = max(grand_range, na.rm=T)) %>%
  ungroup()

static_range <- survey_info %>%
  summarise(hab_min = min(nest_hab, na.rm=T),
            hab_max = max(nest_hab, na.rm=T),
```

```

    shore_min = min(dist_shore, na.rm=T),
    shore_max = max(dist_shore, na.rm=T),
    depth_min = min(depth, na.rm=T),
    depth_max = max(depth, na.rm=T),
    bpi_min = min(bpi, na.rm=T),
    bpi_max = max(bpi, na.rm=T),
    rie_min = min(rie, na.rm=T),
    rie_max = max(rie, na.rm=T))

grid_flags <- grid_info %>%
  rename(month = mon) %>%
  full_join(dynamic_ranges, by = "month") %>%
  # clear NAs
  mutate(par = ifelse(is.na(par), mean(par, na.rm=T), par),
    do = ifelse(is.na(do), mean(do, na.rm=T), do),
    grand_range = ifelse(is.na(grand_range), mean(grand_range, na.rm=T), grand_range),
    dist_shore = ifelse(is.na(dist_shore), mean(dist_shore, na.rm=T), dist_shore),
    nest_hab = ifelse(is.na(nest_hab), mean(nest_hab, na.rm=T), nest_hab),
    bpi = ifelse(is.na(bpi), mean(bpi, na.rm=T), bpi),
    rie = ifelse(is.na(rie), mean(rie, na.rm=T), rie),
    depth = ifelse(is.na(depth), mean(depth, na.rm=T), depth)) %>%
  # Ensure not above max or min
  mutate(flag = ifelse(par > par_max |
    par < par_min |
    do > do_max |
    do < do_min |
    grand_range > gt_max |
    grand_range < gt_min |
    dist_shore > static_range$shore_max |
    dist_shore < static_range$shore_min |
    nest_hab > static_range$hab_max |
    nest_hab < static_range$hab_min |
    bpi > static_range$bpi_max |
    bpi < static_range$bpi_min |
    rie > static_range$rie_max |
    rie < static_range$rie_min |
    depth > static_range$depth_max |
    depth < static_range$depth_min, "CAP", "-") %>%
  select(c(ID, numid, seasNum, month, year, flag)) %>%
  rename(mon = month)

grid_flags_all <- grid_flags %>%
  group_by(ID, numid, mon) %>%
  summarize(flag2 = ifelse(any(flag == "CAP"), "CAP", "-"), .groups="drop") %>%
  ungroup() %>%
  mutate(year = "ALL") %>%
  rbind(grid_flags %>%
    group_by(ID, numid, seasNum) %>%
    summarize(flag2 = ifelse(any(flag == "CAP"), "CAP", "-"), .groups="drop") %>%
    ungroup() %>%
    mutate(year = "ALL") %>%
    rename(mon = seasNum) %>%

```

```

    mutate(mon = ifelse(mon == 1, "B", "NB")) %>%
  rbind(grid_flags %>%
    group_by(ID, numid, seasNum, year) %>%
    summarize(flag2 = ifelse(any(flag == "CAP"), "CAP", "-"), .groups="drop") %>%
    ungroup() %>%
    rename(mon = seasNum) %>%
    mutate(mon = ifelse(mon == 1, "B", "NB")) %>%
  rename(flag = flag2) %>%
  rbind(grid_flags %>%
    select(c(ID, numid, mon, year, flag)))

```

```

all_predictions <- month_ave_den %>%
  rename(mon = month) %>%
  mutate(year = "ALL") %>%
  rbind(month_year_den %>%
    rename(mon = month)) %>%
  rbind(season_year_den %>%
    rename(mon = seasNum) %>%
    mutate(mon = ifelse(mon == 1, "B", "NB"))) %>%
  rbind(B_ave_den %>%
    rename(mon = seasNum) %>%
    mutate(mon = "B", year = "ALL")) %>%
  rbind(NB_ave_den %>%
    rename(mon = seasNum) %>%
    mutate(mon = "NB", year = "ALL")) %>%
  full_join(grid_flags_all, by = c("numid", "mon", "year"))

```

Flag Capped Grids

Total

```

top_5_frequency <- all_predictions %>%
  filter(mon != "B" & mon != "NB" & year != "ALL") %>%
  mutate(top5_cutoff = quantile(MED, probs = 0.95, na.rm = TRUE),
    is_top5 = MED >= top5_cutoff) %>%
  # Count how many times each grid cell is in the top 5%
  group_by(ID) %>%
  summarise(
    n_surveyed = n(),
    n_top5 = sum(is_top5, na.rm = TRUE),
    prop_top5 = n_top5 / n_surveyed, .groups = "drop")

grid_freq_season <- grids %>%
  left_join(top_5_frequency, by = c("ID"))

## total area listed as top 5% at least once
grid_freq_season %>%
  st_drop_geometry() %>%

```

```

filter(prop_top5 >0) %>%
summarise(sum(area))

##      sum(area)
## 1      483.52

# 483.52

scico(5, palette = 'roma')

## [1] "#7E1700" "#B58B31" "#C0E9C2" "#389CC6" "#023198"

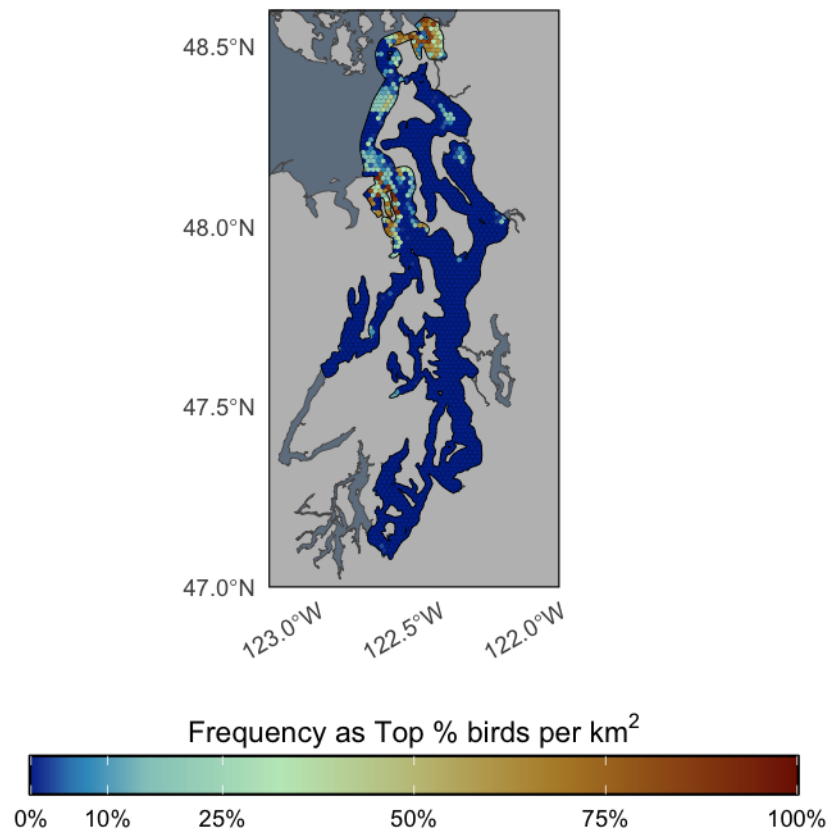
(my_top <-ggplot() +
  geom_sf(data = WA_State, fill ="gray") +
  geom_sf(data=puget, fill="slategray")+
  geom_sf(data = grid_freq_season, fill =NA, color = "black", linewidth=0.4) +
  geom_sf(data = grid_freq_season, aes(fill = prop_top5), color =NA)+
  scale_fill_gradientn(
    # colours = c("#001959", "#215F61", "#818231", "#F19D6B", "#F9CCF9"),
    colours = c("#023198", "#389CC6", "#C0E9C2", "#B58B31", "#7E1700"),
    values = scales::rescale(c(0, 0.05, 0.15, 0.5, 0.75, 1)),
    limits = c(0, 1),
    # name = expression("Birds per km"^2),
    expression("Frequency as Top % birds per km"^2),
    breaks = c(0, 0.1, 0.25, 0.5, 0.75, 1),
    labels = c("0%", "10%", "25%", "50%", "75%", "100%"),
    guide = guide_colourbar(
      ticks = TRUE,
      ticks.colour = "black",
      ticks.linewidth = 0.5,
      barheight = unit(8, "cm"),
      frame.colour = "black"))+
  coord_sf(
    xlim = c(-123.2, -122),
    ylim = c(47, 48.6),
    crs = st_crs(4326),
    expand = FALSE) +
  scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
  scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
  theme_minimal()+
  theme(
    legend.position = "bottom",
    legend.box = "vertical",
    legend.title = element_text(
      hjust = 0.5,
      margin = margin(b = 2)),
    axis.text.x = element_text(angle = 30, hjust = 1),
    #strip.text = element_text(size = 10),
    strip.placement = "outside",
    strip.text = element_blank(),
    strip.background = element_blank(),
    panel.spacing = unit(0, "lines"),

```

```

panel.border = element_rect(color = "black",
                             fill = NA, linewidth = 0.5)) +
guides(fill = guide_colourbar(
  title.position = "top",
  title.hjust = 0.5,
  direction = "horizontal",
  barwidth = unit(4, "in"),
  barheight = unit(0.2, "in"),
  ticks = TRUE,
  frame.colour = "black")))

```



```

#ggsave("RESULTS/MAPS/Freq_plot.pdf", plot = my_top, width = 12, height = 11, units = "in")
#ggsave("RESULTS/MAPS/Freq_plot.png", plot = my_top, width = 12, height = 11, units = "in", dpi = 600)

```

Breeding

```

top_5_frequency_b <- all_predictions %>%
  filter((mon == "MAY" | mon == "JUN" | mon == "JUL") & year != "ALL") %>%
  mutate(top5_cutoff = quantile(MED, probs = 0.95, na.rm = TRUE),
         is_top5 = MED >= top5_cutoff) %>%
  # Count how many times each grid cell is in the top 5%
  group_by(ID) %>%
  summarise(

```

```

n_surveyed = n(),
n_top5 = sum(is_top5, na.rm = TRUE),
prop_top5 = n_top5 / n_surveyed, .groups = "drop")

grid_freq_season <- grids %>%
  left_join(top_5_frequency_b, by = c("ID"))

## total area listed as top 5% at least once
grid_freq_season %>%
  st_drop_geometry() %>%
  filter(prop_top5 > 0) %>%
  summarise(sum(area))

##    sum(area)
## 1      226.34

# 226.34

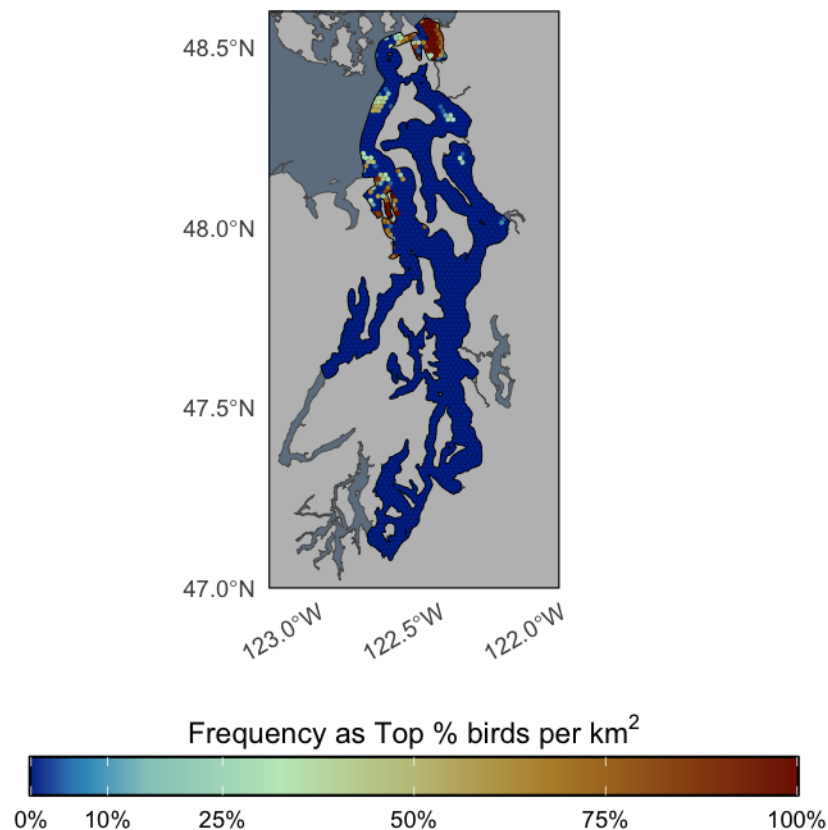
(my_top <-ggplot() +
  geom_sf(data = WA_State, fill = "gray") +
  geom_sf(data=puget, fill="slategray")+
  geom_sf(data = grid_freq_season, fill =NA, color = "black", linewidth=0.4) +
  geom_sf(data = grid_freq_season, aes(fill = prop_top5), color =NA)+
  scale_fill_gradientn(
    colours = c("#023198", "#389CC6", "#C0E9C2", "#B58B31", "#7E1700"),
    values = scales::rescale(c(0, 0.05, 0.15, 0.5, 0.75, 1)),
    limits = c(0, 1),
    # name = expression("Birds per km"^2),
    expression("Frequency as Top % birds per km"^2),
    breaks = c(0, 0.1, 0.25, 0.5, 0.75, 1),
    labels = c("0%", "10%", "25%", "50%", "75%", "100%"),
    guide = guide_colourbar(
      ticks = TRUE,
      ticks.colour = "black",
      ticks.linewidth = 0.5,
      barheight = unit(8, "cm"),
      frame.colour = "black"))+
  coord_sf(
    xlim = c(-123.2, -122),
    ylim = c(47, 48.6),
    crs = st_crs(4326),
    expand = FALSE) +
  scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
  scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
  theme_minimal()+
  theme(
    legend.position = "bottom",
    legend.box = "vertical",
    legend.title = element_text(
      hjust = 0.5,
      margin = margin(b = 2)),
    axis.text.x = element_text(angle = 30, hjust = 1),

```

```

#strip.text = element_text(size = 10),
strip.placement = "outside",
strip.text = element_blank(),
strip.background = element_blank(),
panel.spacing = unit(0, "lines"),
panel.border = element_rect(color = "black",
                             fill = NA, linewidth = 0.5)) +
guides(fill = guide_colourbar(
  title.position = "top",
  title.hjust = 0.5,
  direction = "horizontal",
  barwidth = unit(4, "in"),
  barheight = unit(0.2, "in"),
  ticks = TRUE,
  frame.colour = "black")))

```



```

#ggsave("RESULTS/MAPS/Freq_plot_b.pdf", plot = my_top, width = 12, height = 11, units = "in")
#ggsave("RESULTS/MAPS/Freq_plot_b.png", plot = my_top, width = 12, height = 11, units = "in", dpi = 600)

```

Non-breeding

```

top_5_frequency_nb <- all_predictions %>%
  filter(mon != "MAY" &

```

```

      mon != "JUN" &
      mon != "JUL" & mon != "B" & mon != "NB" & mon != "ALL" & year != "ALL") %>%
mutate(top5_cutoff = quantile(MED, probs = 0.95, na.rm = TRUE),
      is_top5 = MED >= top5_cutoff) %>%
# Count how many times each grid cell is in the top 5%
group_by(ID) %>%
summarise(
  n_surveyed = n(),
  n_top5 = sum(is_top5, na.rm = TRUE),
  prop_top5 = n_top5 / n_surveyed, .groups = "drop")

grid_freq_season <- grids %>%
  left_join(top_5_frequency_nb, by = c("ID"))

## total area listed as top 5% at least once
grid_freq_season %>%
  st_drop_geometry() %>%
  filter(prop_top5 > 0) %>%
  summarise(sum(area))

##    sum(area)
## 1      370.44

```

```

# 370.44

```

```

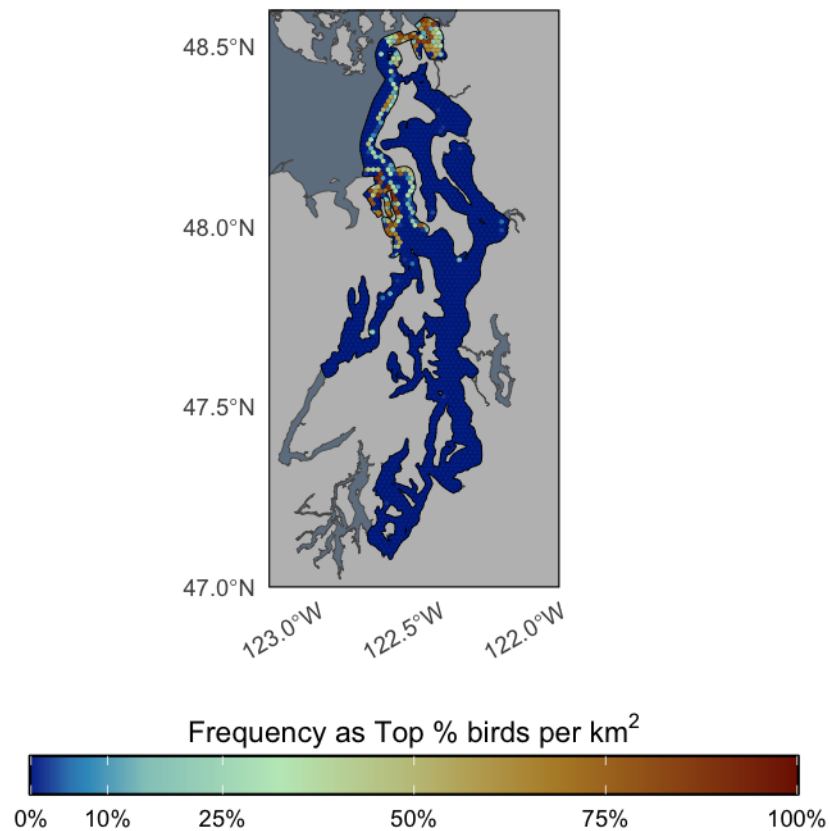
(my_top <- ggplot() +
  geom_sf(data = WA_State, fill = "gray") +
  geom_sf(data = puget, fill = "slategray") +
  geom_sf(data = grid_freq_season, fill = NA, color = "black", linewidth = 0.4) +
  geom_sf(data = grid_freq_season, aes(fill = prop_top5), color = NA) +
  scale_fill_gradientn(
    colours = c("#023198", "#389CC6", "#C0E9C2", "#B58B31", "#7E1700"),
    values = scales::rescale(c(0, 0.05, 0.15, 0.5, 0.75, 1)),
    limits = c(0, 1),
    # name = expression("Birds per km"^2),
    expression("Frequency as Top % birds per km"^2),
    breaks = c(0, 0.1, 0.25, 0.5, 0.75, 1),
    labels = c("0%", "10%", "25%", "50%", "75%", "100%"),
    guide = guide_colourbar(
      ticks = TRUE,
      ticks.colour = "black",
      ticks.linewidth = 0.5,
      barheight = unit(8, "cm"),
      frame.colour = "black")) +
  coord_sf(
    xlim = c(-123.2, -122),
    ylim = c(47, 48.6),
    crs = st_crs(4326),
    expand = FALSE) +
  scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +

```

```

scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
theme_minimal()+
theme(
  legend.position = "bottom",
  legend.box = "vertical",
  legend.title = element_text(
    hjust = 0.5,
    margin = margin(b = 2)),
  axis.text.x = element_text(angle = 30, hjust = 1),
  #strip.text = element_text(size = 10),
  strip.placement = "outside",
  strip.text = element_blank(),
  strip.background = element_blank(),
  panel.spacing = unit(0, "lines"),
  panel.border = element_rect(color = "black",
                                fill = NA, linewidth = 0.5)) +
guides(fill = guide_colourbar(
  title.position = "top",
  title.hjust = 0.5,
  direction = "horizontal",
  barwidth = unit(4, "in"),
  barheight = unit(0.2, "in"),
  ticks = TRUE,
  frame.colour = "black")))

```



```
#ggsave("RESULTS/MAPS/Freq_plot_nb.pdf", plot = my_top, width = 12, height = 11, units = "in")
#ggsave("RESULTS/MAPS/Freq_plot_nb.png", plot = my_top, width = 12, height = 11, units = "in", dpi = 600)
```

Figure 5: Average Monthly Density

JAN

```
jan <- grids %>%
  full_join(all_predictions %>%
    filter(mon == "JAN" & year == "ALL") %>%
    mutate(per_max = MED/max(MED),
           per_bin = case_when(
             per_max < 0.01 ~ "0-1%",
             per_max < 0.25 ~ "1-25%",
             per_max < 0.5  ~ "25-50%",
             TRUE ~ "50-100%"),
           per_bin = factor(per_bin,
                           levels = c("0-1%", "1-25%",
                                         "25-50%", "50-100%"))),
  by = c("numid", "ID"))

min_med <- jan %>%
  mutate(minnn = min(MED, na.rm = TRUE)) %>%
  distinct(minnn) %>%
  pull(minnn)

max_med <- jan %>%
  mutate(maxxx = max(MED, na.rm = TRUE)) %>%
  distinct(maxxx) %>%
  pull(maxxx)

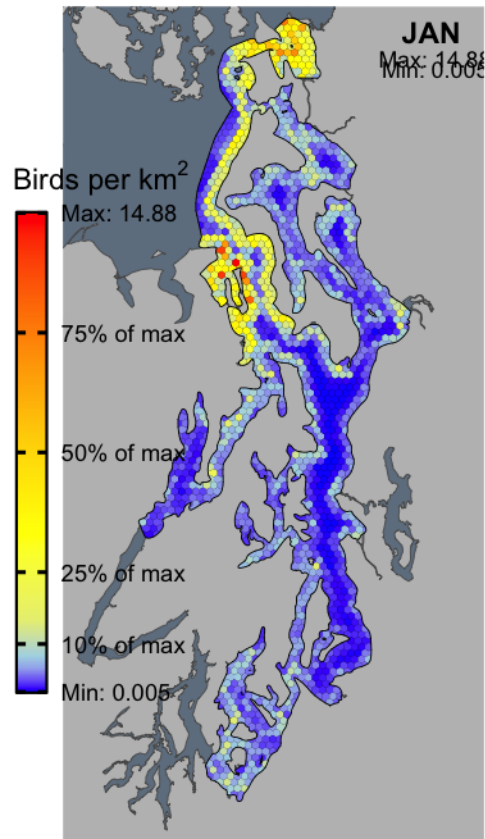
min_label <- paste0("Min: ", formatC(min_med, format = "f", digits = 3))
max_label <- paste0("Max: ", formatC(max_med, format = "f", digits = 2))

(JAN_Plot <- ggplot() +
  geom_sf(data = WA_State, fill = "gray") +
  geom_sf(data = puget, fill = "slategray") +
  geom_sf(data = jan,
          fill = NA, color = "black", linewidth = 0.4) +
  geom_sf(data = jan,
          aes(fill = per_max), color = NA) +
  scale_fill_gradientn(
    colours = c("blue", "lightblue", "yellow", "orange", "red"),
    values = scales::rescale(c(0, 0.05, 0.15, 0.5, 0.75, 1)),
    limits = c(0, 1),
    name = expression("Birds per km"^2),
    breaks = c(0, 0.1, 0.25, 0.5, 0.75, 1),
    labels = c(min_label, "10% of max", "25% of max",
               "50% of max", "75% of max", max_label),
    guide = guide_colourbar(
```

```

direction = "vertical",
title.position = "top",
title.hjust = 0.5,
label.position = "right",
label.hjust = 0,
ticks = TRUE,
ticks.colour = "black",
ticks.linewidth = 0.7,
barwidth = unit(4, "mm"),
barheight = unit(2.5, "in"),
frame.colour = "black",
frame.linewidth = 0.5,
draw.ulim = TRUE,
draw.llim = TRUE)) +
coord_sf(
  xlim = c(-123.2, -122),
  ylim = c(47, 48.6),
  crs = st_crs(4326),
  expand = FALSE) +
scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
theme_minimal() +
theme(
  legend.margin = margin(0, 0, 0, 0),
  # legend.position = "right",
  legend.position = c(0.09, 0.5),
  axis.text = element_blank(),
  axis.title = element_blank(),
  axis.ticks = element_blank())+
annotate("text", x = -122.3, y = 48.5,
  label = max_label, size = 3, hjust = 0) +
annotate("text", x = -122.29, y = 48.5 - 0.02,
  label = min_label, size = 3, hjust = 0)+
annotate("text", x = -122.15, y = 48.55,
  label = "JAN", fontface = "bold"))

```



Jan Main

```
# Histogram data
jan_vals <- jan %>%
  filter(!is.na(per_max)) %>%
  pull(per_max)

hist_obj <- hist(jan_vals, breaks = seq(0, 1, length.out = 100), plot = FALSE)

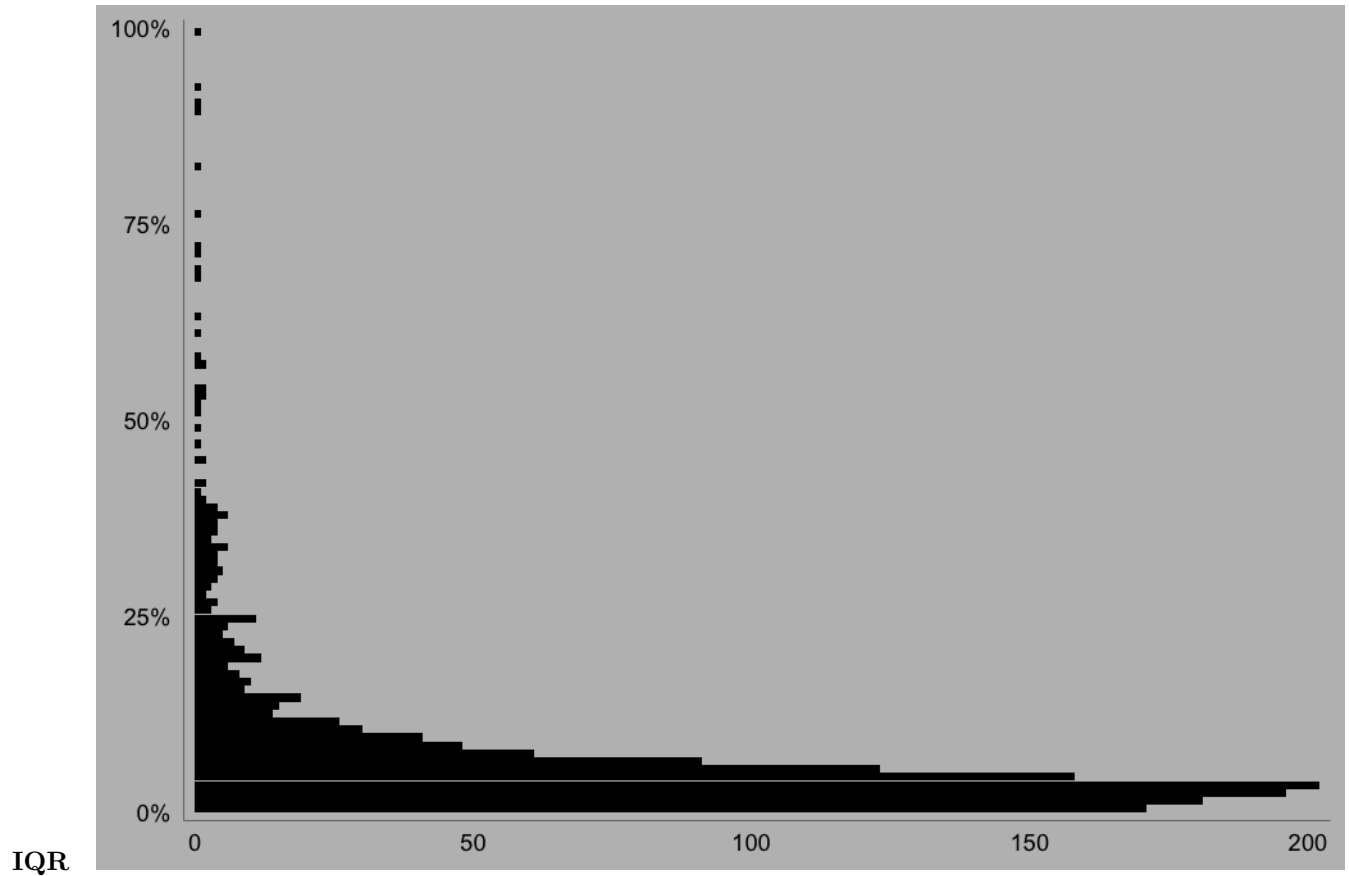
hist_data <- data.frame(
  bin_mid = hist_obj$mids,
  count = hist_obj$counts)

(hist_plot <- ggplot(hist_data, aes(x = bin_mid, y = count)) +
  geom_col(fill = "black", width = 0.01) +
  theme_minimal() +
  coord_flip() +
  theme(
    #plot.margin = margin(2, 2, 2, 2), # top, right, bottom, left in pts
    # axis.text = element_blank(),
    axis.line.x = element_line(linewidth = 0.1, linetype = "solid",
                                colour = "black"),
    axis.line.y = element_line(linewidth = 0.1, linetype = "solid",
                                colour = "black"),
    panel.grid = element_blank(),
```

```

panel.background = element_rect(fill = "gray",color = "gray"),
panel.spacing = unit(0, "pt"),
axis.text = element_text(color = "black"),
axis.title = element_blank(),
plot.background = element_rect(fill = "gray"),
axis.ticks = element_blank() +
scale_x_continuous(expand = c(0.01, 0),
                    labels = scales::percent_format(accuracy = 1)) +
scale_y_continuous(expand = c(0.01, 0))

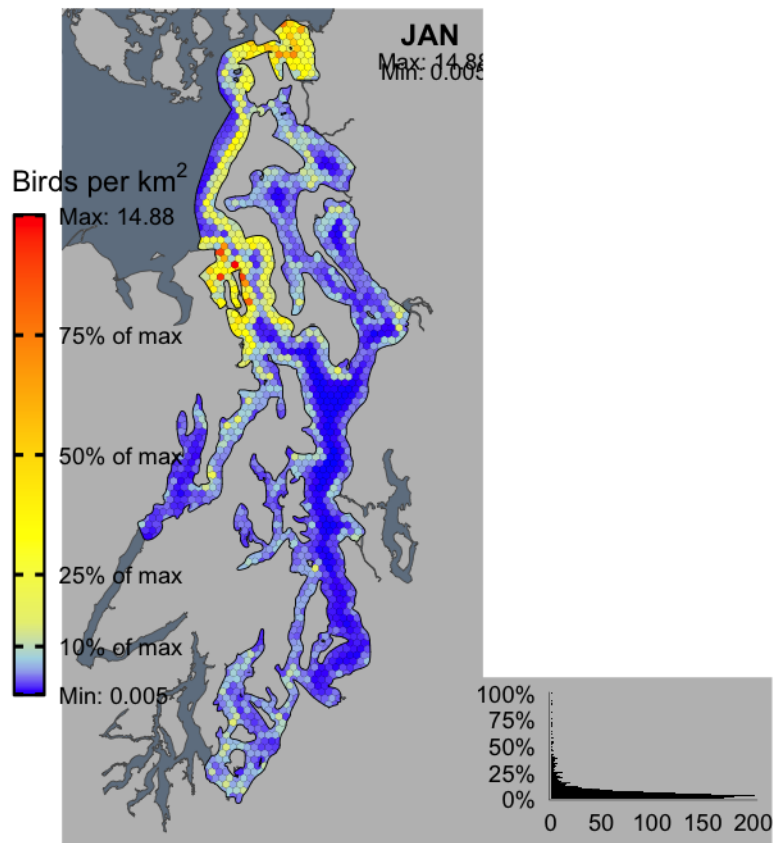
```



```

(jan_plot <- ggdraw(JAN_Plot) +
  draw_plot(hist_plot,
    x = 0.65, y = 0.01,
    width = 0.25, height = 0.2))

```



```
median(jan$IQR)
```

```
## [1] 0.2191259
```

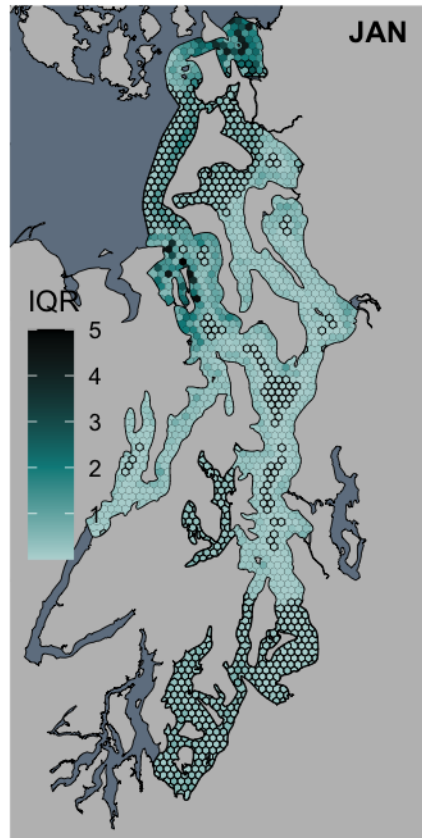
```
flag_jan <- jan %>%
  filter(flag == "CAP")

(jan_iqr <- ggplot() +
  geom_sf(data = WA_State, fill = "gray", color = "black") +
  geom_sf(data = puget, fill = "slategray", color = "black") +
  geom_sf(data = jan, fill = NA, color = "black", linewidth = 0.4) +
  geom_sf(data = jan, aes(fill = IQR), color = NA) +
  geom_sf(data = flag_jan, fill = NA, color = "black") +
  coord_sf(
    xlim = c(-123.2, -122),
    ylim = c(47, 48.6),
    crs = st_crs(4326),
    expand = FALSE) +
  scale_fill_gradient2(low = "white", mid = "cyan4", high = "black",
    midpoint = 2) +
  scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
  scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
  theme_minimal() +
  theme(
    legend.margin = margin(0, 0, 0, 0),
```

```

# legend.position = "right",
legend.position = c(0.13, 0.5),
axis.text = element_blank(),
axis.title = element_blank(),
axis.ticks = element_blank()+
annotate("text", x = -122.15, y = 48.55,
         label = "JAN", fontface = "bold"))

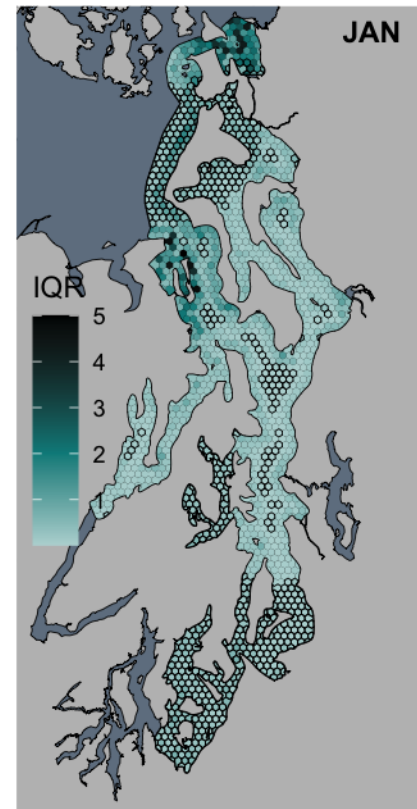
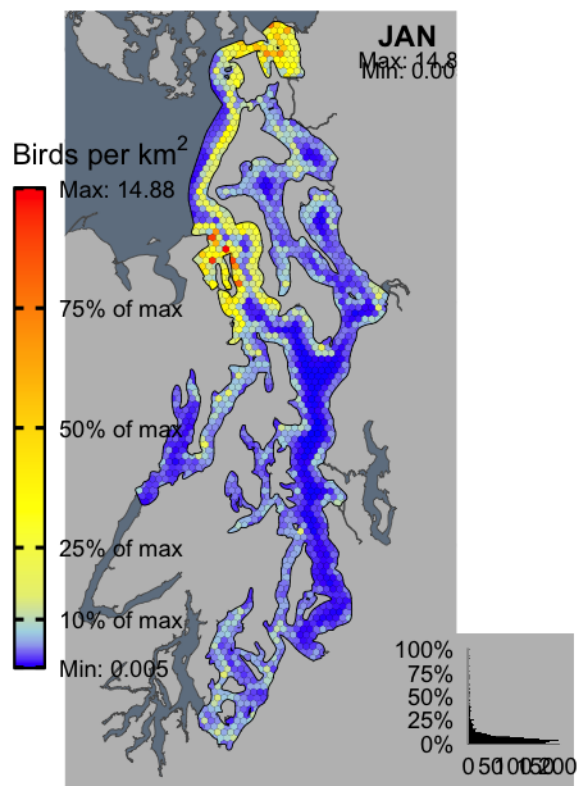
```



```

(jan_plot2 <- jan_plot | jan_iqr)

```



```
#ggsave("RESULTS/MAPS/jan_plot.pdf", plot = jan_plot2, width = 12, height = 11, units = "in")
#ggsave("RESULTS/MAPS/jan_plot.png", plot = jan_plot2, width = 12, height = 11, units = "in", dpi = 600)
```

FEB

```
feb <- grids %>%
  full_join(all_predictions %>%
    filter(mon == "FEB" & year == "ALL") %>%
    mutate(per_max = MED/max(MED),
           per_bin = case_when(
             per_max < 0.01 ~ "0-1%",
             per_max < 0.25 ~ "1-25%",
             per_max < 0.5 ~ "25-50%",
             TRUE ~ "50-100%"),
           per_bin = factor(per_bin,
                           levels = c("0-1%", "1-25%",
                                       "25-50%", "50-100%"))),
    by = c("numid", "ID"))

min_med <- feb %>%
  mutate(minnn = min(MED, na.rm = TRUE)) %>%
  distinct(minnn) %>%
  pull(minnn)
```

```

max_med <- feb %>%
  mutate(maxxx = max(MED, na.rm = TRUE)) %>%
  distinct(maxxx) %>%
  pull(maxxx)

min_label <- paste0("Min: ", formatC(min_med, format = "f", digits = 3))
max_label <- paste0("Max: ", formatC(max_med, format = "f", digits = 2))

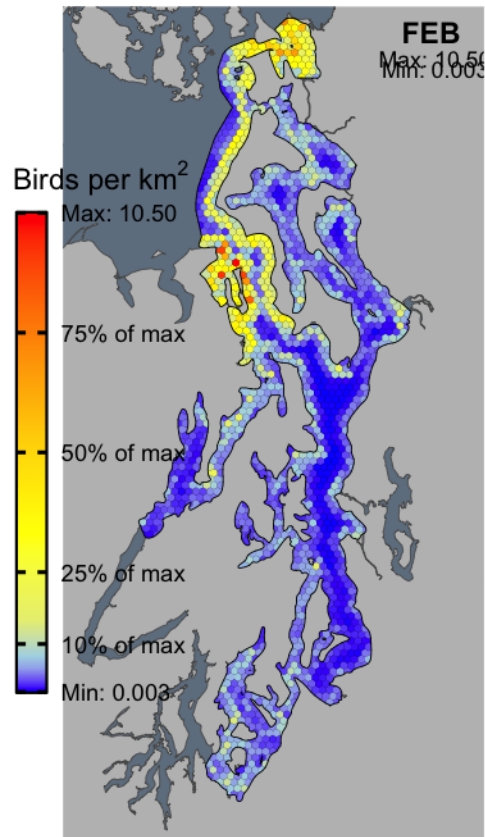
(FEB_Plot <- ggplot() +
  geom_sf(data = WA_State, fill = "gray") +
  geom_sf(data = puget, fill = "slategray") +
  geom_sf(data = feb,
    fill = NA, color = "black", linewidth=0.4) +
  geom_sf(data = feb,
    aes(fill = per_max), color = NA) +
  scale_fill_gradientn(
    colours = c("blue", "lightblue", "yellow", "orange", "red"),
    values = scales::rescale(c(0, 0.05, 0.15, 0.5, 0.75, 1)),
    limits = c(0, 1),
    name = expression("Birds per km"^2),
    breaks = c(0, 0.1, 0.25, 0.5, 0.75, 1),
    labels = c(min_label, "10% of max", "25% of max",
      "50% of max", "75% of max", max_label),
    guide = guide_colourbar(
      direction = "vertical",
      title.position = "top",
      title.hjust = 0.5,
      label.position = "right",
      label.hjust = 0,
      ticks = TRUE,
      ticks.colour = "black",
      ticks.linewidth = 0.7,
      barwidth = unit(4, "mm"),
      barheight = unit(2.5, "in"),
      frame.colour = "black",
      frame.linewidth = 0.5,
      draw.ulim = TRUE,
      draw.llim = TRUE)) +
  coord_sf(
    xlim = c(-123.2, -122),
    ylim = c(47, 48.6),
    crs = st_crs(4326),
    expand = FALSE) +
  scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
  scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
  theme_minimal() +
  theme(
    legend.margin = margin(0, 0, 0, 0),
    # legend.position = "right",
    legend.position = c(0.09, 0.5),
    axis.text = element_blank(),
    axis.title = element_blank(),
    axis.ticks = element_blank())+

```

```

annotate("text", x = -122.3, y = 48.5,
  label = max_label, size = 3, hjust = 0) +
annotate("text", x = -122.29, y = 48.5 - 0.02,
  label = min_label, size = 3, hjust = 0)+
annotate("text", x = -122.15, y = 48.55,
  label = "FEB", fontface = "bold"))

```



Feb Main

```

# Histogram data
feb_vals <- feb %>%
  filter(!is.na(per_max)) %>%
  pull(per_max)

hist_obj <- hist(feb_vals, breaks = seq(0, 1, length.out = 100), plot = FALSE)

hist_data <- data.frame(
  bin_mid = hist_obj$mids,
  count = hist_obj$counts)

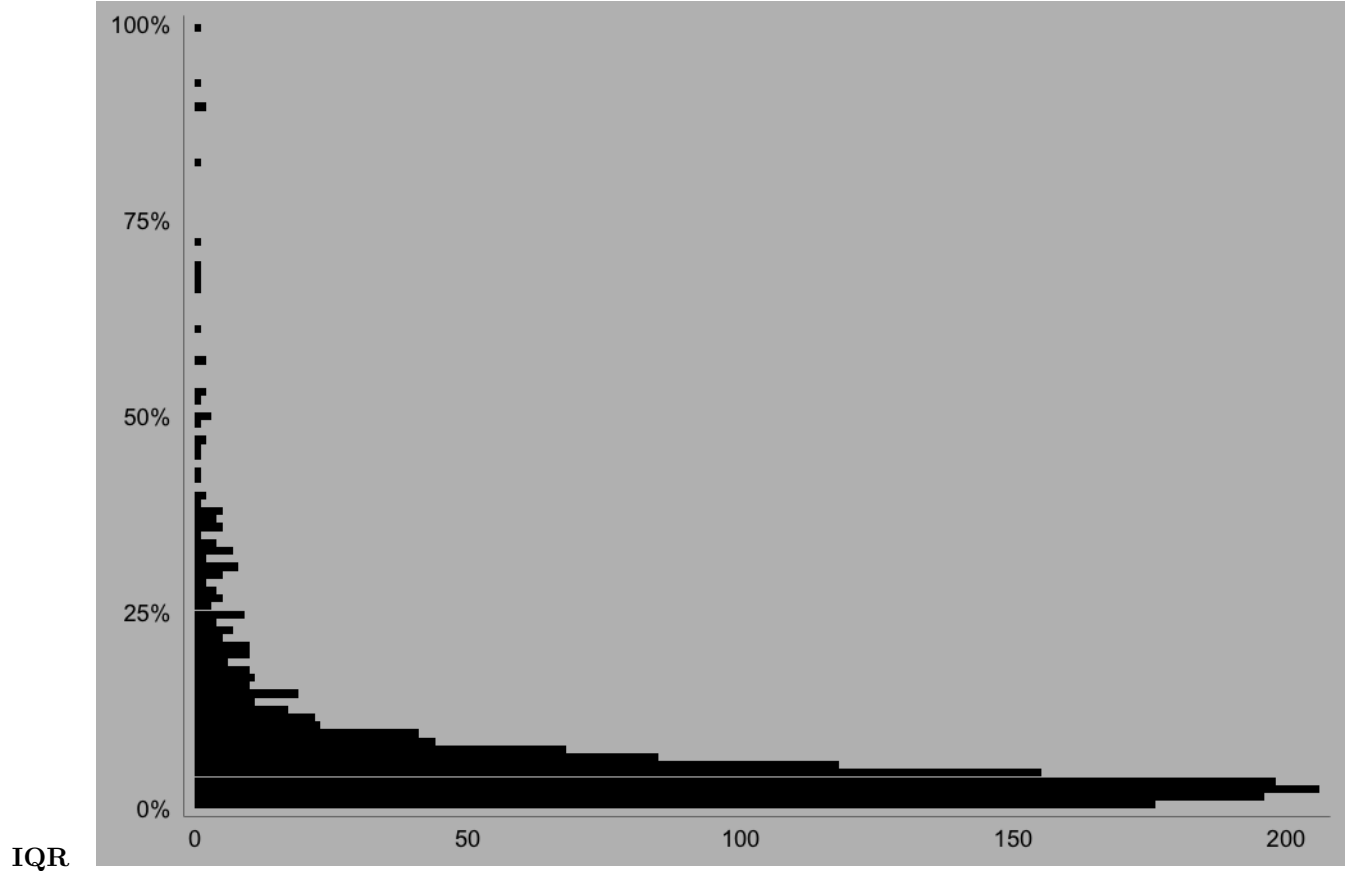
(hist_plot <- ggplot(hist_data, aes(x = bin_mid, y = count)) +
  geom_col(fill = "black", width = 0.01) +
  theme_minimal() +
  coord_flip()+

```

```

theme(
  #plot.margin = margin(2, 2, 2, 2), # top, right, bottom, left in pts
  # axis.text = element_blank(),
  axis.line.x = element_line(linewidth = 0.1, linetype = "solid",
                              colour = "black"),
  axis.line.y = element_line(linewidth = 0.1, linetype = "solid",
                              colour = "black"),
  panel.grid = element_blank(),
  panel.background = element_rect(fill = "gray", color = "gray"),
  panel.spacing = unit(0, "pt"),
  axis.text = element_text(color = "black"),
  axis.title = element_blank(),
  plot.background = element_rect(fill = "gray"),
  axis.ticks = element_blank() +
scale_x_continuous(expand = c(0.01, 0),
                    labels = scales::percent_format(accuracy = 1)) +
scale_y_continuous(expand = c(0.01, 0)))

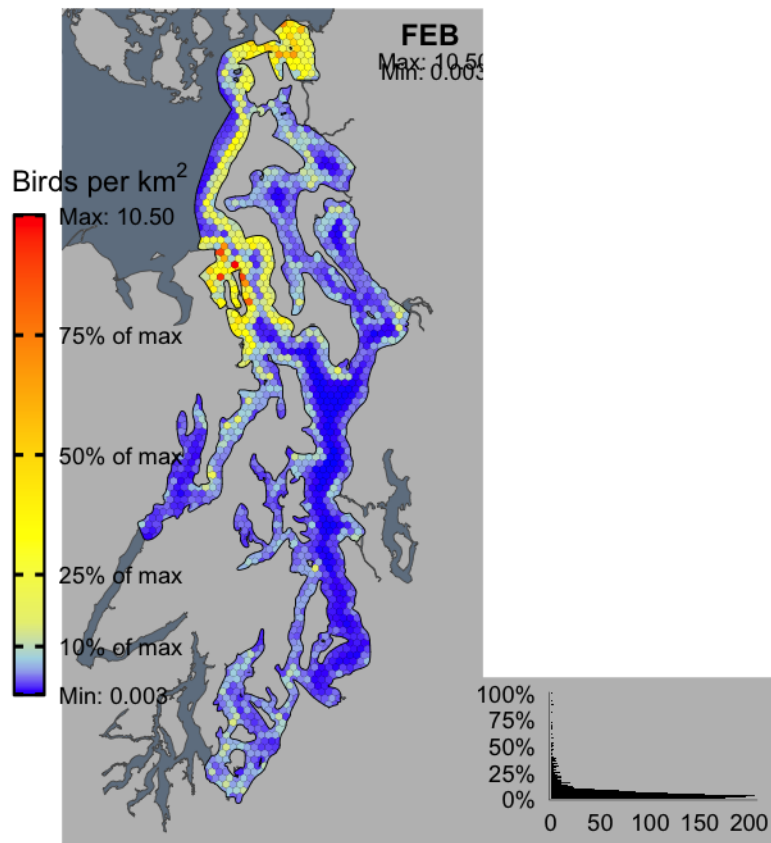
```



```

(feb_plot <- ggdraw(FEB_Plot) +
  draw_plot(hist_plot,
    x = 0.65, y = 0.01,
    width = 0.25, height = 0.2))

```



```
max(feb$IQR)
```

```
## [1] 4.041957
```

```
median(feb$IQR)
```

```
## [1] 0.1727196
```

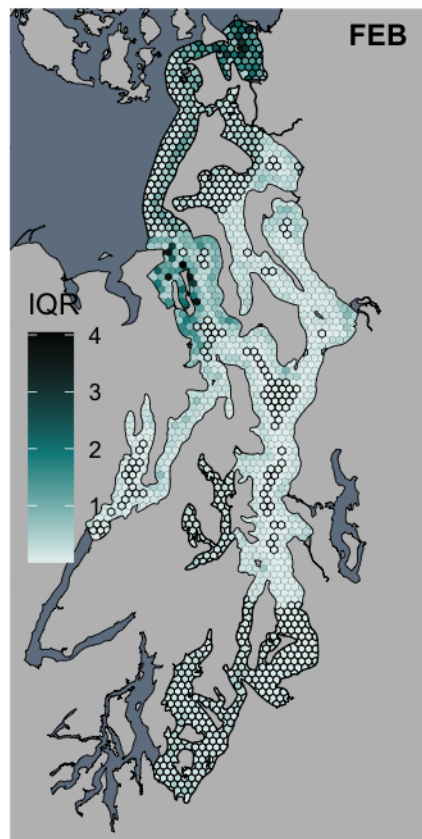
```
flag_feb <- feb %>%
  filter(flag == "CAP")

(feb_iqr <- ggplot() +
  geom_sf(data = WA_State, fill = "gray", color = "black") +
  geom_sf(data = puget, fill = "slategray", color = "black") +
  geom_sf(data = feb, fill = NA, color = "black", linewidth = 0.4) +
  geom_sf(data = feb, aes(fill = IQR), color = NA) +
  geom_sf(data = flag_feb, fill = NA, color = "black") +
  coord_sf(
    xlim = c(-123.2, -122),
    ylim = c(47, 48.6),
    crs = st_crs(4326),
    expand = FALSE) +
  scale_fill_gradient2(low = "white", mid = "cyan4", high = "black",
    midpoint = 1.9) +
```

```

scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
theme_minimal() +
theme(
  legend.margin = margin(0, 0, 0, 0),
  # legend.position = "right",
  legend.position = c(0.13, 0.5),
  axis.text = element_blank(),
  axis.title = element_blank(),
  axis.ticks = element_blank())+
annotate("text", x = -122.15, y = 48.55,
  label = "FEB", fontface = "bold"))

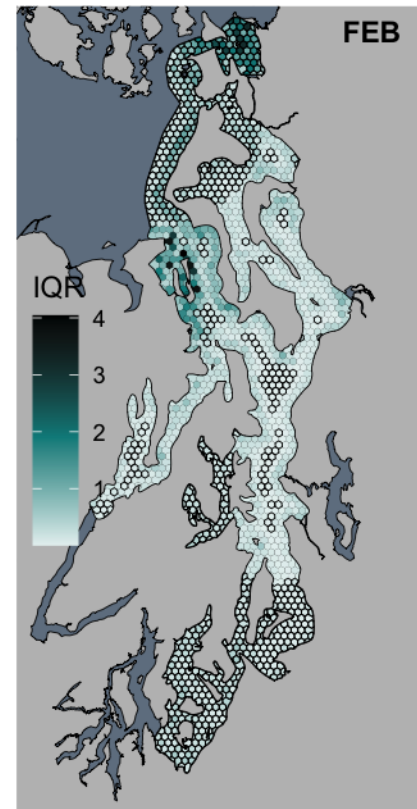
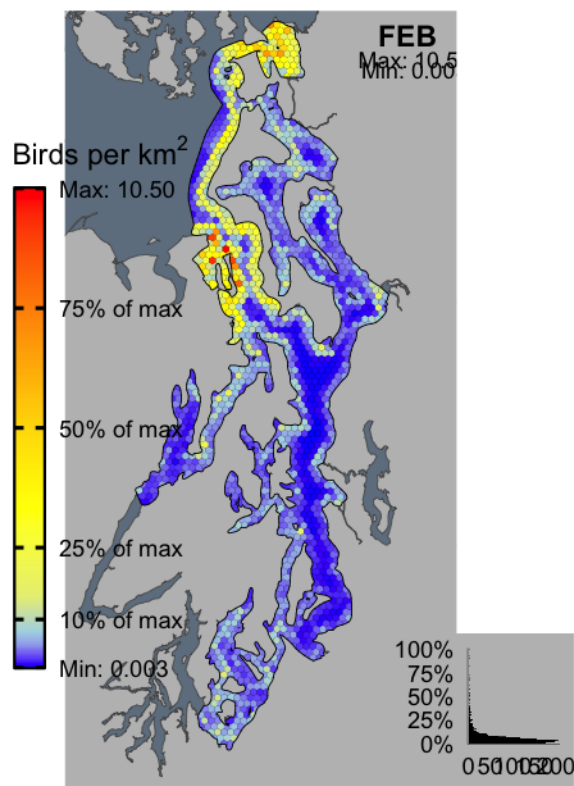
```



```

(feb_plot2 <- feb_plot | feb_iqr)

```



```
#ggsave("RESULTS/MAPS/feb_plot.pdf", plot = feb_plot2, width = 12, height = 11, units = "in")
#ggsave("RESULTS/MAPS/feb_plot.png", plot = feb_plot2, width = 12, height = 11, units = "in", dpi = 600)
```

MAR

```
mar <- grids %>%
  full_join(all_predictions %>%
    filter(mon == "MAR" & year == "ALL") %>%
    mutate(per_max = MED/max(MED),
           per_bin = case_when(
             per_max < 0.01 ~ "0-1%",
             per_max < 0.25 ~ "1-25%",
             per_max < 0.5 ~ "25-50%",
             TRUE ~ "50-100%"),
           per_bin = factor(per_bin,
                           levels = c("0-1%", "1-25%",
                                       "25-50%", "50-100%"))),
    by = c("numid", "ID"))

min_med <- mar %>%
  mutate(minnn = min(MED, na.rm = TRUE)) %>%
  distinct(minnn) %>%
  pull(minnn)
```

```

max_med <- mar %>%
  mutate(maxxx = max(MED, na.rm = TRUE)) %>%
  distinct(maxxx) %>%
  pull(maxxx)

min_label <- paste0("Min: ", formatC(min_med, format = "f", digits = 3))
max_label <- paste0("Max: ", formatC(max_med, format = "f", digits = 2))

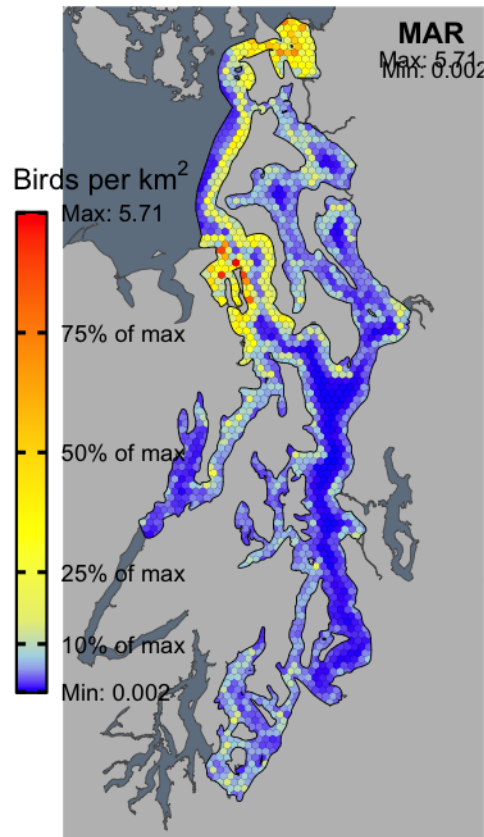
(MAR_Plot <- ggplot() +
  geom_sf(data = WA_State, fill = "gray") +
  geom_sf(data = puget, fill = "slategray") +
  geom_sf(data = mar,
    fill = NA, color = "black", linewidth=0.4) +
  geom_sf(data = mar,
    aes(fill = per_max), color = NA) +
  scale_fill_gradientn(
    colours = c("blue", "lightblue", "yellow", "orange", "red"),
    values = scales::rescale(c(0, 0.05, 0.15, 0.5, 0.75, 1)),
    limits = c(0, 1),
    name = expression("Birds per km"^2),
    breaks = c(0, 0.1, 0.25, 0.5, 0.75, 1),
    labels = c(min_label, "10% of max", "25% of max",
      "50% of max", "75% of max", max_label),
    guide = guide_colourbar(
      direction = "vertical",
      title.position = "top",
      title.hjust = 0.5,
      label.position = "right",
      label.hjust = 0,
      ticks = TRUE,
      ticks.colour = "black",
      ticks.linewidth = 0.7,
      barwidth = unit(4, "mm"),
      barheight = unit(2.5, "in"),
      frame.colour = "black",
      frame.linewidth = 0.5,
      draw.ulim = TRUE,
      draw.llim = TRUE)) +
  coord_sf(
    xlim = c(-123.2, -122),
    ylim = c(47, 48.6),
    crs = st_crs(4326),
    expand = FALSE) +
  scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
  scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
  theme_minimal() +
  theme(
    legend.margin = margin(0, 0, 0, 0),
    # legend.position = "right",
    legend.position = c(0.09, 0.5),
    axis.text = element_blank(),
    axis.title = element_blank(),
    axis.ticks = element_blank())+

```

```

annotate("text", x = -122.3, y = 48.5,
        label = max_label, size = 3, hjust = 0) +
annotate("text", x = -122.29, y = 48.5 - 0.02,
        label = min_label, size = 3, hjust = 0) +
annotate("text", x = -122.15, y = 48.55,
        label = "MAR", fontface = "bold"))

```



Mar Main

```

# Histogram data
mar_vals <- mar %>%
  filter(!is.na(per_max)) %>%
  pull(per_max)

hist_obj <- hist(mar_vals, breaks = seq(0, 1, length.out = 100), plot = FALSE)

hist_data <- data.frame(
  bin_mid = hist_obj$mids,
  count = hist_obj$counts)

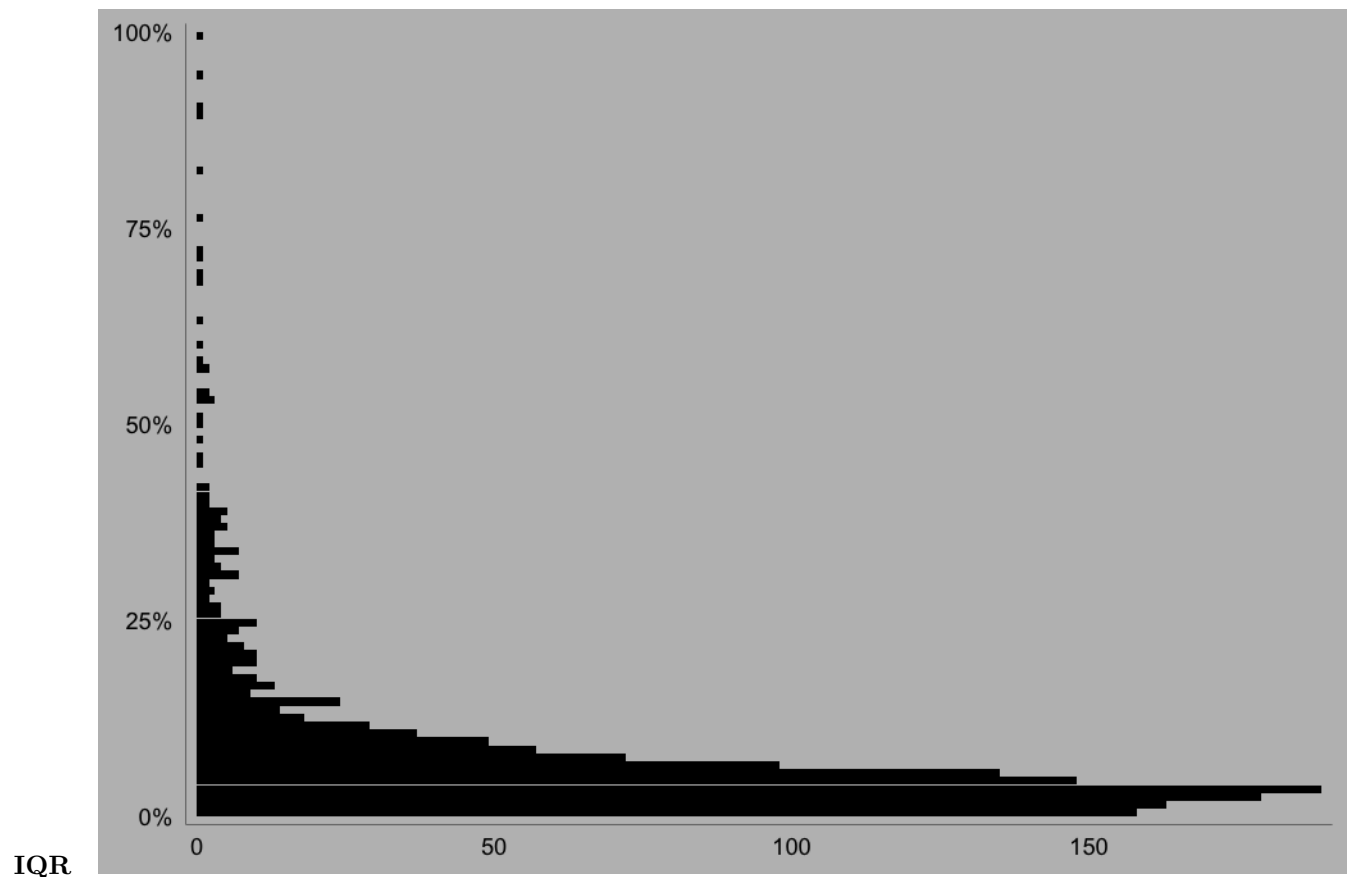
(hist_plot <- ggplot(hist_data, aes(x = bin_mid, y = count)) +
  geom_col(fill = "black", width = 0.01) +
  theme_minimal() +
  coord_flip() +

```

```

theme(
  #plot.margin = margin(2, 2, 2, 2), # top, right, bottom, left in pts
  # axis.text = element_blank(),
  axis.line.x = element_line(linewidth = 0.1, linetype = "solid",
                             colour = "black"),
  axis.line.y = element_line(linewidth = 0.1, linetype = "solid",
                             colour = "black"),
  panel.grid = element_blank(),
  panel.background = element_rect(fill = "gray", color = "gray"),
  panel.spacing = unit(0, "pt"),
  axis.text = element_text(color = "black"),
  axis.title = element_blank(),
  plot.background = element_rect(fill = "gray"),
  axis.ticks = element_blank() +
scale_x_continuous(expand = c(0.01, 0),
                   labels = scales::percent_format(accuracy = 1)) +
scale_y_continuous(expand = c(0.01, 0)))

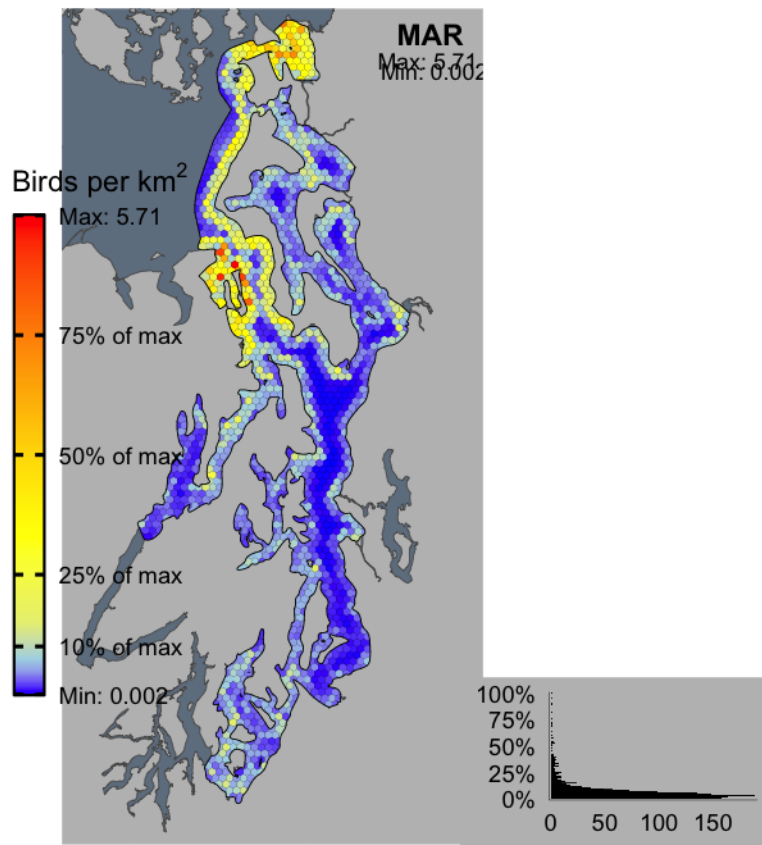
```



```

(mar_plot <- ggdraw(MAR_Plot) +
  draw_plot(hist_plot,
    x = 0.65, y = 0.01,
    width = 0.25, height = 0.2))

```



```
max(mar$IQR)
```

```
## [1] 2.146005
```

```
mean(mar$IQR)
```

```
## [1] 0.1756967
```

```
median(mar$IQR)
```

```
## [1] 0.09817337
```

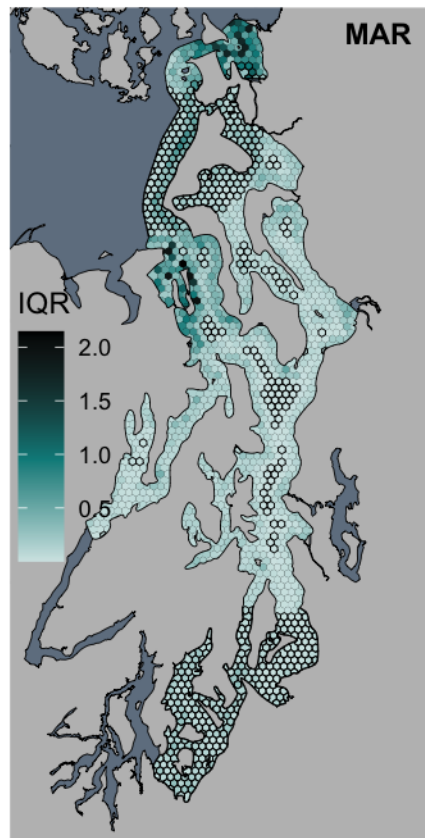
```
flag_mar <- mar %>%
  filter(flag == "CAP")

(mar_iqr <- ggplot() +
  geom_sf(data = WA_State, fill = "gray", color = "black") +
  geom_sf(data = puget, fill = "slategray", color = "black") +
  geom_sf(data = mar, fill = NA, color = "black", linewidth = 0.4) +
  geom_sf(data = mar, aes(fill = IQR), color = NA) +
  geom_sf(data = flag_mar, fill = NA, color = "black") +
  coord_sf(
    xlim = c(-123.2, -122),
```

```

ylim = c(47, 48.6),
crs = st_crs(4326),
expand = FALSE) +
scale_fill_gradient2(low = "white", mid = "cyan4", high = "black",
                     midpoint = 0.95)+
scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
theme_minimal() +
theme(
  legend.margin = margin(0, 0, 0, 0),
  # legend.position = "right",
  legend.position = c(0.13, 0.5),
  axis.text = element_blank(),
  axis.title = element_blank(),
  axis.ticks = element_blank())+
annotate("text", x = -122.15, y = 48.55,
         label = "MAR", fontface = "bold"))

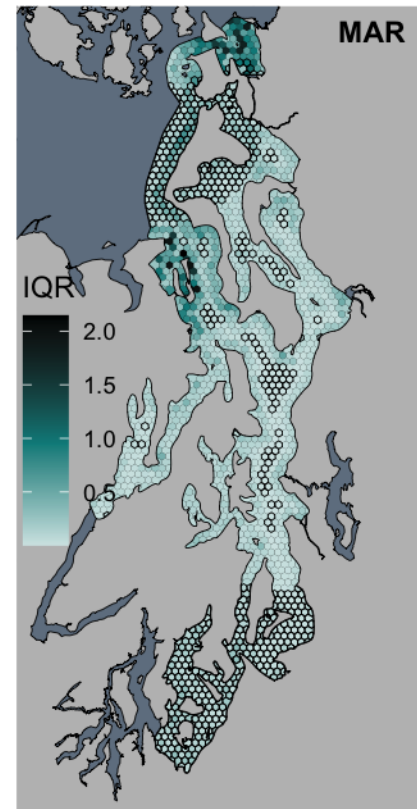
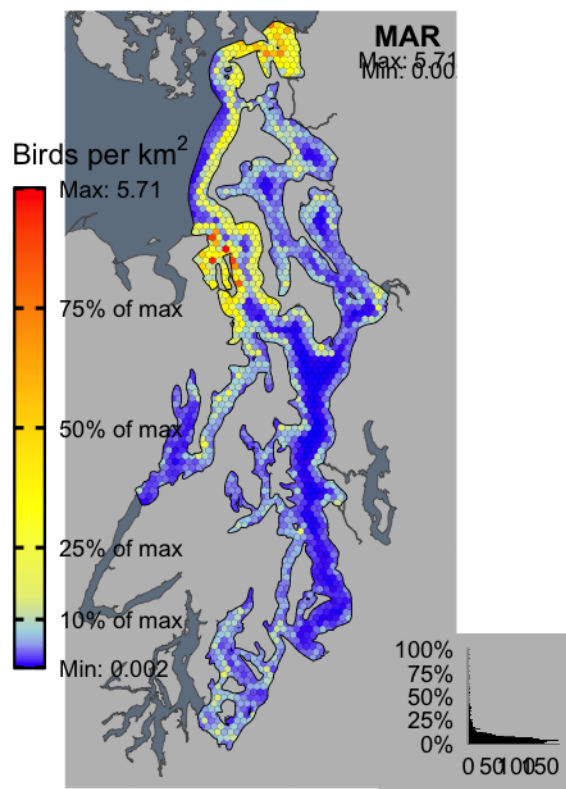
```



```

(mar_plot2 <- mar_plot | mar_iqr)

```



```
#ggsave("RESULTS/MAPS/mar_plot.pdf", plot = mar_plot2, width = 12, height = 11, units = "in")
#ggsave("RESULTS/MAPS/mar_plot.png", plot = mar_plot2, width = 12, height = 11, units = "in", dpi = 600)
```

APR

```
apr <- grids %>%
  full_join(all_predictions %>%
    filter(mon == "APR" & year == "ALL") %>%
    mutate(per_max = MED/max(MED),
           per_bin = case_when(
             per_max < 0.01 ~ "0-1%",
             per_max < 0.25 ~ "1-25%",
             per_max < 0.5 ~ "25-50%",
             TRUE ~ "50-100%"),
           per_bin = factor(per_bin,
                           levels = c("0-1%", "1-25%",
                                       "25-50%", "50-100%"))),
    by = c("numid", "ID"))

min_med <- apr %>%
  mutate(minnn = min(MED, na.rm = TRUE)) %>%
  distinct(minnn) %>%
  pull(minnn)
```

```

max_med <- apr %>%
  mutate(maxxx = max(MED, na.rm = TRUE)) %>%
  distinct(maxxx) %>%
  pull(maxxx)

min_label <- paste0("Min: ", formatC(min_med, format = "f", digits = 3))
max_label <- paste0("Max: ", formatC(max_med, format = "f", digits = 2))

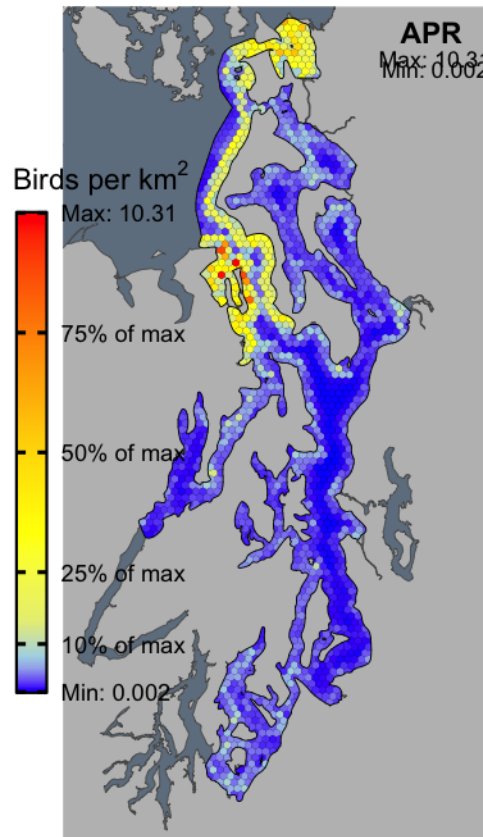
(APR_Plot <- ggplot() +
  geom_sf(data = WA_State, fill = "gray") +
  geom_sf(data = puget, fill = "slategray") +
  geom_sf(data = apr,
    fill = NA, color = "black", linewidth=0.4) +
  geom_sf(data = apr,
    aes(fill = per_max), color = NA) +
  scale_fill_gradientn(
    colours = c("blue", "lightblue", "yellow", "orange", "red"),
    values = scales::rescale(c(0, 0.05, 0.15, 0.5, 0.75, 1)),
    limits = c(0, 1),
    name = expression("Birds per km"^2),
    breaks = c(0, 0.1, 0.25, 0.5, 0.75, 1),
    labels = c(min_label, "10% of max", "25% of max",
      "50% of max", "75% of max", max_label),
    guide = guide_colourbar(
      direction = "vertical",
      title.position = "top",
      title.hjust = 0.5,
      label.position = "right",
      label.hjust = 0,
      ticks = TRUE,
      ticks.colour = "black",
      ticks.linewidth = 0.7,
      barwidth = unit(4, "mm"),
      barheight = unit(2.5, "in"),
      frame.colour = "black",
      frame.linewidth = 0.5,
      draw.ulim = TRUE,
      draw.llim = TRUE)) +
  coord_sf(
    xlim = c(-123.2, -122),
    ylim = c(47, 48.6),
    crs = st_crs(4326),
    expand = FALSE) +
  scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
  scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
  theme_minimal() +
  theme(
    legend.margin = margin(0, 0, 0, 0),
    # legend.position = "right",
    legend.position = c(0.09, 0.5),
    axis.text = element_blank(),
    axis.title = element_blank(),
    axis.ticks = element_blank())+

```

```

annotate("text", x = -122.3, y = 48.5,
        label = max_label, size = 3, hjust = 0) +
annotate("text", x = -122.29, y = 48.5 - 0.02,
        label = min_label, size = 3, hjust = 0) +
annotate("text", x = -122.15, y = 48.55,
        label = "APR", fontface = "bold"))

```



Apr Main

```

# Histogram data
apr_vals <- apr %>%
  filter(!is.na(per_max)) %>%
  pull(per_max)

hist_obj <- hist(apr_vals, breaks = seq(0, 1, length.out = 100), plot = FALSE)

hist_data <- data.frame(
  bin_mid = hist_obj$mids,
  count = hist_obj$counts)

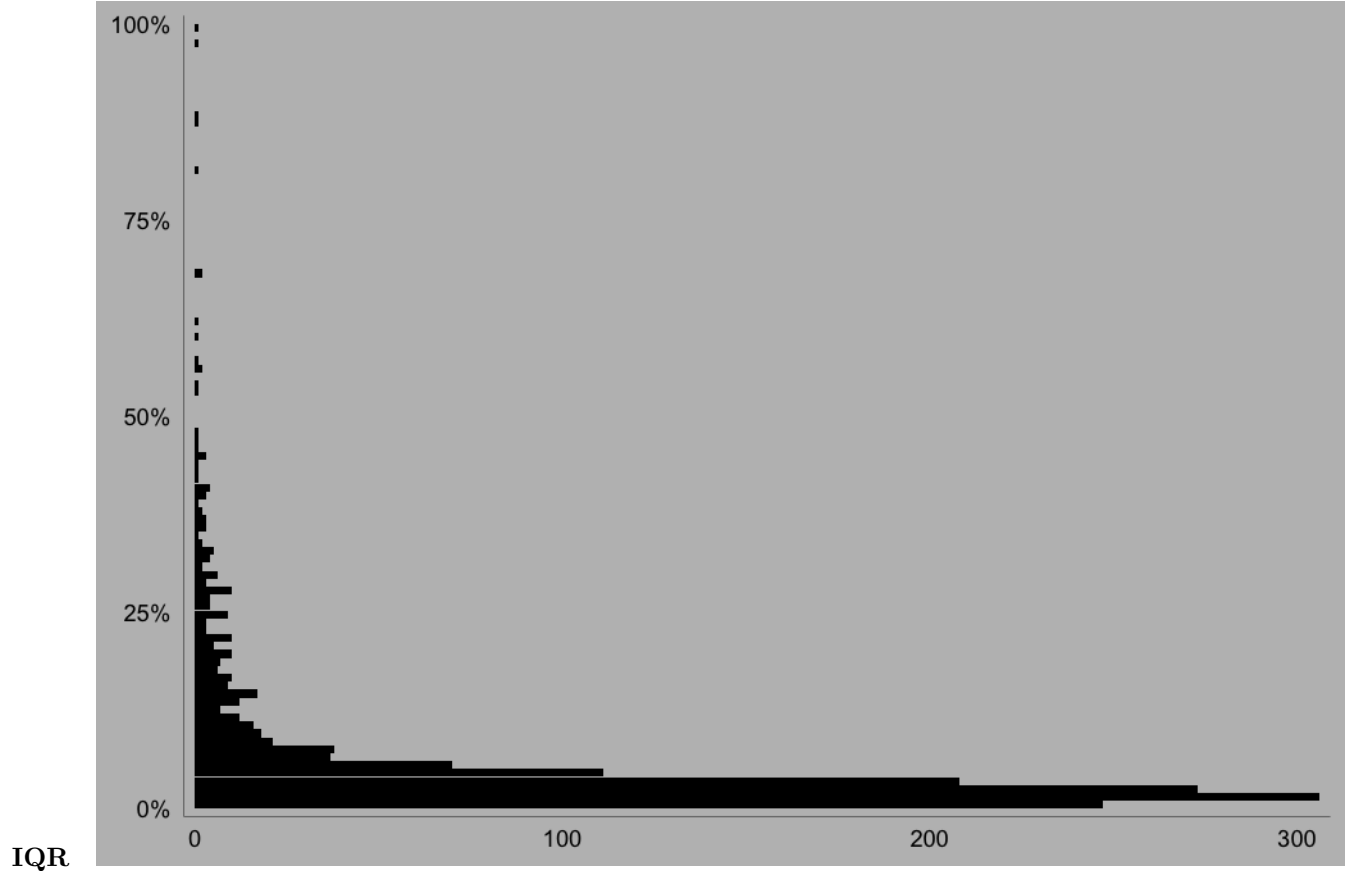
(hist_plot <- ggplot(hist_data, aes(x = bin_mid, y = count)) +
  geom_col(fill = "black", width = 0.01) +
  theme_minimal() +
  coord_flip() +

```

```

theme(
  #plot.margin = margin(2, 2, 2, 2), # top, right, bottom, left in pts
  # axis.text = element_blank(),
  axis.line.x = element_line(linewidth = 0.1, linetype = "solid",
                              colour = "black"),
  axis.line.y = element_line(linewidth = 0.1, linetype = "solid",
                              colour = "black"),
  panel.grid = element_blank(),
  panel.background = element_rect(fill = "gray", color = "gray"),
  panel.spacing = unit(0, "pt"),
  axis.text = element_text(color = "black"),
  axis.title = element_blank(),
  plot.background = element_rect(fill = "gray"),
  axis.ticks = element_blank()) +
scale_x_continuous(expand = c(0.01, 0),
                   labels = scales::percent_format(accuracy = 1)) +
scale_y_continuous(expand = c(0.01, 0))

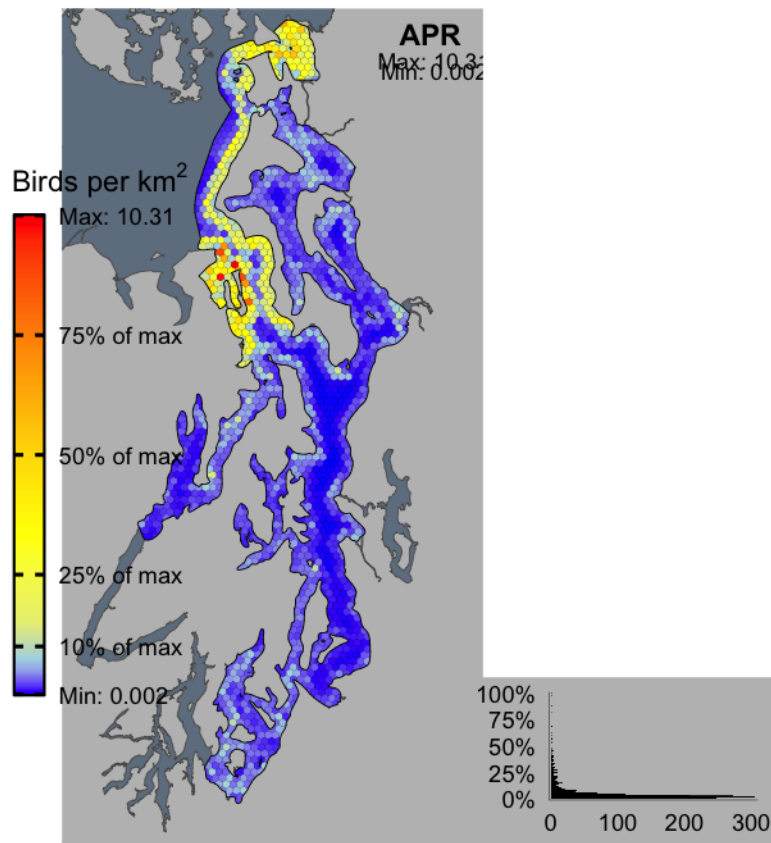
```



```

(apr_plot <- ggdraw(APR_Plot) +
  draw_plot(hist_plot,
            x = 0.65, y = 0.01,
            width = 0.25, height = 0.2))

```



```
max(apr$IQR)
```

```
## [1] 4.386139
```

```
mean(apr$IQR)
```

```
## [1] 0.2723756
```

```
median(apr$IQR)
```

```
## [1] 0.1241323
```

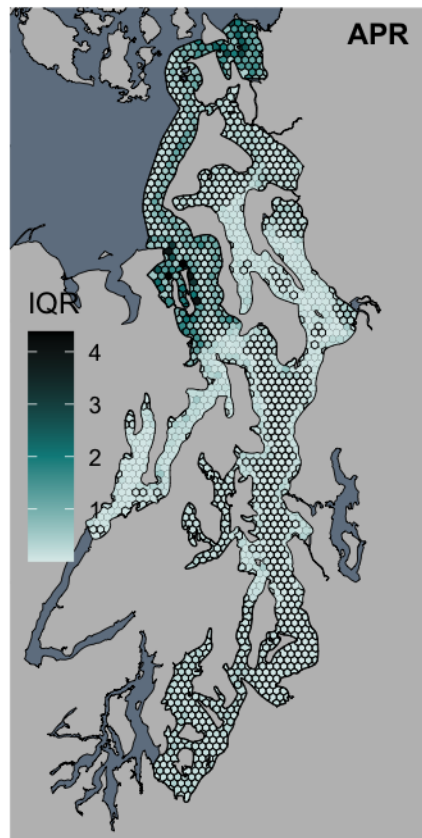
```
flag_apr <- apr %>%
  filter(flag == "CAP")

(apr_iqr <- ggplot() +
  geom_sf(data = WA_State, fill = "gray", color = "black") +
  geom_sf(data = puget, fill = "slategray", color = "black") +
  geom_sf(data = apr, fill = NA, color = "black", linewidth=0.4) +
  geom_sf(data = apr, aes(fill = IQR), color = NA) +
  geom_sf(data = flag_apr, fill = NA, color = "black") +
  coord_sf(
    xlim = c(-123.2, -122),
```

```

ylim = c(47, 48.6),
crs = st_crs(4326),
expand = FALSE) +
scale_fill_gradient2(low = "white", mid = "cyan4", high = "black",
                     midpoint = 2)+
scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
theme_minimal() +
theme(
  legend.margin = margin(0, 0, 0, 0),
  # legend.position = "right",
  legend.position = c(0.13, 0.5),
  axis.text = element_blank(),
  axis.title = element_blank(),
  axis.ticks = element_blank())+
annotate("text", x = -122.15, y = 48.55,
         label = "APR", fontface = "bold"))

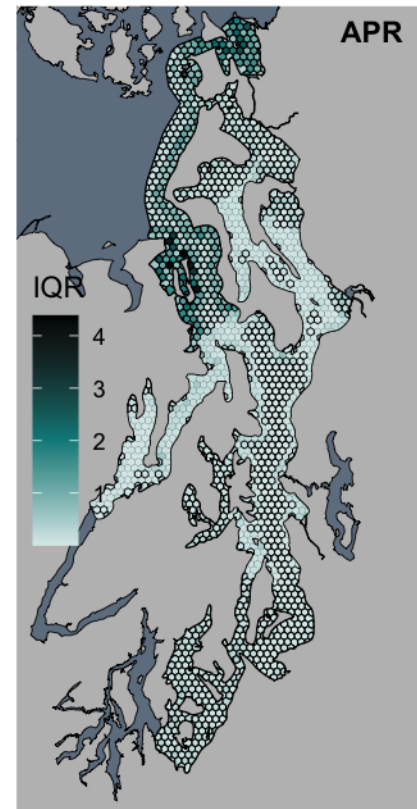
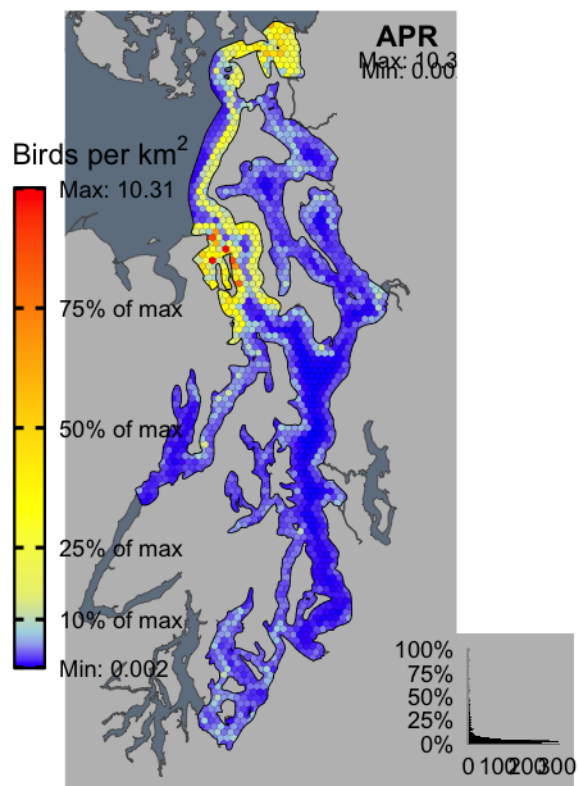
```



```

(apr_plot2 <- apr_plot | apr_iqr)

```



```
#ggsave("RESULTS/MAPS/apr_plot.pdf", plot = apr_plot2, width = 12, height = 11, units = "in")
#ggsave("RESULTS/MAPS/apr_plot.png", plot = apr_plot2, width = 12, height = 11, units = "in", dpi = 600)
```

MAY

```
may <- grids %>%
  full_join(all_predictions %>%
    filter(mon == "MAY" & year == "ALL") %>%
    mutate(per_max = MED/max(MED),
           per_bin = case_when(
             per_max < 0.01 ~ "0-1%",
             per_max < 0.25 ~ "1-25%",
             per_max < 0.5 ~ "25-50%",
             TRUE ~ "50-100%"),
           per_bin = factor(per_bin,
                           levels = c("0-1%", "1-25%",
                                       "25-50%", "50-100%"))),
    by = c("numid", "ID"))

min_med <- may %>%
  mutate(minnn = min(MED, na.rm = TRUE)) %>%
  distinct(minnn) %>%
  pull(minnn)
```

```

max_med <- may %>%
  mutate(maxxxx =max(MED, na.rm = TRUE)) %>%
  distinct(maxxxx) %>%
  pull(maxxxx)

min_label <- paste0("Min: ",formatC(min_med, format = "f", digits = 3))
max_label <- paste0("Max: ",formatC(max_med, format = "f", digits = 2))

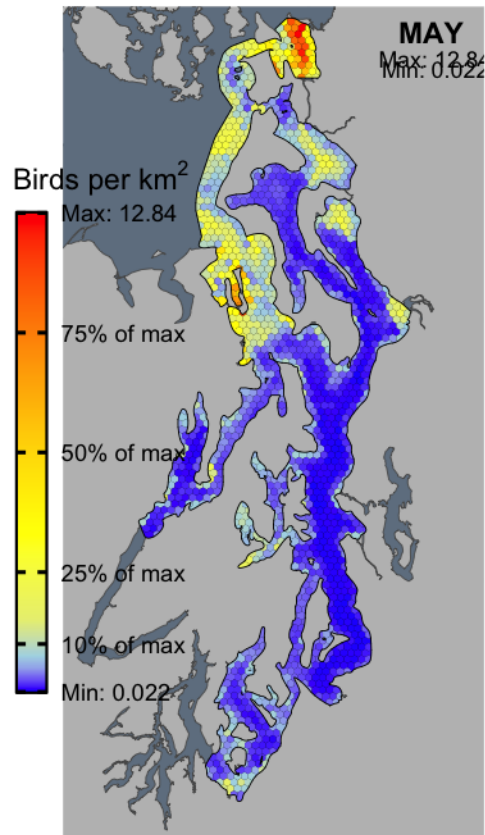
(MAY_Plot <- ggplot() +
  geom_sf(data = WA_State, fill = "gray") +
  geom_sf(data = puget, fill = "slategray") +
  geom_sf(data = may,
    fill =NA, color = "black", linewidth=0.4) +
  geom_sf(data = may,
    aes(fill = per_max), color = NA) +
  scale_fill_gradientn(
    colours = c("blue", "lightblue", "yellow", "orange", "red"),
    values = scales::rescale(c(0, 0.05, 0.15, 0.5, 0.75, 1)),
    limits = c(0, 1),
    name = expression("Birds per km"^2),
    breaks = c(0,0.1,0.25, 0.5, 0.75, 1),
    labels = c(min_label,"10% of max", "25% of max",
      "50% of max", "75% of max", max_label),
    guide = guide_colourbar(
      direction = "vertical",
      title.position = "top",
      title.hjust = 0.5,
      label.position = "right",
      label.hjust = 0,
      ticks = TRUE,
      ticks.colour = "black",
      ticks.linewidth = 0.7,
      barwidth = unit(4, "mm"),
      barheight = unit(2.5, "in"),
      frame.colour = "black",
      frame.linewidth = 0.5,
      draw.ulim = TRUE,
      draw.llim = TRUE)) +
  coord_sf(
    xlim = c(-123.2, -122),
    ylim = c(47, 48.6),
    crs = st_crs(4326),
    expand = FALSE) +
  scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
  scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
  theme_minimal() +
  theme(
    legend.margin = margin(0, 0, 0, 0),
    # legend.position = "right",
    legend.position = c(0.09, 0.5),
    axis.text = element_blank(),
    axis.title = element_blank(),
    axis.ticks = element_blank())+

```

```

annotate("text", x = -122.3, y = 48.5,
  label = max_label, size = 3, hjust = 0) +
annotate("text", x = -122.29, y = 48.5 - 0.02,
  label = min_label, size = 3, hjust = 0)+
annotate("text", x = -122.15, y = 48.55,
  label = "MAY", fontface = "bold"))

```



May Main

```

# Histogram data
may_vals <- may %>%
  filter(!is.na(per_max)) %>%
  pull(per_max)

hist_obj <- hist(may_vals, breaks = seq(0, 1, length.out = 100), plot = FALSE)

hist_data <- data.frame(
  bin_mid = hist_obj$mids,
  count = hist_obj$counts)

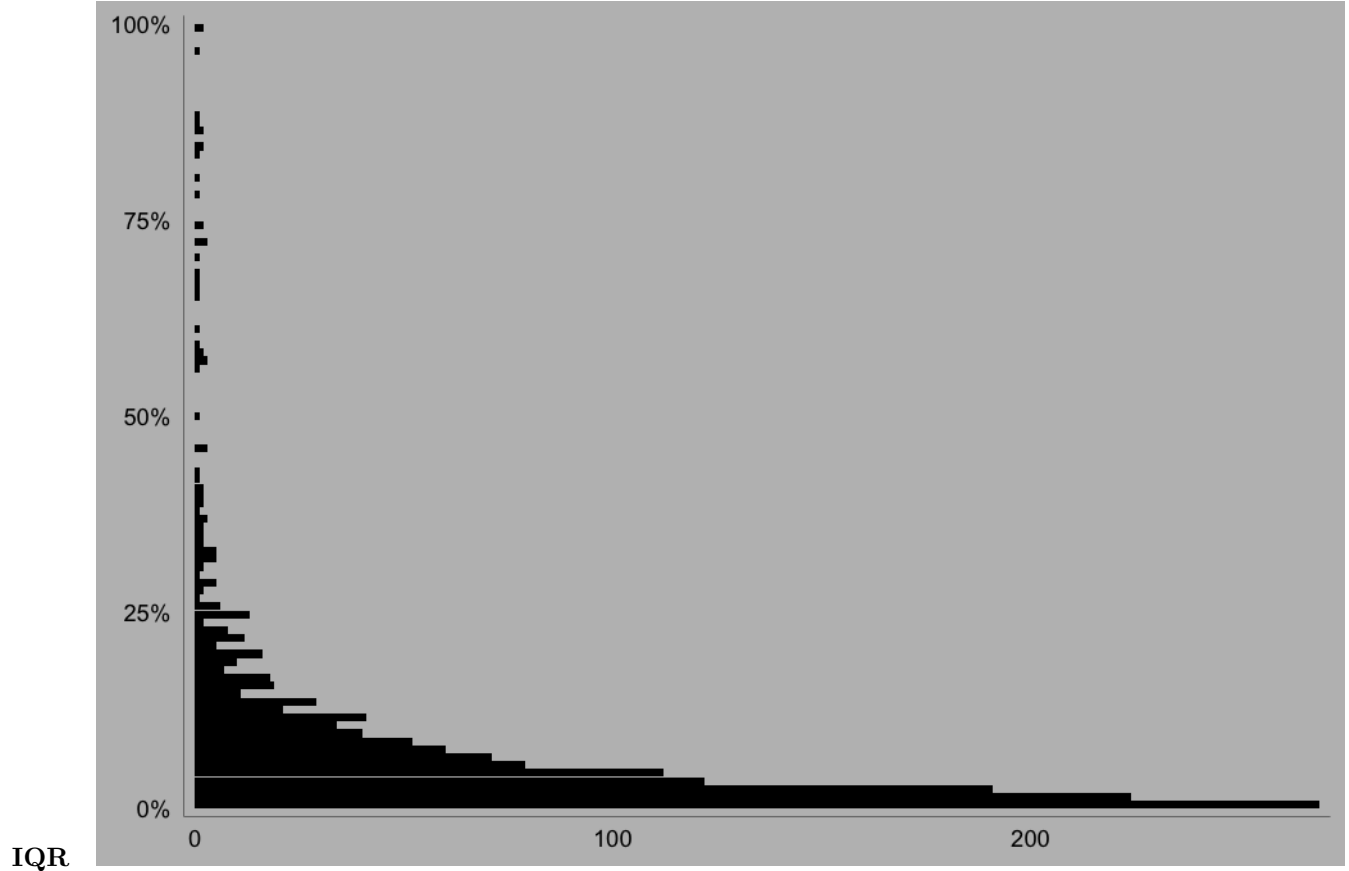
(hist_plot <- ggplot(hist_data, aes(x = bin_mid, y = count)) +
  geom_col(fill = "black", width = 0.01) +
  theme_minimal() +
  coord_flip()+

```

```

theme(
  #plot.margin = margin(2, 2, 2, 2), # top, right, bottom, left in pts
  # axis.text = element_blank(),
  axis.line.x = element_line(linewidth = 0.1, linetype = "solid",
                             colour = "black"),
  axis.line.y = element_line(linewidth = 0.1, linetype = "solid",
                             colour = "black"),
  panel.grid = element_blank(),
  panel.background = element_rect(fill = "gray", color = "gray"),
  panel.spacing = unit(0, "pt"),
  axis.text = element_text(color = "black"),
  axis.title = element_blank(),
  plot.background = element_rect(fill = "gray"),
  axis.ticks = element_blank() +
scale_x_continuous(expand = c(0.01, 0),
                   labels = scales::percent_format(accuracy = 1)) +
scale_y_continuous(expand = c(0.01, 0)))

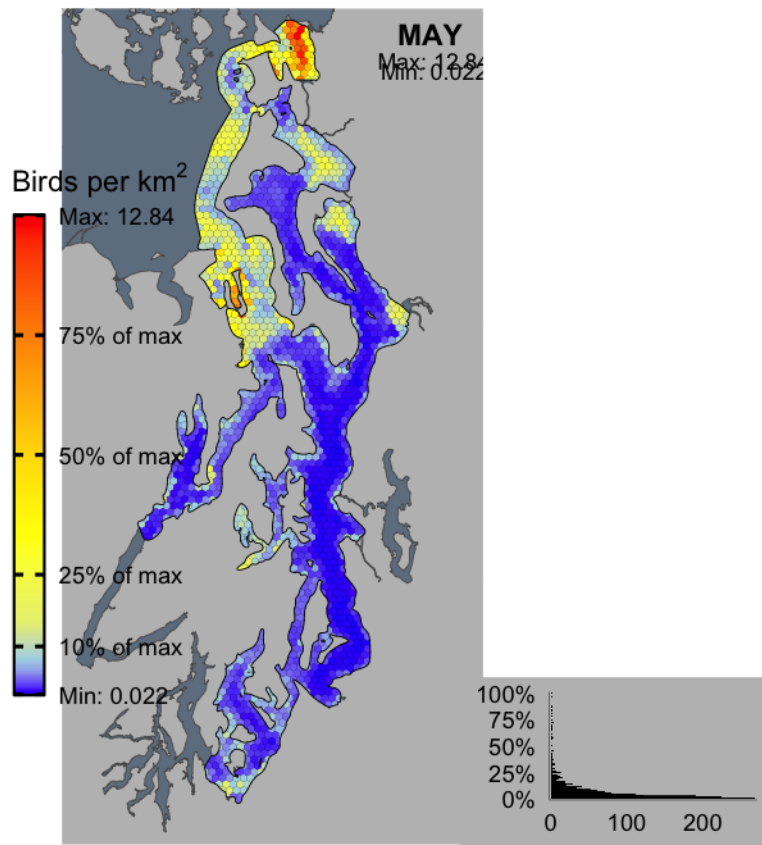
```



```

(may_plot <- ggdraw(MAY_Plot) +
  draw_plot(hist_plot,
    x = 0.65, y = 0.01,
    width = 0.25, height = 0.2))

```



```
max(may$IQR)
```

```
## [1] 18.84004
```

```
mean(may$IQR)
```

```
## [1] 1.055933
```

```
median(may$IQR)
```

```
## [1] 0.2769899
```

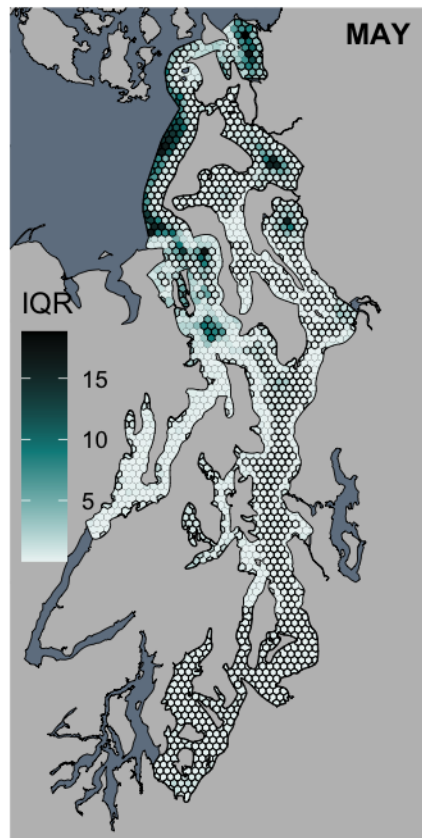
```
flag_may <- may %>%
  filter(flag == "CAP")

(may_iqr <- ggplot() +
  geom_sf(data = WA_State, fill = "gray", color = "black") +
  geom_sf(data = puget, fill = "slategray", color = "black") +
  geom_sf(data = may, fill = NA, color = "black", linewidth=0.4) +
  geom_sf(data = may, aes(fill = IQR), color = NA) +
  geom_sf(data = flag_may, fill= NA, color = "black") +
  coord_sf(
    xlim = c(-123.2, -122),
```

```

ylim = c(47, 48.6),
crs = st_crs(4326),
expand = FALSE) +
scale_fill_gradient2(low = "white", mid = "cyan4", high = "black",
                     midpoint = 9)+
scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
theme_minimal() +
theme(
  legend.margin = margin(0, 0, 0, 0),
  # legend.position = "right",
  legend.position = c(0.13, 0.5),
  axis.text = element_blank(),
  axis.title = element_blank(),
  axis.ticks = element_blank())+
annotate("text", x = -122.15, y = 48.55,
        label = "MAY", fontface = "bold"))

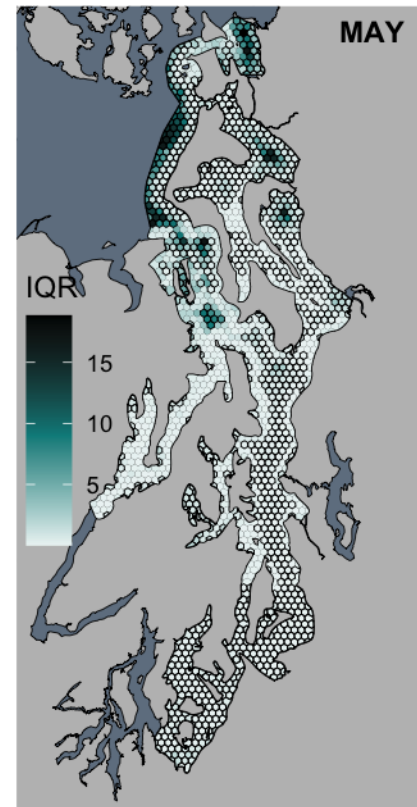
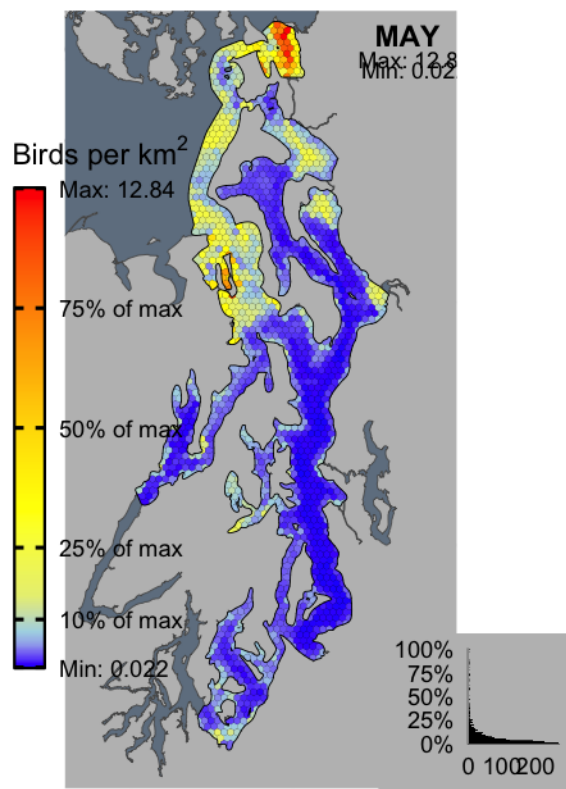
```



```

(may_plot2 <- may_plot | may_iqr)

```



```
#ggsave("RESULTS/MAPS/may_plot.pdf", plot = may_plot2, width = 12, height = 11, units = "in")
#ggsave("RESULTS/MAPS/may_plot.png", plot = may_plot2, width = 12, height = 11, units = "in", dpi = 600)
```

JUN

```
jun <- grids %>%
  full_join(all_predictions %>%
    filter(mon == "JUN" & year == "ALL") %>%
    mutate(per_max = MED/max(MED),
           per_bin = case_when(
             per_max < 0.01 ~ "0-1%",
             per_max < 0.25 ~ "1-25%",
             per_max < 0.5 ~ "25-50%",
             TRUE ~ "50-100%"),
           per_bin = factor(per_bin,
                           levels = c("0-1%", "1-25%",
                                       "25-50%", "50-100%"))),
    by = c("numid", "ID"))

min_med <- jun %>%
  mutate(minnn = min(MED, na.rm = TRUE)) %>%
  distinct(minnn) %>%
  pull(minnn)
```

```

max_med <- jun %>%
  mutate(maxxxx =max(MED, na.rm = TRUE)) %>%
  distinct(maxxxx) %>%
  pull(maxxxx)

min_label <- paste0("Min: ",formatC(min_med, format = "f", digits = 3))
max_label <- paste0("Max: ",formatC(max_med, format = "f", digits = 2))

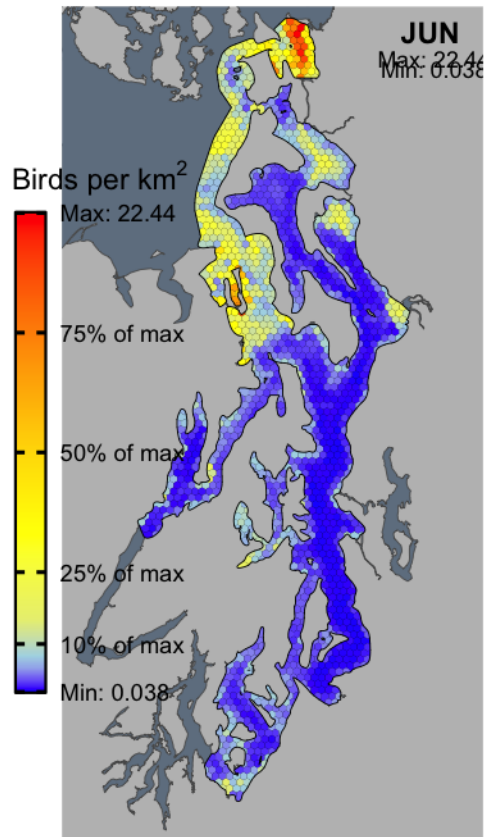
(JUN_Plot <- ggplot() +
  geom_sf(data = WA_State, fill = "gray") +
  geom_sf(data = puget, fill = "slategray") +
  geom_sf(data = jun,
    fill =NA, color = "black", linewidth=0.4) +
  geom_sf(data = jun,
    aes(fill = per_max), color = NA) +
  scale_fill_gradientn(
    colours = c("blue", "lightblue", "yellow", "orange", "red"),
    values = scales::rescale(c(0, 0.05, 0.15, 0.5, 0.75, 1)),
    limits = c(0, 1),
    name = expression("Birds per km"^2),
    breaks = c(0,0.1,0.25, 0.5, 0.75, 1),
    labels = c(min_label,"10% of max", "25% of max",
      "50% of max", "75% of max", max_label),
    guide = guide_colourbar(
      direction = "vertical",
      title.position = "top",
      title.hjust = 0.5,
      label.position = "right",
      label.hjust = 0,
      ticks = TRUE,
      ticks.colour = "black",
      ticks.linewidth = 0.7,
      barwidth = unit(4, "mm"),
      barheight = unit(2.5, "in"),
      frame.colour = "black",
      frame.linewidth = 0.5,
      draw.ulim = TRUE,
      draw.llim = TRUE)) +
  coord_sf(
    xlim = c(-123.2, -122),
    ylim = c(47, 48.6),
    crs = st_crs(4326),
    expand = FALSE) +
  scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
  scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
  theme_minimal() +
  theme(
    legend.margin = margin(0, 0, 0, 0),
    # legend.position = "right",
    legend.position = c(0.09, 0.5),
    axis.text = element_blank(),
    axis.title = element_blank(),
    axis.ticks = element_blank())+

```

```

annotate("text", x = -122.3, y = 48.5,
  label = max_label, size = 3, hjust = 0) +
annotate("text", x = -122.29, y = 48.5 - 0.02,
  label = min_label, size = 3, hjust = 0)+
annotate("text", x = -122.15, y = 48.55,
  label = "JUN", fontface = "bold"))

```



Jun Main

```

# Histogram data
jun_vals <- jun %>%
  filter(!is.na(per_max)) %>%
  pull(per_max)

hist_obj <- hist(jun_vals, breaks = seq(0, 1, length.out = 100), plot = FALSE)

hist_data <- data.frame(
  bin_mid = hist_obj$mids,
  count = hist_obj$counts)

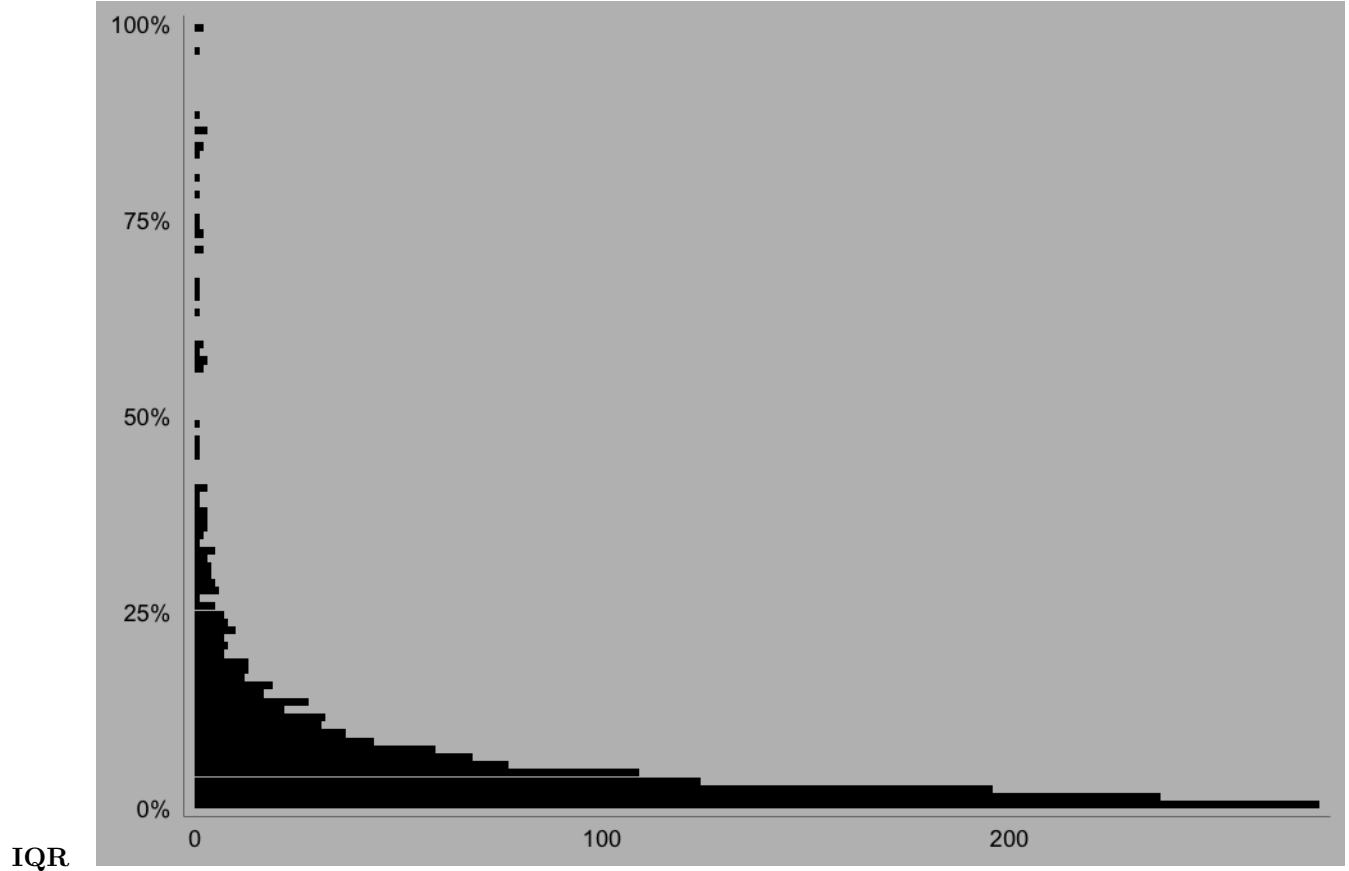
(hist_plot <- ggplot(hist_data, aes(x = bin_mid, y = count)) +
  geom_col(fill = "black", width = 0.01) +
  theme_minimal() +
  coord_flip()+

```

```

theme(
  #plot.margin = margin(2, 2, 2, 2), # top, right, bottom, left in pts
  # axis.text = element_blank(),
  axis.line.x = element_line(linewidth = 0.1, linetype = "solid",
                              colour = "black"),
  axis.line.y = element_line(linewidth = 0.1, linetype = "solid",
                              colour = "black"),
  panel.grid = element_blank(),
  panel.background = element_rect(fill = "gray", color = "gray"),
  panel.spacing = unit(0, "pt"),
  axis.text = element_text(color = "black"),
  axis.title = element_blank(),
  plot.background = element_rect(fill = "gray"),
  axis.ticks = element_blank() +
scale_x_continuous(expand = c(0.01, 0),
                    labels = scales::percent_format(accuracy = 1)) +
scale_y_continuous(expand = c(0.01, 0)))

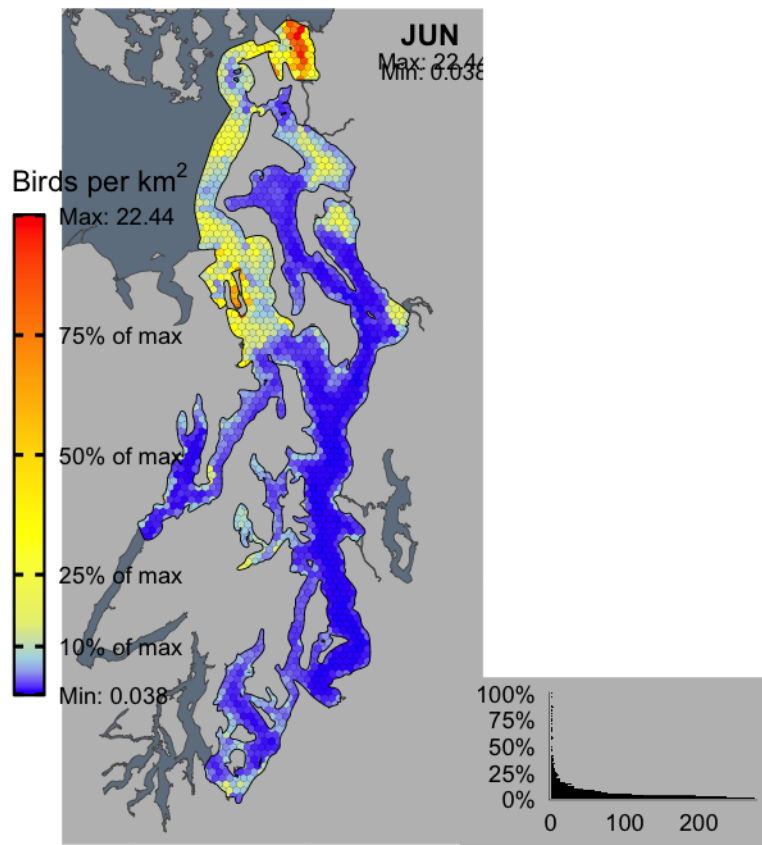
```



```

(jun_plot <- ggdraw(JUN_Plot) +
  draw_plot(hist_plot,
    x = 0.65, y = 0.01,
    width = 0.25, height = 0.2))

```



```
max(jun$IQR)
```

```
## [1] 32.26321
```

```
mean(jun$IQR)
```

```
## [1] 1.792497
```

```
median(jun$IQR)
```

```
## [1] 0.4001109
```

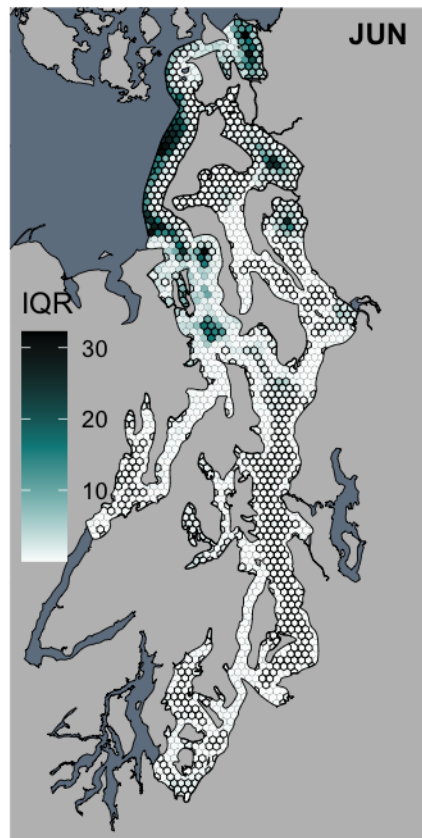
```
flag_jun <- jun %>%
  filter(flag == "CAP")

(jun_iqr <- ggplot() +
  geom_sf(data = WA_State, fill = "gray", color = "black") +
  geom_sf(data = puget, fill = "slategray", color = "black") +
  geom_sf(data = jun, fill = NA, color = "black", linewidth=0.4) +
  geom_sf(data = jun, aes(fill = IQR), color = NA) +
  geom_sf(data = flag_jun, fill = NA, color = "black") +
  coord_sf(
    xlim = c(-123.2, -122),
```

```

ylim = c(47, 48.6),
crs = st_crs(4326),
expand = FALSE) +
scale_fill_gradient2(low = "white", mid = "cyan4", high = "black",
                     midpoint = 16)+
scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
theme_minimal() +
theme(
  legend.margin = margin(0, 0, 0, 0),
  # legend.position = "right",
  legend.position = c(0.13, 0.5),
  axis.text = element_blank(),
  axis.title = element_blank(),
  axis.ticks = element_blank())+
annotate("text", x = -122.15, y = 48.55,
         label = "JUN", fontface = "bold"))

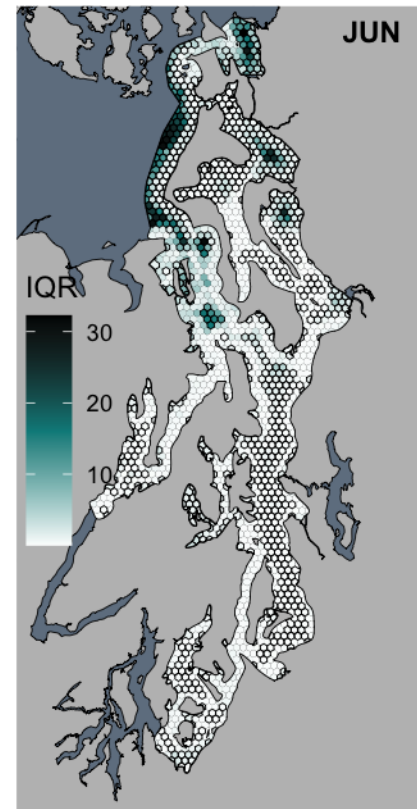
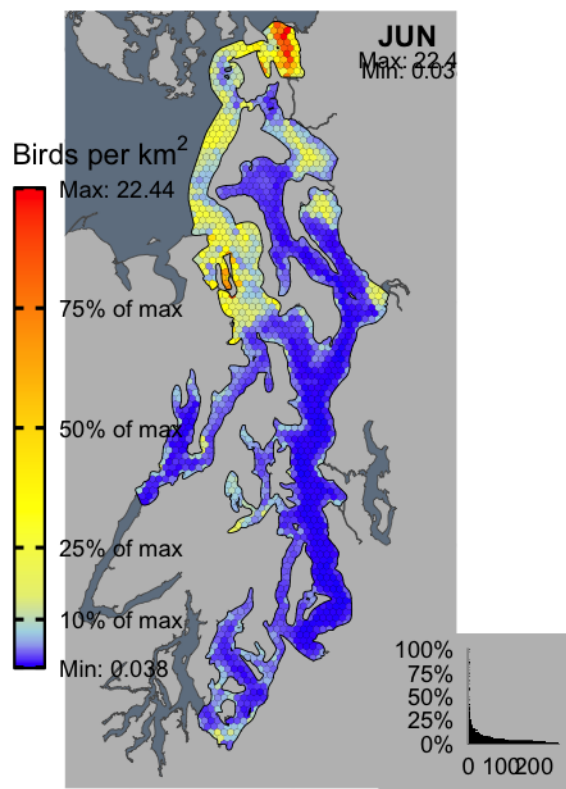
```



```

(jun_plot2 <- jun_plot | jun_iqr)

```



```
#ggsave("RESULTS/MAPS/jun_plot.pdf", plot = jun_plot2, width = 12, height = 11, units = "in")
#ggsave("RESULTS/MAPS/jun_plot.png", plot = jun_plot2, width = 12, height = 11, units = "in", dpi = 600)
```

JUL

```
jul <- grids %>%
  full_join(all_predictions %>%
    filter(mon == "JUL" & year == "ALL") %>%
    mutate(per_max = MED/max(MED),
           per_bin = case_when(
             per_max < 0.01 ~ "0-1%",
             per_max < 0.25 ~ "1-25%",
             per_max < 0.5 ~ "25-50%",
             TRUE ~ "50-100%"),
           per_bin = factor(per_bin,
                           levels = c("0-1%", "1-25%",
                                       "25-50%", "50-100%"))),
    by = c("numid", "ID"))

min_med <- jul %>%
  mutate(minnn = min(MED, na.rm = TRUE)) %>%
  distinct(minnn) %>%
  pull(minnn)
```

```

max_med <- jul %>%
  mutate(maxxxx =max(MED, na.rm = TRUE)) %>%
  distinct(maxxxx) %>%
  pull(maxxxx)

min_label <- paste0("Min: ",formatC(min_med, format = "f", digits = 3))
max_label <- paste0("Max: ",formatC(max_med, format = "f", digits = 2))

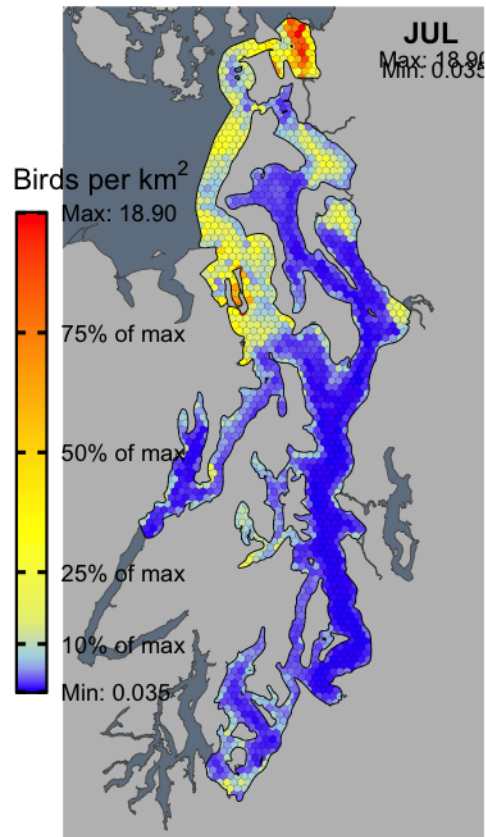
(JUL_Plot <- ggplot() +
  geom_sf(data = WA_State, fill = "gray") +
  geom_sf(data = puget, fill = "slategray") +
  geom_sf(data = jul,
    fill =NA, color = "black", linewidth=0.4) +
  geom_sf(data = jul,
    aes(fill = per_max), color = NA) +
  scale_fill_gradientn(
    colours = c("blue", "lightblue", "yellow", "orange", "red"),
    values = scales::rescale(c(0, 0.05, 0.15, 0.5, 0.75, 1)),
    limits = c(0, 1),
    name = expression("Birds per km"^2),
    breaks = c(0,0.1,0.25, 0.5, 0.75, 1),
    labels = c(min_label,"10% of max", "25% of max",
      "50% of max", "75% of max", max_label),
    guide = guide_colourbar(
      direction = "vertical",
      title.position = "top",
      title.hjust = 0.5,
      label.position = "right",
      label.hjust = 0,
      ticks = TRUE,
      ticks.colour = "black",
      ticks.linewidth = 0.7,
      barwidth = unit(4, "mm"),
      barheight = unit(2.5, "in"),
      frame.colour = "black",
      frame.linewidth = 0.5,
      draw.ulim = TRUE,
      draw.llim = TRUE)) +
  coord_sf(
    xlim = c(-123.2, -122),
    ylim = c(47, 48.6),
    crs = st_crs(4326),
    expand = FALSE) +
  scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
  scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
  theme_minimal() +
  theme(
    legend.margin = margin(0, 0, 0, 0),
    # legend.position = "right",
    legend.position = c(0.09, 0.5),
    axis.text = element_blank(),
    axis.title = element_blank(),
    axis.ticks = element_blank())+

```

```

annotate("text", x = -122.3, y = 48.5,
  label = max_label, size = 3, hjust = 0) +
annotate("text", x = -122.29, y = 48.5 - 0.02,
  label = min_label, size = 3, hjust = 0)+
annotate("text", x = -122.15, y = 48.55,
  label = "JUL", fontface = "bold"))

```



Jul Main

```

# Histogram data
jul_vals <- jul %>%
  filter(!is.na(per_max)) %>%
  pull(per_max)

hist_obj <- hist(jul_vals, breaks = seq(0, 1, length.out = 100), plot = FALSE)

hist_data <- data.frame(
  bin_mid = hist_obj$mids,
  count = hist_obj$counts)

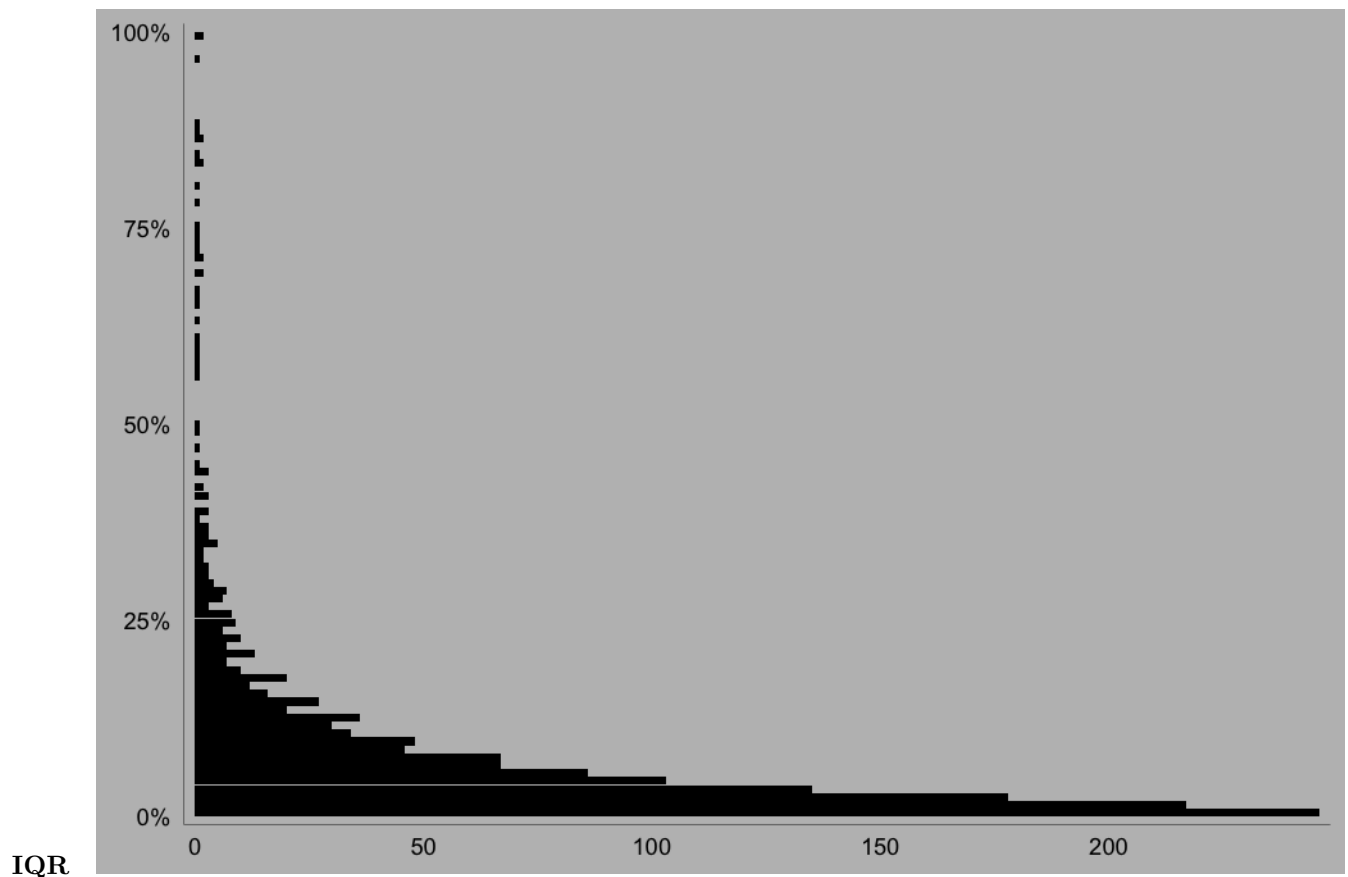
(hist_plot <- ggplot(hist_data, aes(x = bin_mid, y = count)) +
  geom_col(fill = "black", width = 0.01) +
  theme_minimal() +
  coord_flip()+

```

```

theme(
  #plot.margin = margin(2, 2, 2, 2), # top, right, bottom, left in pts
  # axis.text = element_blank(),
  axis.line.x = element_line(linewidth = 0.1, linetype = "solid",
                             colour = "black"),
  axis.line.y = element_line(linewidth = 0.1, linetype = "solid",
                             colour = "black"),
  panel.grid = element_blank(),
  panel.background = element_rect(fill = "gray", color = "gray"),
  panel.spacing = unit(0, "pt"),
  axis.text = element_text(color = "black"),
  axis.title = element_blank(),
  plot.background = element_rect(fill = "gray"),
  axis.ticks = element_blank() +
scale_x_continuous(expand = c(0.01, 0),
                   labels = scales::percent_format(accuracy = 1)) +
scale_y_continuous(expand = c(0.01, 0)))

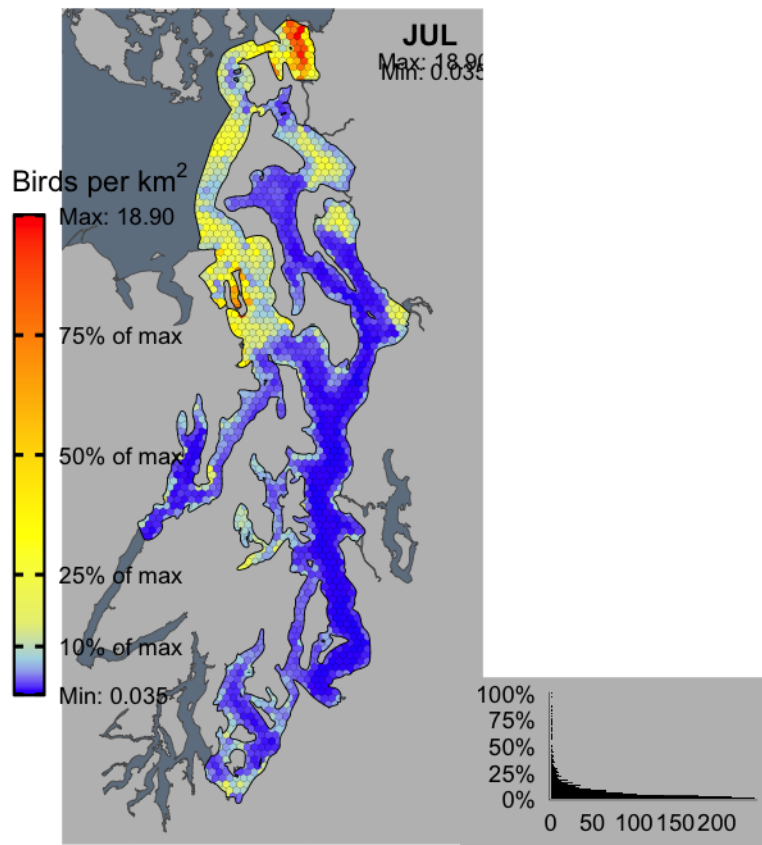
```



```

(jul_plot <- ggdraw(JUL_Plot) +
  draw_plot(hist_plot,
    x = 0.65, y = 0.01,
    width = 0.25, height = 0.2))

```



```
max(jul$IQR)
```

```
## [1] 28.41988
```

```
mean(jul$IQR)
```

```
## [1] 1.581857
```

```
median(jul$IQR)
```

```
## [1] 0.3770601
```

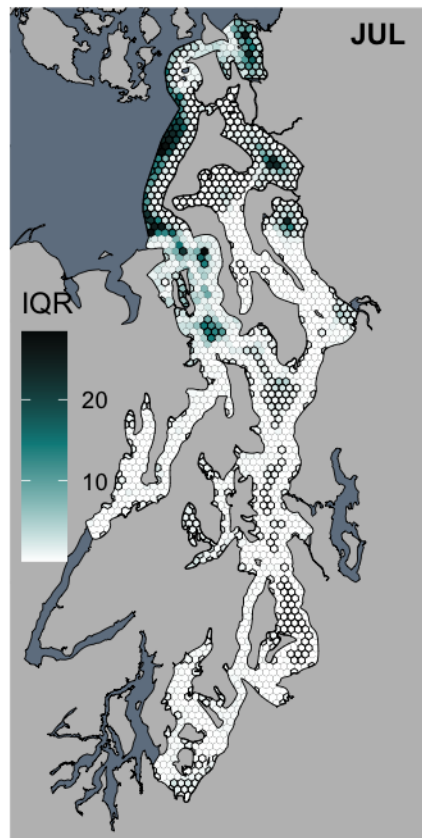
```
flag_jul <- jul %>%
  filter(flag == "CAP")

(jul_iqr <- ggplot() +
  geom_sf(data = WA_State, fill = "gray", color = "black") +
  geom_sf(data = puget, fill = "slategray", color = "black") +
  geom_sf(data = jul, fill = NA, color = "black", linewidth=0.4) +
  geom_sf(data = jul, aes(fill = IQR), color = NA) +
  geom_sf(data = flag_jul, fill = NA, color = "black") +
  coord_sf(
    xlim = c(-123.2, -122),
```

```

ylim = c(47, 48.6),
crs = st_crs(4326),
expand = FALSE) +
scale_fill_gradient2(low = "white", mid = "cyan4", high = "black",
                     midpoint = 14.5)+
scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
theme_minimal() +
theme(
  legend.margin = margin(0, 0, 0, 0),
  # legend.position = "right",
  legend.position = c(0.13, 0.5),
  axis.text = element_blank(),
  axis.title = element_blank(),
  axis.ticks = element_blank())+
annotate("text", x = -122.15, y = 48.55,
         label = "JUL", fontface = "bold"))

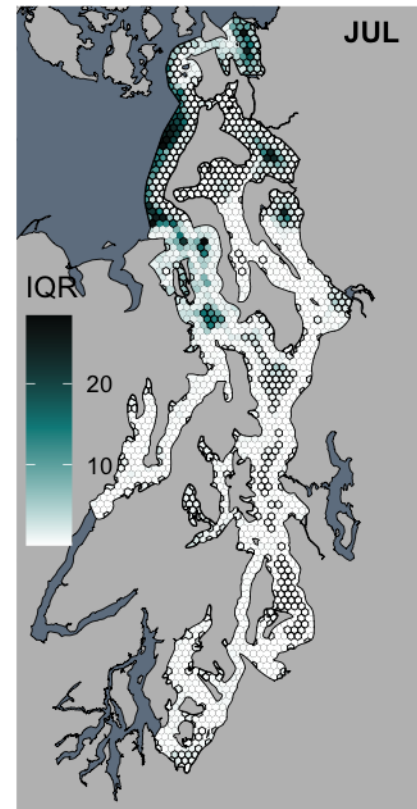
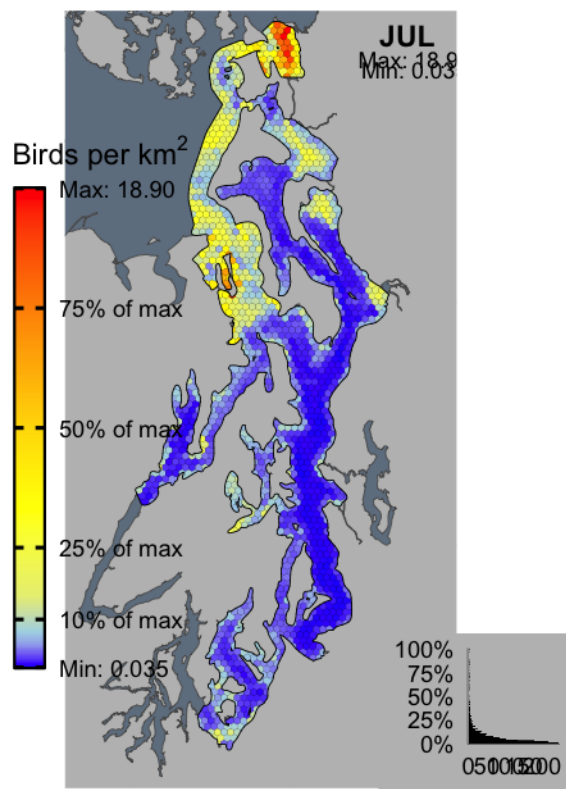
```



```

(jul_plot2 <- jul_plot | jul_iqr)

```



```
#ggsave("RESULTS/MAPS/jul_plot.pdf", plot = jul_plot2, width = 12, height = 11, units = "in")
#ggsave("RESULTS/MAPS/jul_plot.png", plot = jul_plot2, width = 12, height = 11, units = "in", dpi = 600)
```

SEP

```
sep <- grids %>%
  full_join(all_predictions %>%
    filter(mon == "SEP" & year == "ALL") %>%
    mutate(per_max = MED/max(MED),
           per_bin = case_when(
             per_max < 0.01 ~ "0-1%",
             per_max < 0.25 ~ "1-25%",
             per_max < 0.5 ~ "25-50%",
             TRUE ~ "50-100%"),
           per_bin = factor(per_bin,
                           levels = c("0-1%", "1-25%",
                                         "25-50%", "50-100%"))),
    by = c("numid", "ID"))

min_med <- sep %>%
  mutate(minnn = min(MED, na.rm = TRUE)) %>%
  distinct(minnn) %>%
  pull(minnn)
```

```

max_med <- sep %>%
  mutate(maxxx = max(MED, na.rm = TRUE)) %>%
  distinct(maxxx) %>%
  pull(maxxx)

min_label <- paste0("Min: ", formatC(min_med, format = "f", digits = 3))
max_label <- paste0("Max: ", formatC(max_med, format = "f", digits = 2))

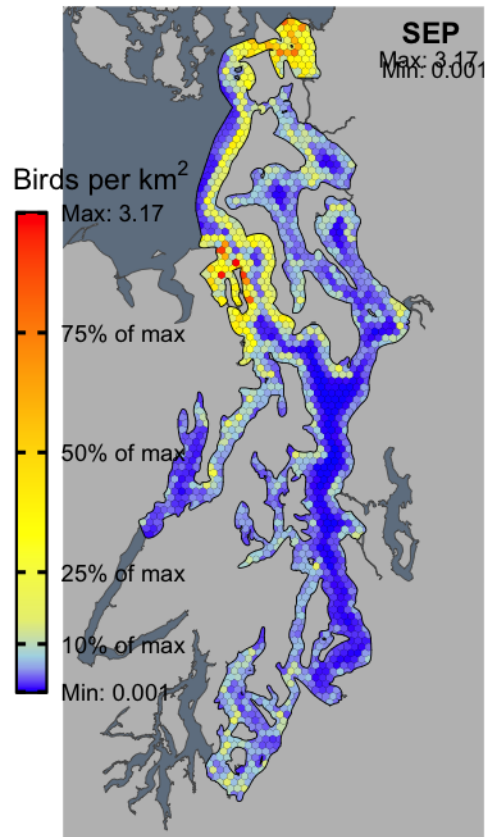
(SEP_Plot <- ggplot() +
  geom_sf(data = WA_State, fill = "gray") +
  geom_sf(data = puget, fill = "slategray") +
  geom_sf(data = sep,
    fill = NA, color = "black", linewidth = 0.4) +
  geom_sf(data = sep,
    aes(fill = per_max), color = NA) +
  scale_fill_gradientn(
    colours = c("blue", "lightblue", "yellow", "orange", "red"),
    values = scales::rescale(c(0, 0.05, 0.15, 0.5, 0.75, 1)),
    limits = c(0, 1),
    name = expression("Birds per km"2),
    breaks = c(0, 0.1, 0.25, 0.5, 0.75, 1),
    labels = c(min_label, "10% of max", "25% of max",
      "50% of max", "75% of max", max_label),
    guide = guide_colourbar(
      direction = "vertical",
      title.position = "top",
      title.hjust = 0.5,
      label.position = "right",
      label.hjust = 0,
      ticks = TRUE,
      ticks.colour = "black",
      ticks.linewidth = 0.7,
      barwidth = unit(4, "mm"),
      barheight = unit(2.5, "in"),
      frame.colour = "black",
      frame.linewidth = 0.5,
      draw.ulim = TRUE,
      draw.llim = TRUE)) +
  coord_sf(
    xlim = c(-123.2, -122),
    ylim = c(47, 48.6),
    crs = st_crs(4326),
    expand = FALSE) +
  scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
  scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
  theme_minimal() +
  theme(
    legend.margin = margin(0, 0, 0, 0),
    # legend.position = "right",
    legend.position = c(0.09, 0.5),
    axis.text = element_blank(),
    axis.title = element_blank(),
    axis.ticks = element_blank()) +

```

```

annotate("text", x = -122.3, y = 48.5,
  label = max_label, size = 3, hjust = 0) +
annotate("text", x = -122.29, y = 48.5 - 0.02,
  label = min_label, size = 3, hjust = 0)+
annotate("text", x = -122.15, y = 48.55,
  label = "SEP", fontface = "bold"))

```



Sep Main

```

# Histogram data
sep_vals <- sep %>%
  filter(!is.na(per_max)) %>%
  pull(per_max)

hist_obj <- hist(sep_vals, breaks = seq(0, 1, length.out = 100), plot = FALSE)

hist_data <- data.frame(
  bin_mid = hist_obj$mids,
  count = hist_obj$counts)

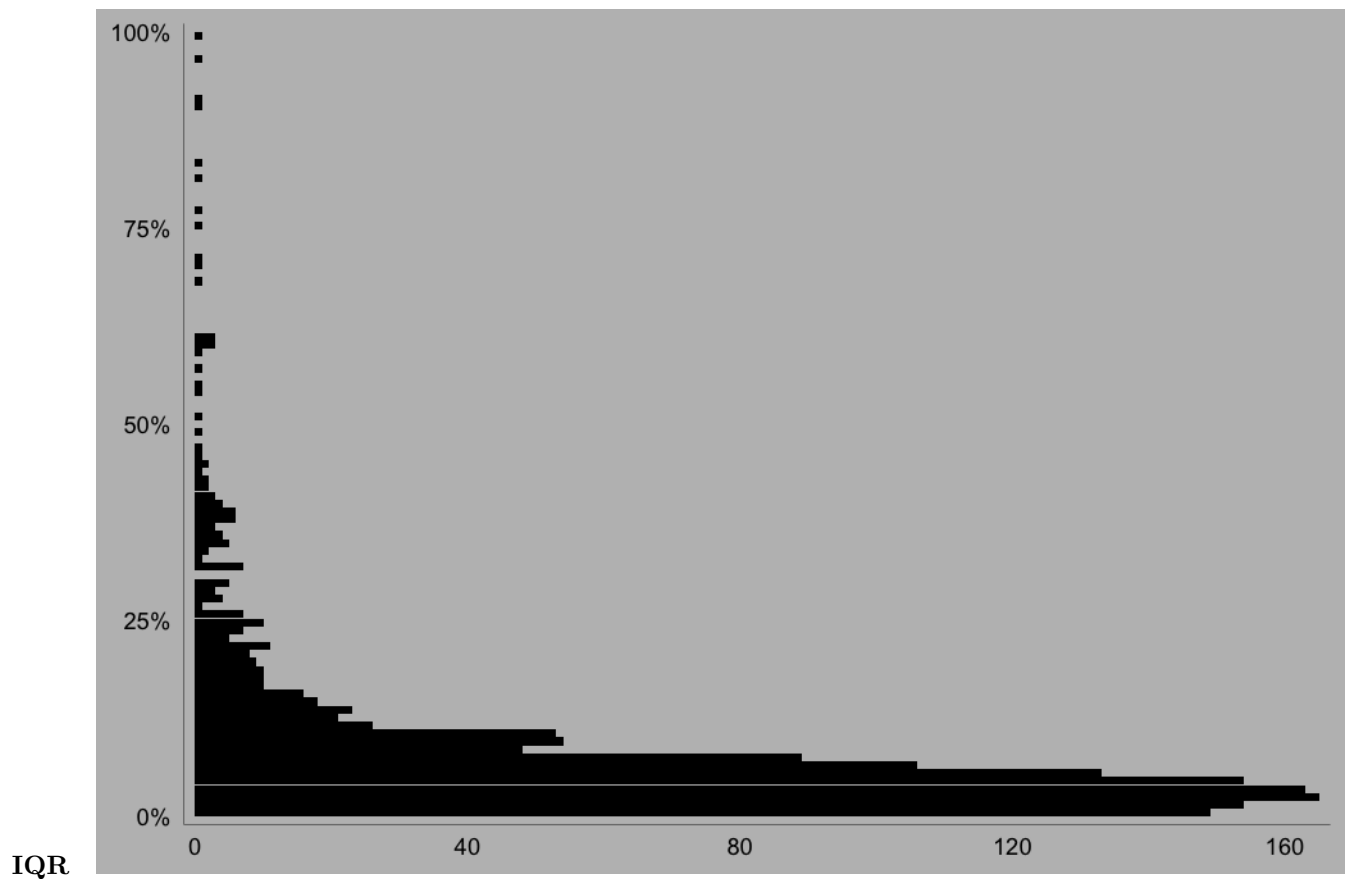
(hist_plot <- ggplot(hist_data, aes(x = bin_mid, y = count)) +
  geom_col(fill = "black", width = 0.01) +
  theme_minimal() +
  coord_flip()+

```

```

theme(
  #plot.margin = margin(2, 2, 2, 2), # top, right, bottom, left in pts
  # axis.text = element_blank(),
  axis.line.x = element_line(linewidth = 0.1, linetype = "solid",
                             colour = "black"),
  axis.line.y = element_line(linewidth = 0.1, linetype = "solid",
                             colour = "black"),
  panel.grid = element_blank(),
  panel.background = element_rect(fill = "gray", color = "gray"),
  panel.spacing = unit(0, "pt"),
  axis.text = element_text(color = "black"),
  axis.title = element_blank(),
  plot.background = element_rect(fill = "gray"),
  axis.ticks = element_blank()) +
scale_x_continuous(expand = c(0.01, 0),
                   labels = scales::percent_format(accuracy = 1)) +
scale_y_continuous(expand = c(0.01, 0))

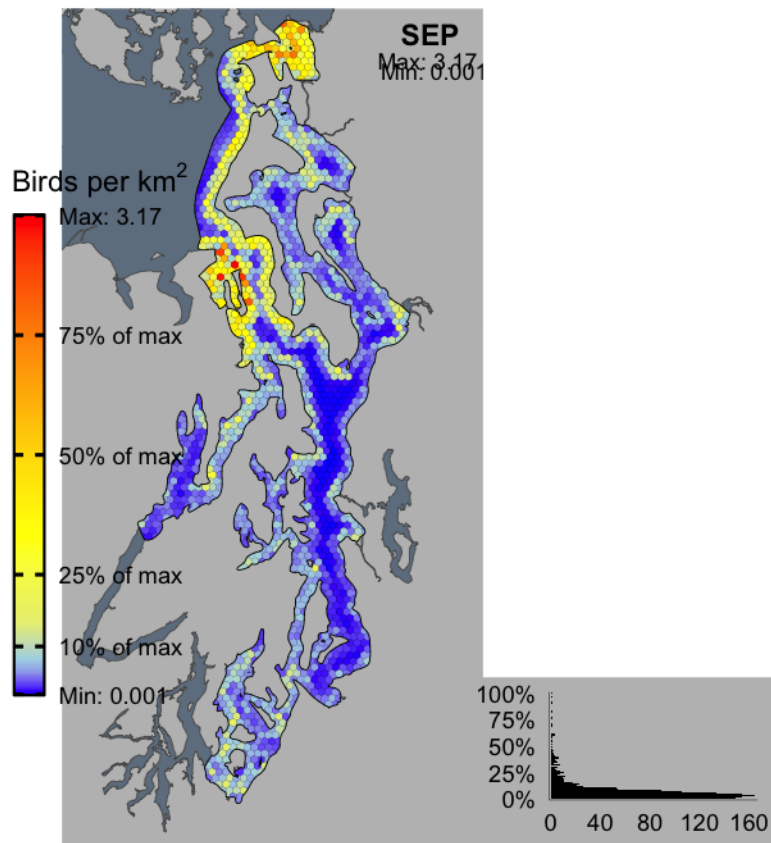
```



```

(sep_plot <- ggdraw(SEP_Plot) +
  draw_plot(hist_plot,
    x = 0.65, y = 0.01,
    width = 0.25, height = 0.2))

```



```
max(sep$IQR)
```

```
## [1] 1.446485
```

```
mean(sep$IQR)
```

```
## [1] 0.1245002
```

```
median(sep$IQR)
```

```
## [1] 0.07374629
```

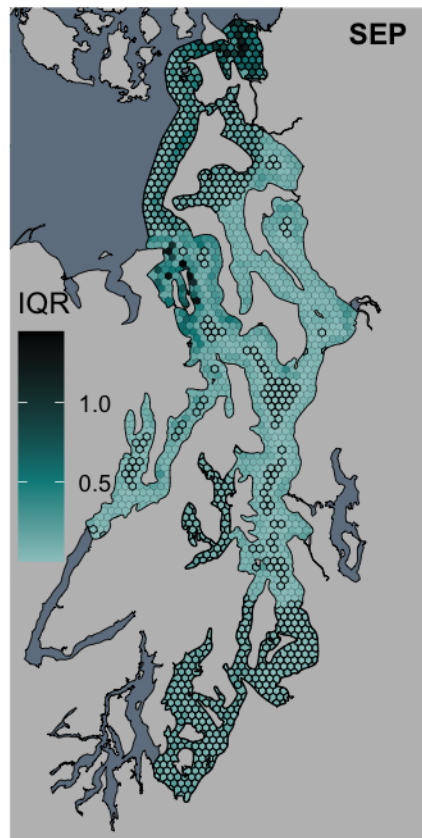
```
flag_sep <- sep %>%
  filter(flag == "CAP")

(sep_iqr <- ggplot() +
  geom_sf(data = WA_State, fill = "gray", color = "black") +
  geom_sf(data = puget, fill = "slategray", color = "black") +
  geom_sf(data = sep, fill = NA, color = "black", linewidth=0.4) +
  geom_sf(data = sep, aes(fill = IQR), color = NA) +
  geom_sf(data = flag_sep, fill = NA, color = "black") +
  coord_sf(
    xlim = c(-123.2, -122),
```

```

ylim = c(47, 48.6),
crs = st_crs(4326),
expand = FALSE) +
scale_fill_gradient2(low = "white", mid = "cyan4", high = "black",
                     midpoint = 0.5)+
scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
theme_minimal() +
theme(
  legend.margin = margin(0, 0, 0, 0),
  # legend.position = "right",
  legend.position = c(0.13, 0.5),
  axis.text = element_blank(),
  axis.title = element_blank(),
  axis.ticks = element_blank())+
annotate("text", x = -122.15, y = 48.55,
         label = "SEP", fontface = "bold"))

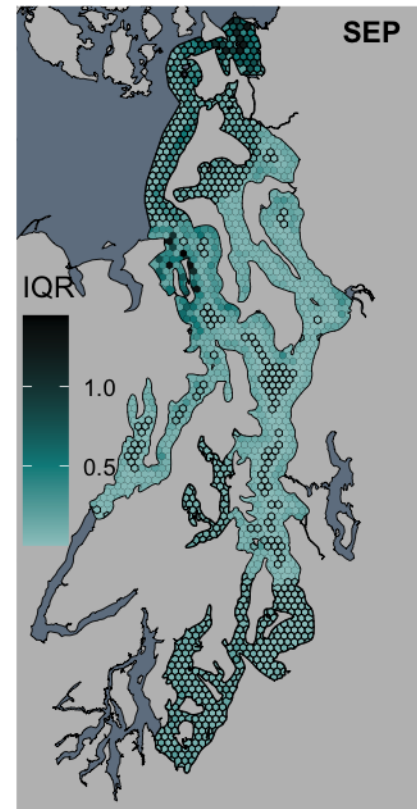
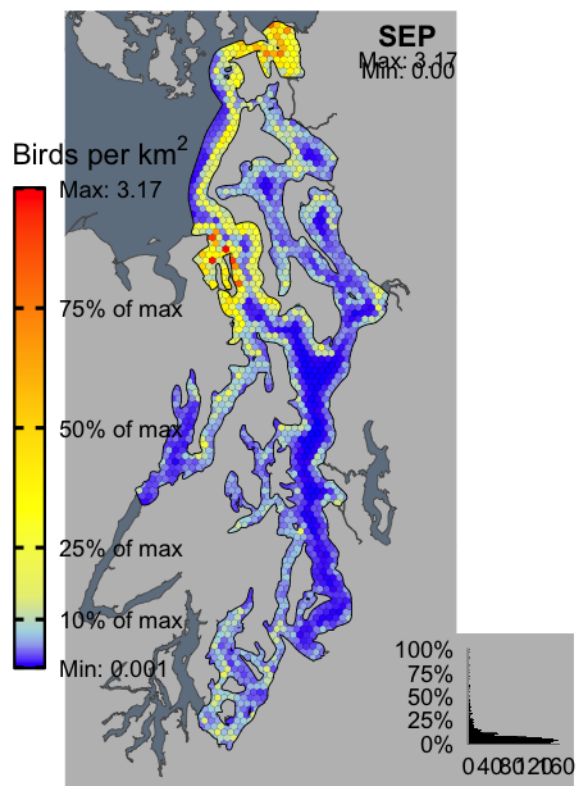
```



```

(sep_plot2 <- sep_plot | sep_iqr)

```



```
#ggsave("RESULTS/MAPS/sep_plot.pdf", plot = sep_plot2, width = 12, height = 11, units = "in")
#ggsave("RESULTS/MAPS/sep_plot.png", plot = sep_plot2, width = 12, height = 11, units = "in", dpi = 600)
```

OCT

```
oct <- grids %>%
  full_join(all_predictions %>%
    filter(mon == "OCT" & year == "ALL") %>%
    mutate(per_max = MED/max(MED),
           per_bin = case_when(
             per_max < 0.01 ~ "0-1%",
             per_max < 0.25 ~ "1-25%",
             per_max < 0.5 ~ "25-50%",
             TRUE ~ "50-100%"),
           per_bin = factor(per_bin,
                           levels = c("0-1%", "1-25%",
                                       "25-50%", "50-100%"))),
    by = c("numid", "ID"))

min_med <- oct %>%
  mutate(minnn = min(MED, na.rm = TRUE)) %>%
  distinct(minnn) %>%
  pull(minnn)
```

```

max_med <- oct %>%
  mutate(maxxx = max(MED, na.rm = TRUE)) %>%
  distinct(maxxx) %>%
  pull(maxxx)

min_label <- paste0("Min: ", formatC(min_med, format = "f", digits = 3))
max_label <- paste0("Max: ", formatC(max_med, format = "f", digits = 2))

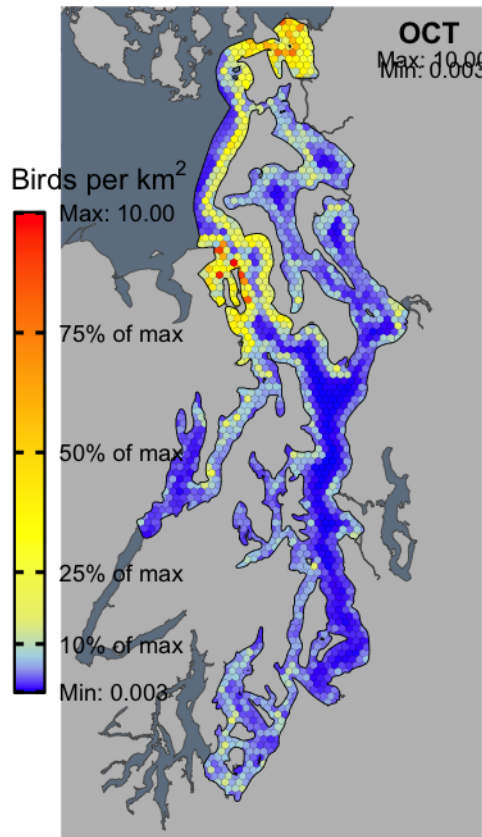
(OCT_Plot <- ggplot() +
  geom_sf(data = WA_State, fill = "gray") +
  geom_sf(data = puget, fill = "slategray") +
  geom_sf(data = oct,
    fill = NA, color = "black", linewidth=0.4) +
  geom_sf(data = oct,
    aes(fill = per_max), color = NA) +
  scale_fill_gradientn(
    colours = c("blue", "lightblue", "yellow", "orange", "red"),
    values = scales::rescale(c(0, 0.05, 0.15, 0.5, 0.75, 1)),
    limits = c(0, 1),
    name = expression("Birds per km"^2),
    breaks = c(0, 0.1, 0.25, 0.5, 0.75, 1),
    labels = c(min_label, "10% of max", "25% of max",
      "50% of max", "75% of max", max_label),
    guide = guide_colourbar(
      direction = "vertical",
      title.position = "top",
      title.hjust = 0.5,
      label.position = "right",
      label.hjust = 0,
      ticks = TRUE,
      ticks.colour = "black",
      ticks.linewidth = 0.7,
      barwidth = unit(4, "mm"),
      barheight = unit(2.5, "in"),
      frame.colour = "black",
      frame.linewidth = 0.5,
      draw.ulim = TRUE,
      draw.llim = TRUE)) +
  coord_sf(
    xlim = c(-123.2, -122),
    ylim = c(47, 48.6),
    crs = st_crs(4326),
    expand = FALSE) +
  scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
  scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
  theme_minimal() +
  theme(
    legend.margin = margin(0, 0, 0, 0),
    # legend.position = "right",
    legend.position = c(0.09, 0.5),
    axis.text = element_blank(),
    axis.title = element_blank(),
    axis.ticks = element_blank())+

```

```

annotate("text", x = -122.3, y = 48.5,
  label = max_label, size = 3, hjust = 0) +
annotate("text", x = -122.29, y = 48.5 - 0.02,
  label = min_label, size = 3, hjust = 0)+
annotate("text", x = -122.15, y = 48.55,
  label = "OCT", fontface = "bold"))

```



Oct Main

```

# Histogram data
oct_vals <- oct %>%
  filter(!is.na(per_max)) %>%
  pull(per_max)

hist_obj <- hist(oct_vals, breaks = seq(0, 1, length.out = 100), plot = FALSE)

hist_data <- data.frame(
  bin_mid = hist_obj$mids,
  count = hist_obj$counts)

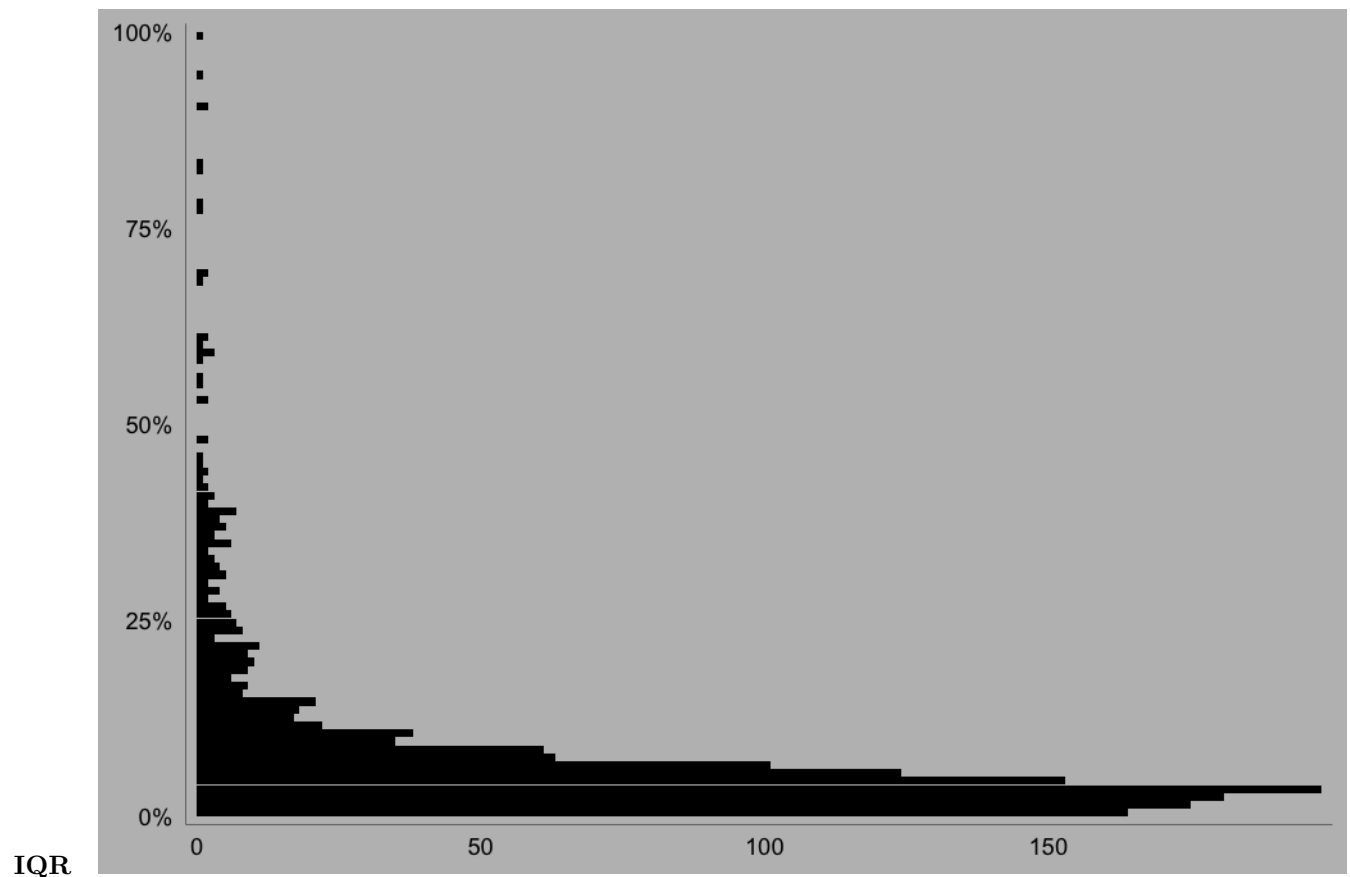
(hist_plot <- ggplot(hist_data, aes(x = bin_mid, y = count)) +
  geom_col(fill = "black", width = 0.01) +
  theme_minimal() +
  coord_flip()+

```

```

theme(
  #plot.margin = margin(2, 2, 2, 2), # top, right, bottom, left in pts
  # axis.text = element_blank(),
  axis.line.x = element_line(linewidth = 0.1, linetype = "solid",
                              colour = "black"),
  axis.line.y = element_line(linewidth = 0.1, linetype = "solid",
                              colour = "black"),
  panel.grid = element_blank(),
  panel.background = element_rect(fill = "gray", color = "gray"),
  panel.spacing = unit(0, "pt"),
  axis.text = element_text(color = "black"),
  axis.title = element_blank(),
  plot.background = element_rect(fill = "gray"),
  axis.ticks = element_blank() +
scale_x_continuous(expand = c(0.01, 0),
                    labels = scales::percent_format(accuracy = 1)) +
scale_y_continuous(expand = c(0.01, 0)))

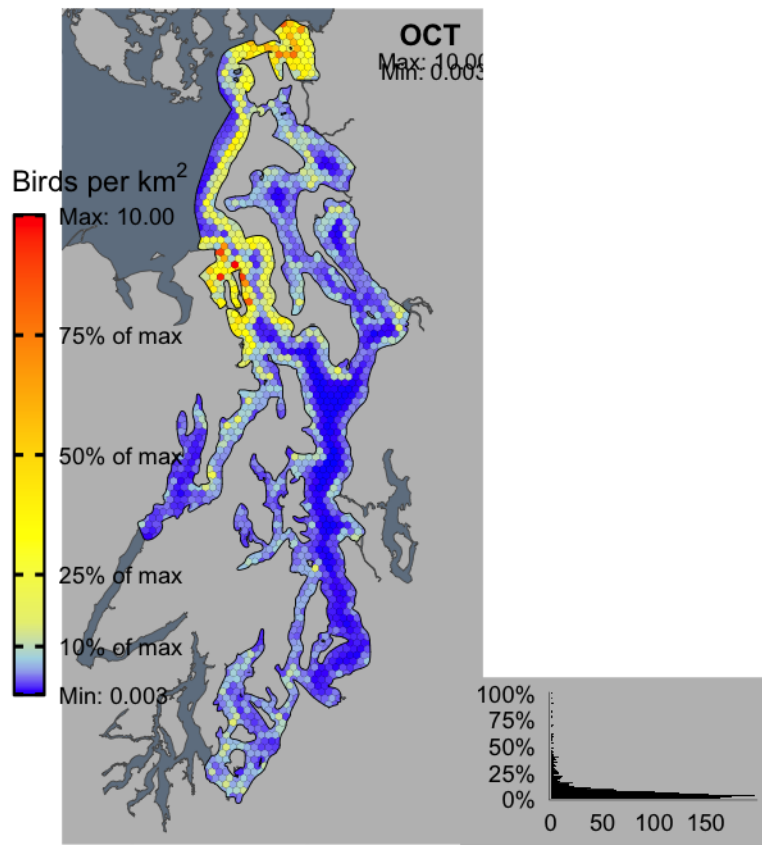
```



```

(oct_plot <- ggdraw(OCT_Plot) +
  draw_plot(hist_plot,
    x = 0.65, y = 0.01,
    width = 0.25, height = 0.2))

```



```
max(oct$IQR)
```

```
## [1] 3.556748
```

```
mean(oct$IQR)
```

```
## [1] 0.3040337
```

```
median(oct$IQR)
```

```
## [1] 0.1644312
```

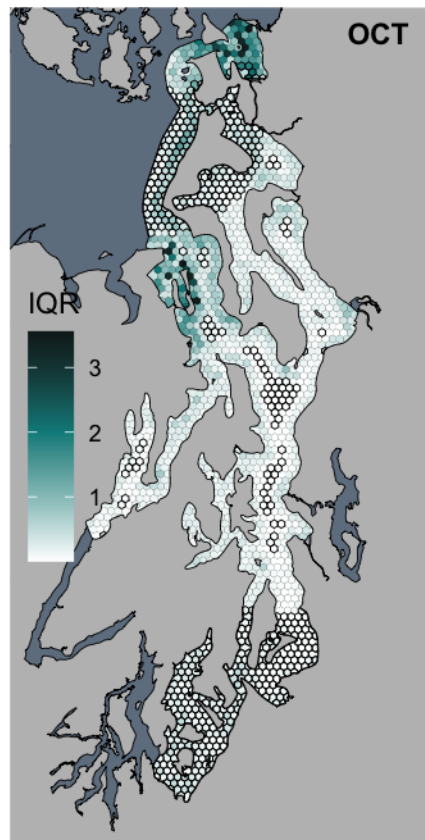
```
flag_oct <- oct %>%
  filter(flag == "CAP")

(oct_iqr <- ggplot() +
  geom_sf(data = WA_State, fill = "gray", color = "black") +
  geom_sf(data = puget, fill = "slategray", color = "black") +
  geom_sf(data = oct, fill = NA, color = "black", linewidth=0.4) +
  geom_sf(data = oct, aes(fill = IQR), color = NA) +
  geom_sf(data = flag_oct, fill = NA, color = "black") +
  coord_sf(
    xlim = c(-123.2, -122),
```

```

ylim = c(47, 48.6),
crs = st_crs(4326),
expand = FALSE) +
scale_fill_gradient2(low = "white", mid = "cyan4", high = "black",
                     midpoint = 2)+
scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
theme_minimal() +
theme(
  legend.margin = margin(0, 0, 0, 0),
  # legend.position = "right",
  legend.position = c(0.13, 0.5),
  axis.text = element_blank(),
  axis.title = element_blank(),
  axis.ticks = element_blank())+
annotate("text", x = -122.15, y = 48.55,
         label = "OCT", fontface = "bold"))

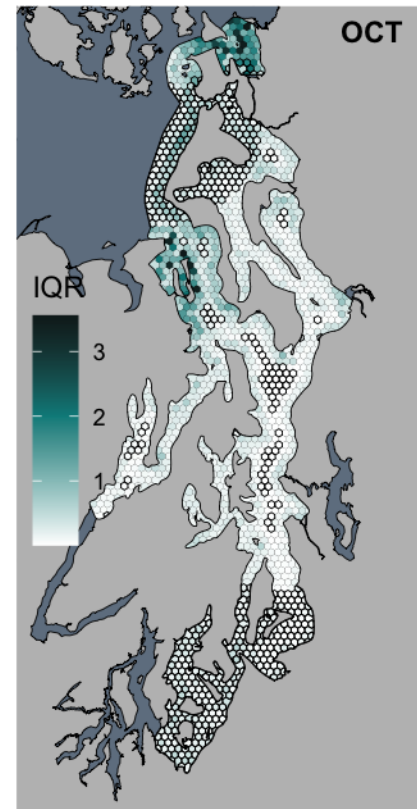
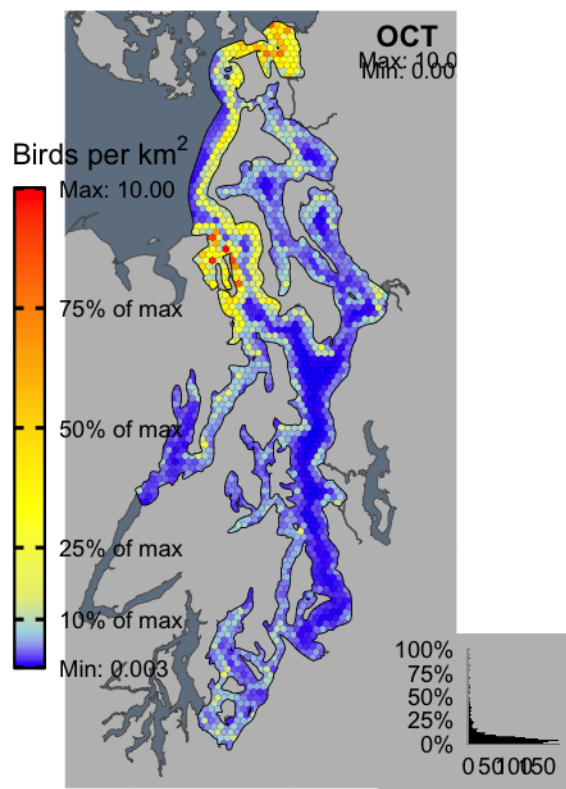
```



```

(oct_plot2 <- oct_plot | oct_iqr)

```



```
#ggsave("RESULTS/MAPS/oct_plot.pdf", plot = oct_plot2, width = 12, height = 11, units = "in")
#ggsave("RESULTS/MAPS/oct_plot.png", plot = oct_plot2, width = 12, height = 11, units = "in", dpi = 600)
```

NOV

```
nov <- grids %>%
  full_join(all_predictions %>%
    filter(mon == "NOV" & year == "ALL") %>%
    mutate(per_max = MED/max(MED),
           per_bin = case_when(
             per_max < 0.01 ~ "0-1%",
             per_max < 0.25 ~ "1-25%",
             per_max < 0.5 ~ "25-50%",
             TRUE ~ "50-100%"),
           per_bin = factor(per_bin,
                           levels = c("0-1%", "1-25%",
                                         "25-50%", "50-100%"))),
    by = c("numid", "ID"))

min_med <- nov %>%
  mutate(minnn = min(MED, na.rm = TRUE)) %>%
  distinct(minnn) %>%
  pull(minnn)
```

```

max_med <- nov %>%
  mutate(maxxx = max(MED, na.rm = TRUE)) %>%
  distinct(maxxx) %>%
  pull(maxxx)

min_label <- paste0("Min: ", formatC(min_med, format = "f", digits = 3))
max_label <- paste0("Max: ", formatC(max_med, format = "f", digits = 2))

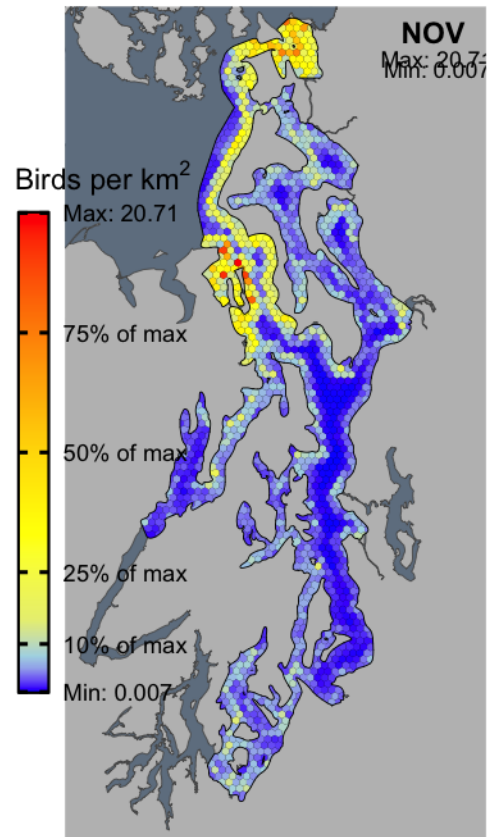
(NOV_Plot <- ggplot() +
  geom_sf(data = WA_State, fill = "gray") +
  geom_sf(data = puget, fill = "slategray") +
  geom_sf(data = nov,
    fill = NA, color = "black", linewidth=0.4) +
  geom_sf(data = nov,
    aes(fill = per_max), color = NA) +
  scale_fill_gradientn(
    colours = c("blue", "lightblue", "yellow", "orange", "red"),
    values = scales::rescale(c(0, 0.05, 0.15, 0.5, 0.75, 1)),
    limits = c(0, 1),
    name = expression("Birds per km"^2),
    breaks = c(0, 0.1, 0.25, 0.5, 0.75, 1),
    labels = c(min_label, "10% of max", "25% of max",
      "50% of max", "75% of max", max_label),
    guide = guide_colourbar(
      direction = "vertical",
      title.position = "top",
      title.hjust = 0.5,
      label.position = "right",
      label.hjust = 0,
      ticks = TRUE,
      ticks.colour = "black",
      ticks.linewidth = 0.7,
      barwidth = unit(4, "mm"),
      barheight = unit(2.5, "in"),
      frame.colour = "black",
      frame.linewidth = 0.5,
      draw.ulim = TRUE,
      draw.llim = TRUE)) +
  coord_sf(
    xlim = c(-123.2, -122),
    ylim = c(47, 48.6),
    crs = st_crs(4326),
    expand = FALSE) +
  scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
  scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
  theme_minimal() +
  theme(
    legend.margin = margin(0, 0, 0, 0),
    # legend.position = "right",
    legend.position = c(0.09, 0.5),
    axis.text = element_blank(),
    axis.title = element_blank(),
    axis.ticks = element_blank())+

```

```

annotate("text", x = -122.3, y = 48.5,
         label = max_label, size = 3, hjust = 0) +
annotate("text", x = -122.29, y = 48.5 - 0.02,
         label = min_label, size = 3, hjust = 0) +
annotate("text", x = -122.15, y = 48.55,
         label = "NOV", fontface = "bold"))

```



Nov Main

```

# Histogram data
nov_vals <- nov %>%
  filter(!is.na(per_max)) %>%
  pull(per_max)

hist_obj <- hist(nov_vals, breaks = seq(0, 1, length.out = 100), plot = FALSE)

hist_data <- data.frame(
  bin_mid = hist_obj$mids,
  count = hist_obj$counts)

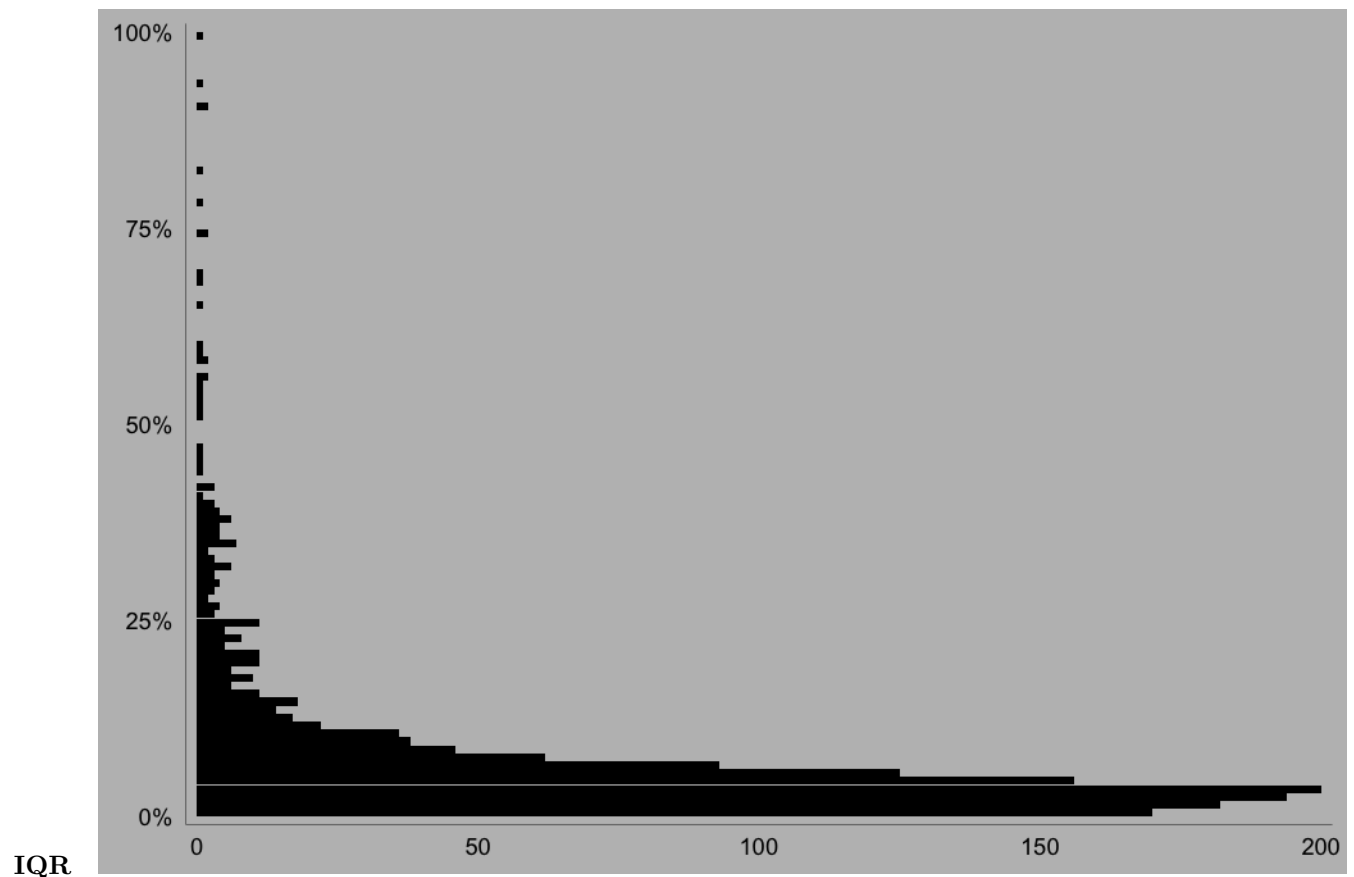
(hist_plot <- ggplot(hist_data, aes(x = bin_mid, y = count)) +
  geom_col(fill = "black", width = 0.01) +
  theme_minimal() +
  coord_flip() +

```

```

theme(
  #plot.margin = margin(2, 2, 2, 2), # top, right, bottom, left in pts
  # axis.text = element_blank(),
  axis.line.x = element_line(linewidth = 0.1, linetype = "solid",
                             colour = "black"),
  axis.line.y = element_line(linewidth = 0.1, linetype = "solid",
                             colour = "black"),
  panel.grid = element_blank(),
  panel.background = element_rect(fill = "gray", color = "gray"),
  panel.spacing = unit(0, "pt"),
  axis.text = element_text(color = "black"),
  axis.title = element_blank(),
  plot.background = element_rect(fill = "gray"),
  axis.ticks = element_blank() +
scale_x_continuous(expand = c(0.01, 0),
                   labels = scales::percent_format(accuracy = 1)) +
scale_y_continuous(expand = c(0.01, 0)))

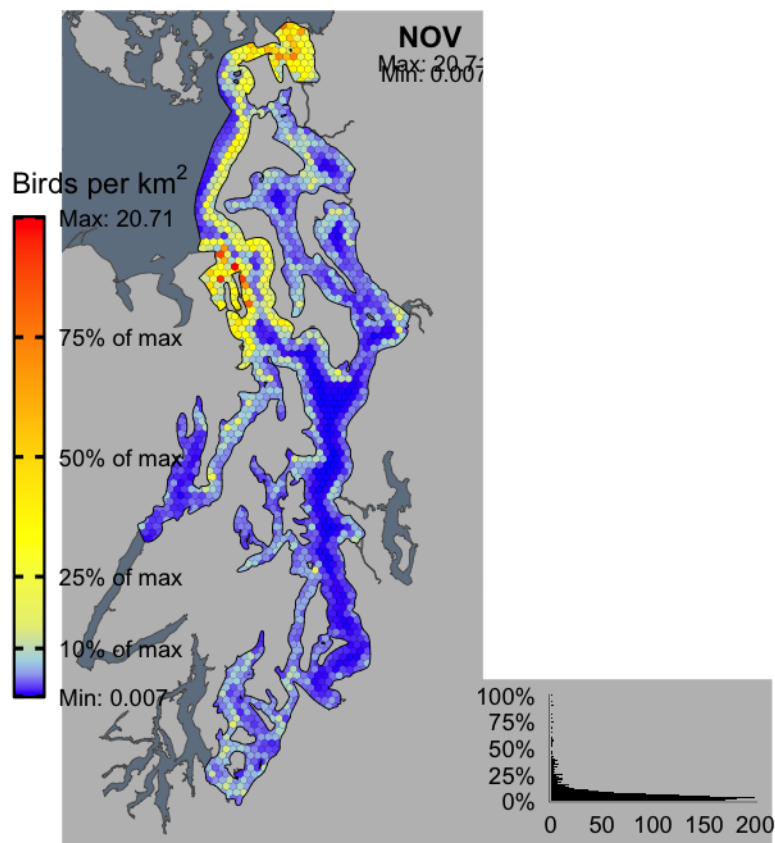
```



```

(nov_plot <- ggdraw(NOV_Plot) +
  draw_plot(hist_plot,
    x = 0.65, y = 0.01,
    width = 0.25, height = 0.2))

```



```
max(nov$IQR)
```

```
## [1] 6.820253
```

```
mean(nov$IQR)
```

```
## [1] 0.5489295
```

```
median(nov$IQR)
```

```
## [1] 0.2774492
```

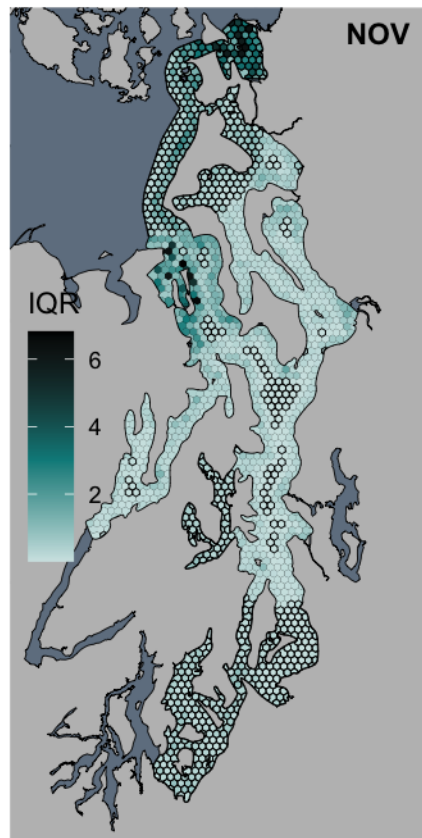
```
flag_nov <- nov %>%
  filter(flag == "CAP")

(nov_iqr <- ggplot() +
  geom_sf(data = WA_State, fill = "gray", color = "black") +
  geom_sf(data = puget, fill = "slategray", color = "black") +
  geom_sf(data = nov, fill = NA, color = "black", linewidth=0.4) +
  geom_sf(data = nov, aes(fill = IQR), color = NA) +
  geom_sf(data = flag_nov, fill = NA, color = "black") +
  coord_sf(
    xlim = c(-123.2, -122),
```

```

ylim = c(47, 48.6),
crs = st_crs(4326),
expand = FALSE) +
scale_fill_gradient2(low = "white", mid = "cyan4", high = "black",
                     midpoint = 3)+
scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
theme_minimal() +
theme(
  legend.margin = margin(0, 0, 0, 0),
  # legend.position = "right",
  legend.position = c(0.13, 0.5),
  axis.text = element_blank(),
  axis.title = element_blank(),
  axis.ticks = element_blank())+
annotate("text", x = -122.15, y = 48.55,
         label = "NOV", fontface = "bold"))

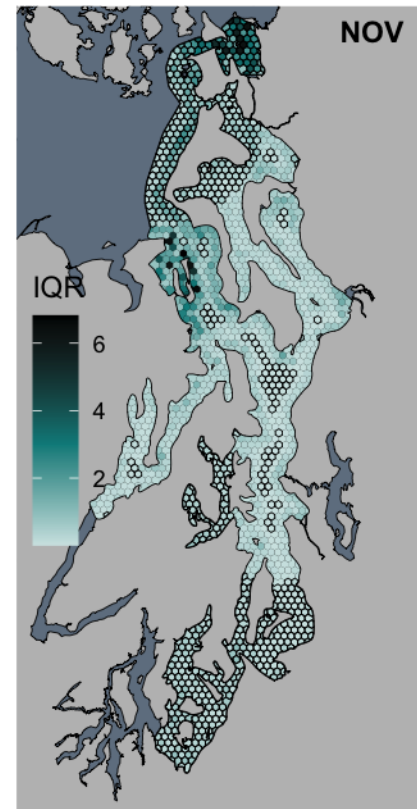
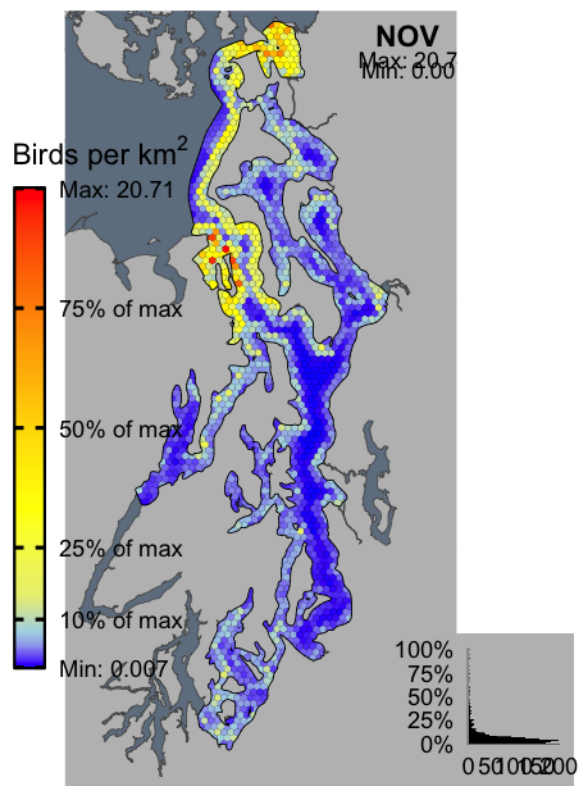
```



```

(nov_plot2 <- nov_plot | nov_iqr)

```



```
#ggsave("RESULTS/MAPS/nov_plot.pdf", plot = nov_plot2, width = 12, height = 11, units = "in")
#ggsave("RESULTS/MAPS/nov_plot.png", plot = nov_plot2, width = 12, height = 11, units = "in", dpi = 600)
```

DEC

```
dec <- grids %>%
  full_join(all_predictions %>%
    filter(mon == "DEC" & year == "ALL") %>%
    mutate(per_max = MED/max(MED),
           per_bin = case_when(
             per_max < 0.01 ~ "0-1%",
             per_max < 0.25 ~ "1-25%",
             per_max < 0.5 ~ "25-50%",
             TRUE ~ "50-100%"),
           per_bin = factor(per_bin,
                           levels = c("0-1%", "1-25%",
                                       "25-50%", "50-100%"))),
    by = c("numid", "ID"))

min_med <- dec %>%
  mutate(minnn = min(MED, na.rm = TRUE)) %>%
  distinct(minnn) %>%
  pull(minnn)
```

```

max_med <- dec %>%
  mutate(maxxxx =max(MED, na.rm = TRUE)) %>%
  distinct(maxxxx) %>%
  pull(maxxxx)

min_label <- paste0("Min: ",formatC(min_med, format = "f", digits = 3))
max_label <- paste0("Max: ",formatC(max_med, format = "f", digits = 2))

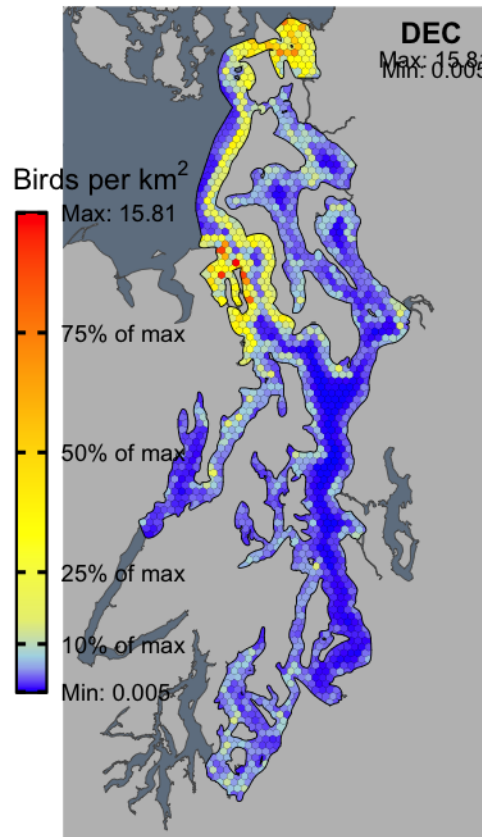
(DEC_Plot <- ggplot() +
  geom_sf(data = WA_State, fill = "gray") +
  geom_sf(data = puget, fill = "slategray") +
  geom_sf(data = dec,
    fill =NA, color = "black", linewidth=0.4) +
  geom_sf(data = dec,
    aes(fill = per_max), color = NA) +
  scale_fill_gradientn(
    colours = c("blue", "lightblue", "yellow", "orange", "red"),
    values = scales::rescale(c(0, 0.05, 0.15, 0.5, 0.75, 1)),
    limits = c(0, 1),
    name = expression("Birds per km"^2),
    breaks = c(0,0.1,0.25, 0.5, 0.75, 1),
    labels = c(min_label,"10% of max", "25% of max",
      "50% of max", "75% of max", max_label),
    guide = guide_colourbar(
      direction = "vertical",
      title.position = "top",
      title.hjust = 0.5,
      label.position = "right",
      label.hjust = 0,
      ticks = TRUE,
      ticks.colour = "black",
      ticks.linewidth = 0.7,
      barwidth = unit(4, "mm"),
      barheight = unit(2.5, "in"),
      frame.colour = "black",
      frame.linewidth = 0.5,
      draw.ulim = TRUE,
      draw.llim = TRUE)) +
  coord_sf(
    xlim = c(-123.2, -122),
    ylim = c(47, 48.6),
    crs = st_crs(4326),
    expand = FALSE) +
  scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
  scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
  theme_minimal() +
  theme(
    legend.margin = margin(0, 0, 0, 0),
    # legend.position = "right",
    legend.position = c(0.09, 0.5),
    axis.text = element_blank(),
    axis.title = element_blank(),
    axis.ticks = element_blank())+

```

```

annotate("text", x = -122.3, y = 48.5,
  label = max_label, size = 3, hjust = 0) +
annotate("text", x = -122.29, y = 48.5 - 0.02,
  label = min_label, size = 3, hjust = 0)+
annotate("text", x = -122.15, y = 48.55,
  label = "DEC", fontface = "bold"))

```



Dec Main

```

# Histogram data
dec_vals <- dec %>%
  filter(!is.na(per_max)) %>%
  pull(per_max)

hist_obj <- hist(dec_vals, breaks = seq(0, 1, length.out = 100), plot = FALSE)

hist_data <- data.frame(
  bin_mid = hist_obj$mids,
  count = hist_obj$counts)

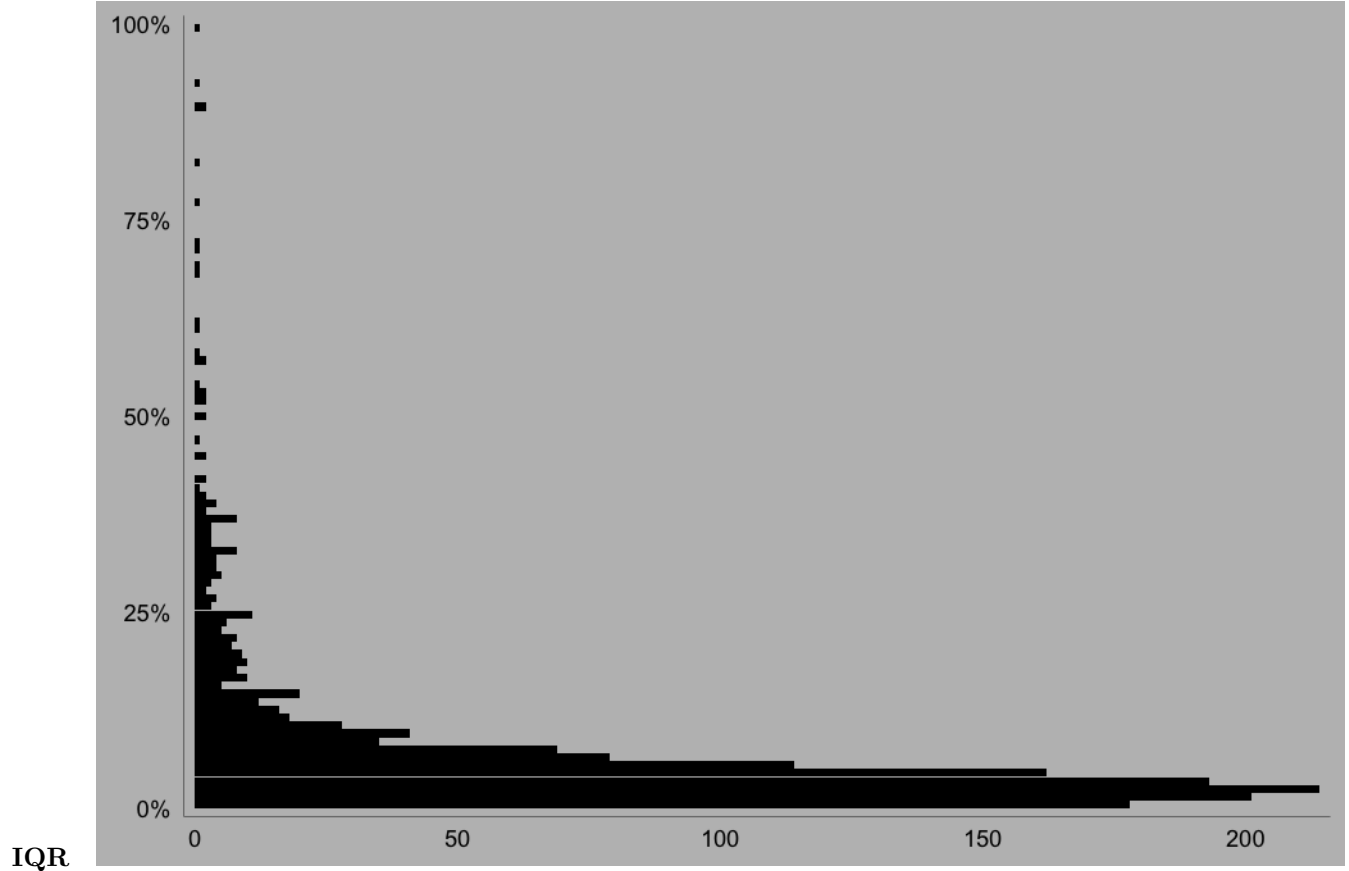
(hist_plot <- ggplot(hist_data, aes(x = bin_mid, y = count)) +
  geom_col(fill = "black", width = 0.01) +
  theme_minimal() +
  coord_flip()+

```

```

theme(
  #plot.margin = margin(2, 2, 2, 2), # top, right, bottom, left in pts
  # axis.text = element_blank(),
  axis.line.x = element_line(linewidth = 0.1, linetype = "solid",
                             colour = "black"),
  axis.line.y = element_line(linewidth = 0.1, linetype = "solid",
                             colour = "black"),
  panel.grid = element_blank(),
  panel.background = element_rect(fill = "gray", color = "gray"),
  panel.spacing = unit(0, "pt"),
  axis.text = element_text(color = "black"),
  axis.title = element_blank(),
  plot.background = element_rect(fill = "gray"),
  axis.ticks = element_blank()) +
scale_x_continuous(expand = c(0.01, 0),
                   labels = scales::percent_format(accuracy = 1)) +
scale_y_continuous(expand = c(0.01, 0))

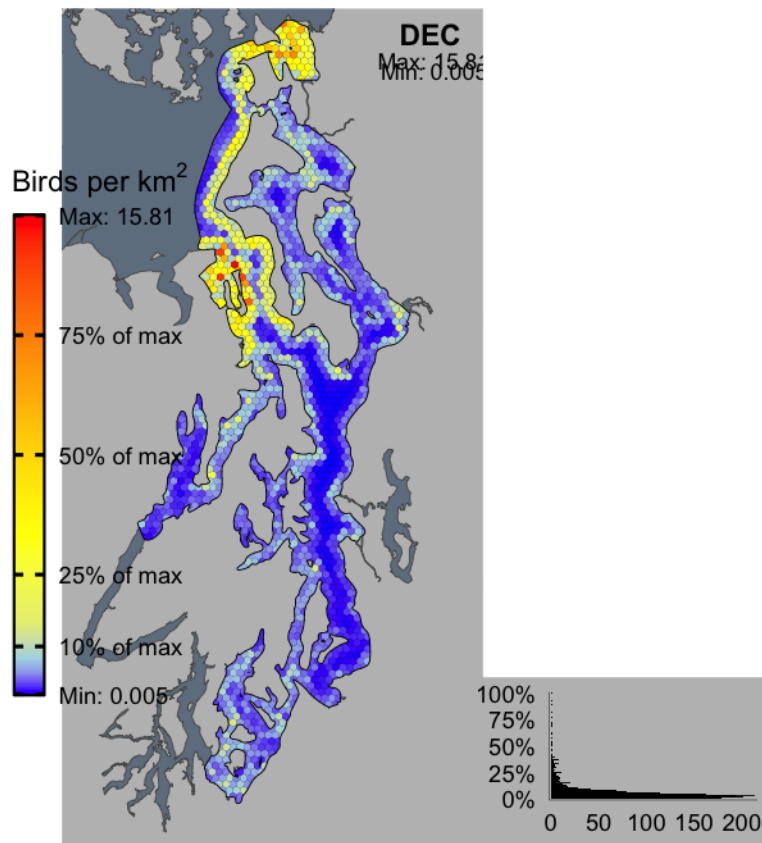
```



```

(dec_plot <- ggdraw(DEC_Plot) +
  draw_plot(hist_plot,
            x = 0.65, y = 0.01,
            width = 0.25, height = 0.2))

```



```
max(dec$IQR)
```

```
## [1] 5.058553
```

```
mean(dec$IQR)
```

```
## [1] 0.4005423
```

```
median(dec$IQR)
```

```
## [1] 0.2052401
```

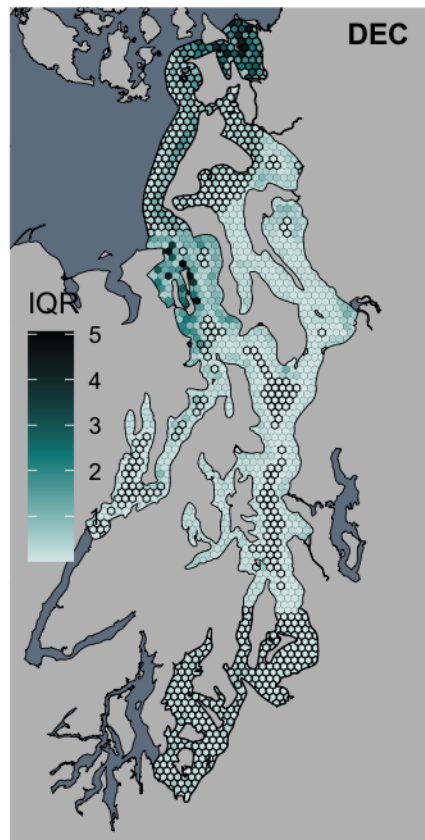
```
flag_dec <- dec %>%
  filter(flag == "CAP")

(dec_iqr <- ggplot() +
  geom_sf(data = WA_State, fill = "gray", color = "black") +
  geom_sf(data = puget, fill = "slategray", color = "black") +
  geom_sf(data = dec, fill = NA, color = "black", linewidth=0.4) +
  geom_sf(data = dec, aes(fill = IQR), color = NA) +
  geom_sf(data = flag_dec, fill = NA, color = "black") +
  coord_sf(
    xlim = c(-123.2, -122),
```

```

ylim = c(47, 48.6),
crs = st_crs(4326),
expand = FALSE) +
scale_fill_gradient2(low = "white", mid = "cyan4", high = "black",
                     midpoint = 2.3)+
scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
theme_minimal() +
theme(
  legend.margin = margin(0, 0, 0, 0),
  # legend.position = "right",
  legend.position = c(0.13, 0.5),
  axis.text = element_blank(),
  axis.title = element_blank(),
  axis.ticks = element_blank())+
annotate("text", x = -122.15, y = 48.55,
        label = "DEC", fontface = "bold"))

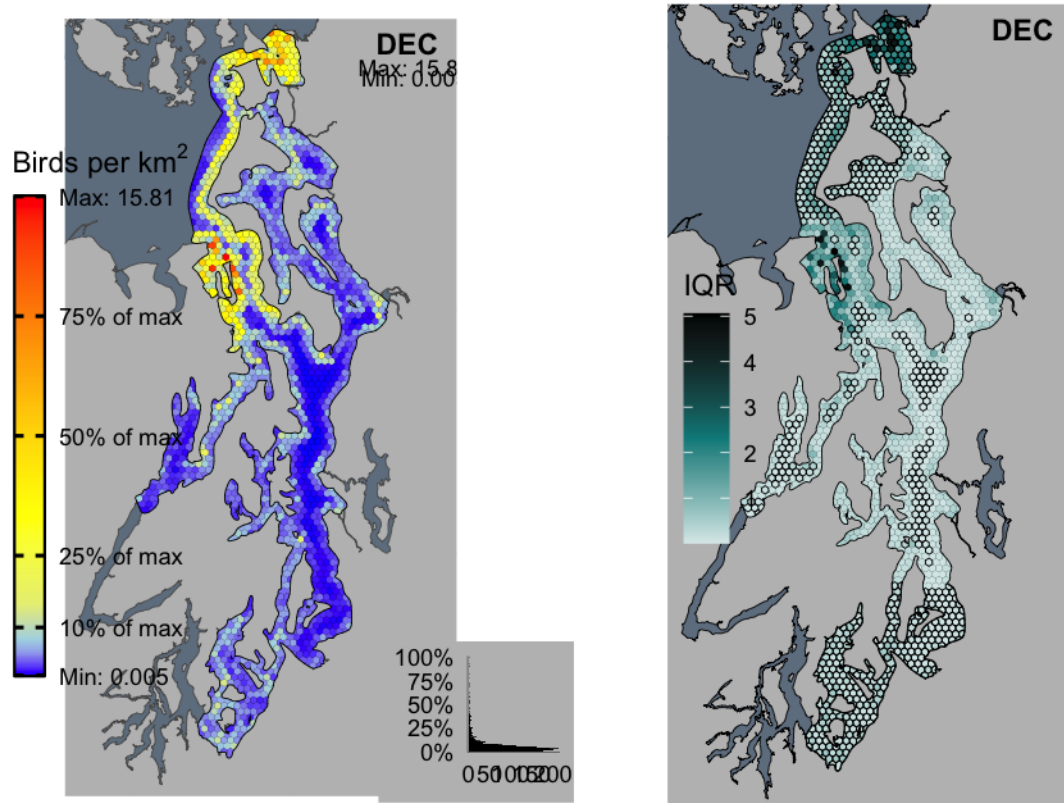
```



```

(dec_plot2 <- dec_plot | dec_iqr)

```



```
#ggsave("RESULTS/MAPS/dec_plot.pdf", plot = dec_plot2, width = 12, height = 11, units = "in")
#ggsave("RESULTS/MAPS/dec_plot.png", plot = dec_plot2, width = 12, height = 11, units = "in", dpi = 600)
```

Figure 6: Seasonal Density

B

```
bred <- grids %>%
  full_join(all_predictions %>%
    filter(mon == "B" & year == "ALL") %>%
    mutate(per_max = MED/max(MED),
           per_bin = case_when(
             per_max < 0.01 ~ "0-1%",
             per_max < 0.25 ~ "1-25%",
             per_max < 0.5 ~ "25-50%",
             TRUE ~ "50-100%"),
           per_bin = factor(per_bin,
                           levels = c("0-1%", "1-25%",
                                         "25-50%", "50-100%"))),
    by = c("numid", "ID"))

min_med <- bred %>%
```

```

mutate(minnnn =min(MED, na.rm = TRUE)) %>%
distinct(minnnn) %>%
pull(minnnn)

max_med <- bred %>%
mutate(maxxxx =max(MED, na.rm = TRUE)) %>%
distinct(maxxxx) %>%
pull(maxxxx)

min_label <- paste0("Min: ",formatC(min_med, format = "f", digits = 3))
max_label <- paste0("Max: ",formatC(max_med, format = "f", digits = 2))

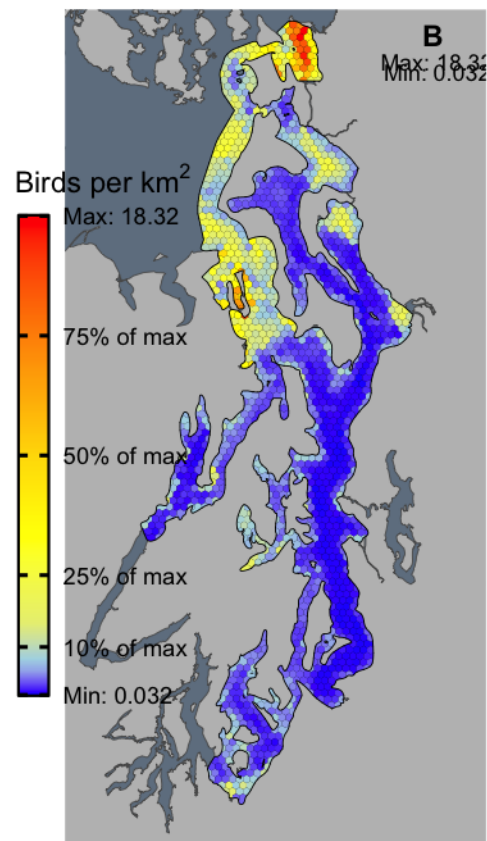
(B_Plot <- ggplot() +
  geom_sf(data = WA_State, fill = "gray") +
  geom_sf(data = puget, fill = "slategray") +
  geom_sf(data = bred,
    fill =NA, color = "black", linewidth=0.4) +
  geom_sf(data = bred,
    aes(fill = per_max), color = NA) +
  scale_fill_gradientn(
    colours = c("blue", "lightblue", "yellow", "orange", "red"),
    values = scales::rescale(c(0, 0.05, 0.15, 0.5, 0.75, 1)),
    limits = c(0, 1),
    name = expression("Birds per km"^2),
    breaks = c(0,0.1,0.25, 0.5, 0.75, 1),
    labels = c(min_label,"10% of max", "25% of max",
      "50% of max", "75% of max", max_label),
    guide = guide_colourbar(
      direction = "vertical",
      title.position = "top",
      title.hjust = 0.5,
      label.position = "right",
      label.hjust = 0,
      ticks = TRUE,
      ticks.colour = "black",
      ticks.linewidth = 0.7,
      barwidth = unit(4, "mm"),
      barheight = unit(2.5, "in"),
      frame.colour = "black",
      frame.linewidth = 0.5,
      draw.ulim = TRUE,
      draw.llim = TRUE)) +
  coord_sf(
    xlim = c(-123.2, -122),
    ylim = c(47, 48.6),
    crs = st_crs(4326),
    expand = FALSE) +
  scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
  scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
  theme_minimal() +
  theme(
    legend.margin = margin(0, 0, 0, 0),
    # legend.position = "right",

```

```

legend.position = c(0.09, 0.5),
axis.text = element_blank(),
axis.title = element_blank(),
axis.ticks = element_blank()+
annotate("text", x = -122.3, y = 48.5,
        label = max_label, size = 3, hjust = 0) +
annotate("text", x = -122.29, y = 48.5 - 0.02,
        label = min_label, size = 3, hjust = 0)+
annotate("text", x = -122.15, y = 48.55,
        label = "B", fontface = "bold"))

```



B Main

```

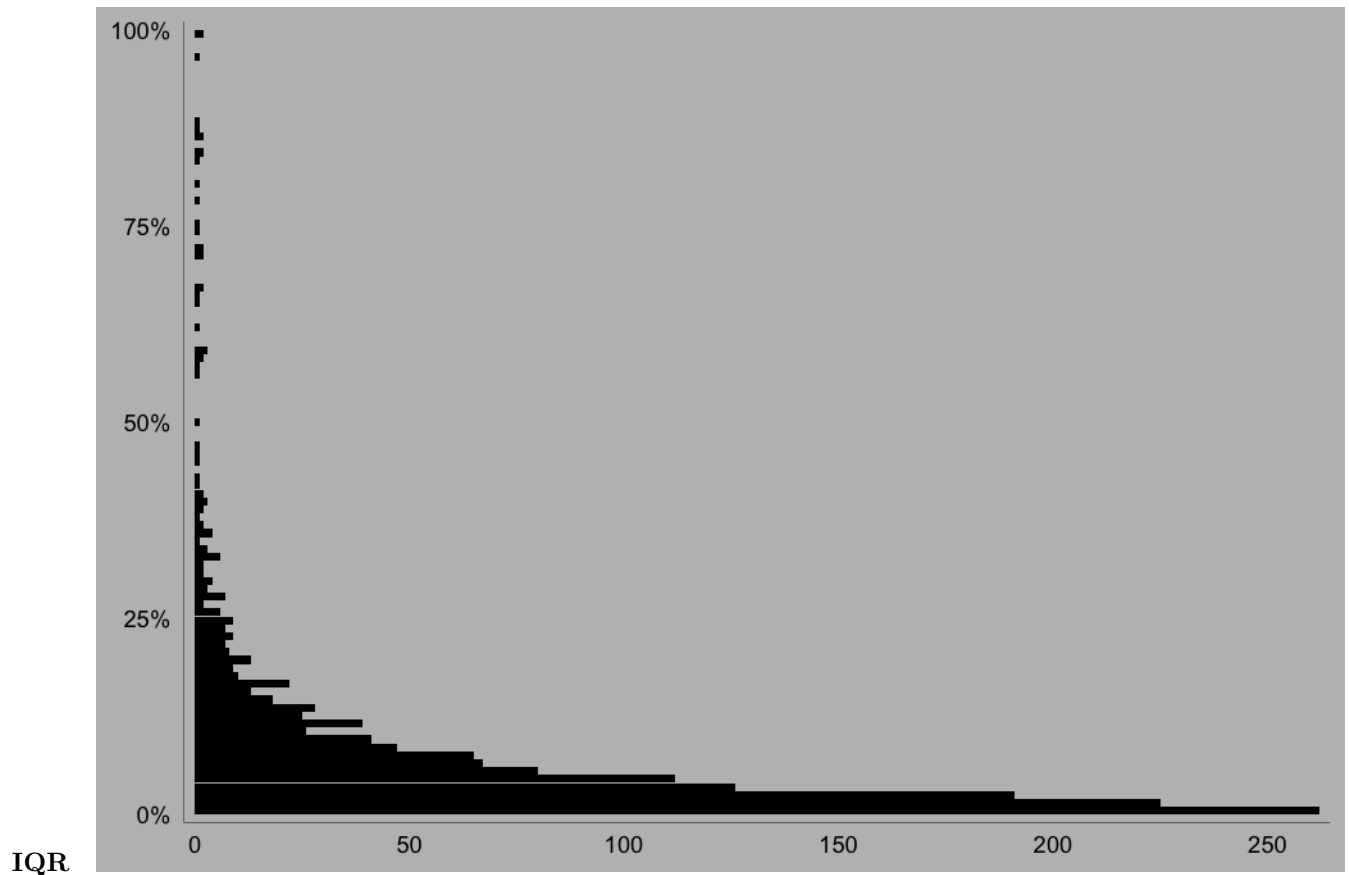
# Histogram data
bred_vals <- bred %>%
  filter(!is.na(per_max)) %>%
  pull(per_max)

hist_obj <- hist(bred_vals, breaks = seq(0, 1, length.out = 100), plot = FALSE)

hist_data <- data.frame(
  bin_mid = hist_obj$mids,
  count = hist_obj$counts)

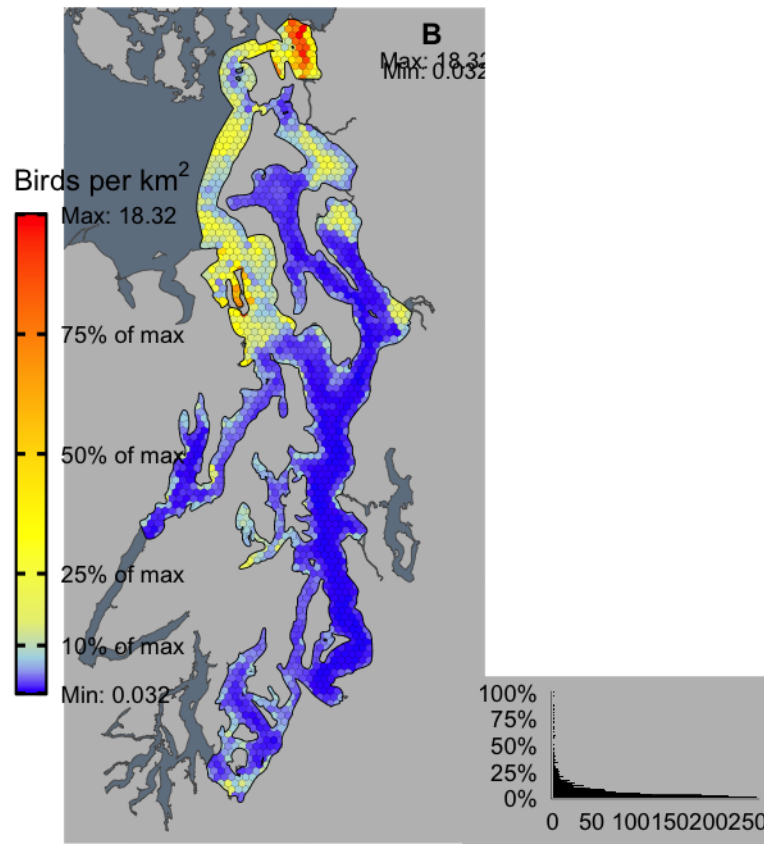
```

```
(hist_plot <- ggplot(hist_data, aes(x = bin_mid, y = count)) +
  geom_col(fill = "black", width = 0.01) +
  theme_minimal() +
  coord_flip() +
  theme(
    #plot.margin = margin(2, 2, 2, 2), # top, right, bottom, left in pts
    # axis.text = element_blank(),
    axis.line.x = element_line(linewidth = 0.1, linetype = "solid",
                               colour = "black"),
    axis.line.y = element_line(linewidth = 0.1, linetype = "solid",
                               colour = "black"),
    panel.grid = element_blank(),
    panel.background = element_rect(fill = "gray", color = "gray"),
    panel.spacing = unit(0, "pt"),
    axis.text = element_text(color = "black"),
    axis.title = element_blank(),
    plot.background = element_rect(fill = "gray"),
    axis.ticks = element_blank() +
  scale_x_continuous(expand = c(0.01, 0),
                    labels = scales::percent_format(accuracy = 1)) +
  scale_y_continuous(expand = c(0.01, 0)))
```



```
(bred_plot <- ggdraw(B_Plot) +
  draw_plot(hist_plot,
    x = 0.65, y = 0.01,
```

```
width = 0.25, height = 0.2))
```



```
max(bred$IQR)
```

```
## [1] 28.58314
```

```
mean(bred$IQR)
```

```
## [1] 1.731936
```

```
median(bred$IQR)
```

```
## [1] 0.4956382
```

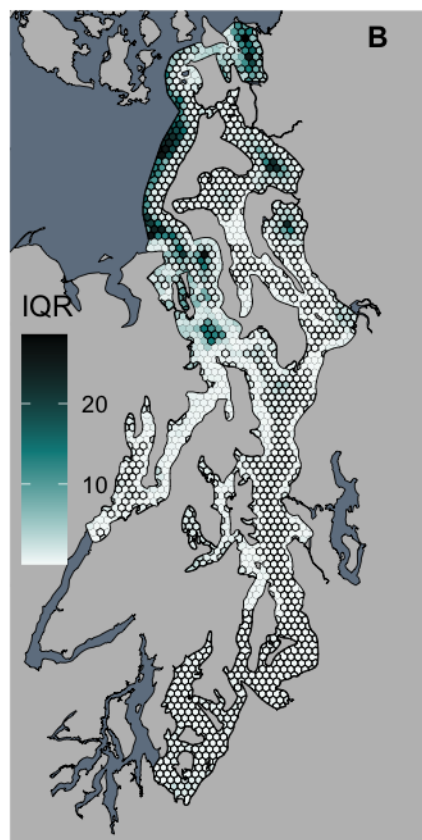
```
flag_bred <- bred %>%  
  filter(flag == "CAP")
```

```
(bred_iqr <- ggplot() +  
  geom_sf(data = WA_State, fill = "gray", color = "black") +  
  geom_sf(data = puget, fill = "slategray", color = "black") +  
  geom_sf(data = bred, fill = NA, color = "black", linewidth = 0.4) +  
  geom_sf(data = bred, aes(fill = IQR), color = NA) +  
  geom_sf(data = flag_bred, fill = NA, color = "black") +
```

```

coord_sf(
  xlim = c(-123.2, -122),
  ylim = c(47, 48.6),
  crs = st_crs(4326),
  expand = FALSE) +
scale_fill_gradient2(low = "white", mid = "cyan4", high = "black",
  midpoint = 14)+
scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
theme_minimal() +
theme(
  legend.margin = margin(0, 0, 0, 0),
  # legend.position = "right",
  legend.position = c(0.13, 0.5),
  axis.text = element_blank(),
  axis.title = element_blank(),
  axis.ticks = element_blank())+
annotate("text", x = -122.15, y = 48.55,
  label = "B", fontface = "bold"))

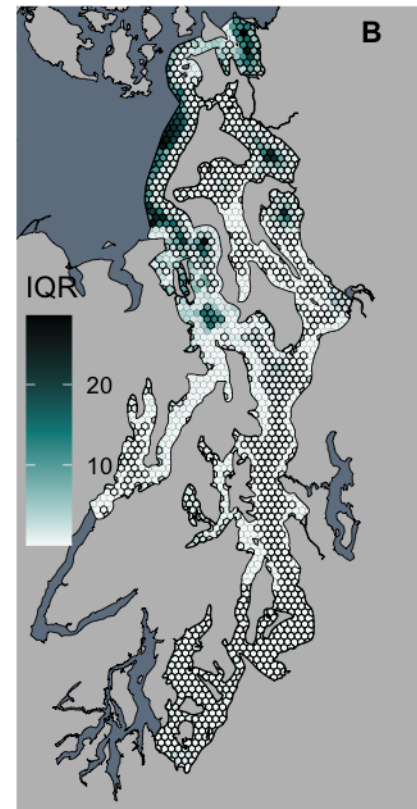
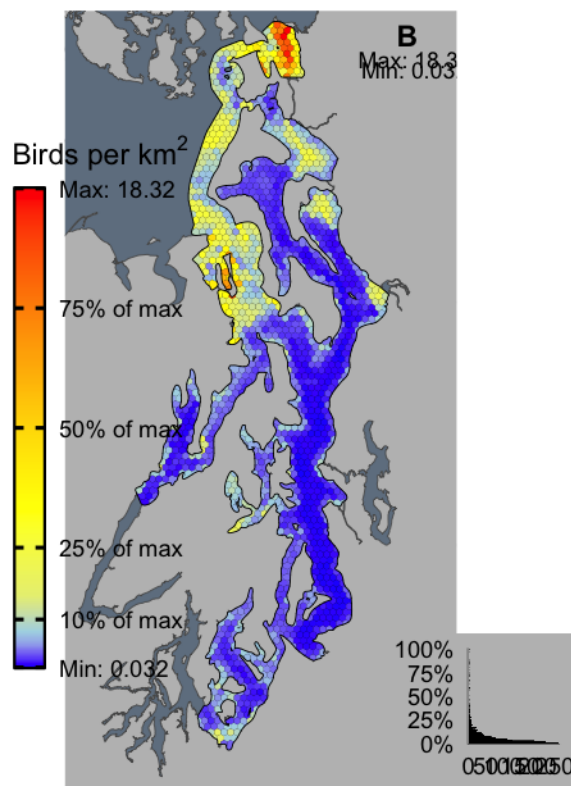
```



```

(bred_plot2 <- bred_plot | bred_iqr)

```



```
#ggsave("RESULTS/MAPS/bred_plot.pdf", plot = bred_plot2, width = 12, height = 11, units = "in")
#ggsave("RESULTS/MAPS/bred_plot.png", plot = bred_plot2, width = 12, height = 11, units = "in", dpi = 60)
```

NB

```
nbred <- grids %>%
  full_join(all_predictions %>%
    filter(mon == "NB" & year == "ALL") %>%
    mutate(per_max = MED/max(MED),
           per_bin = case_when(
             per_max < 0.01 ~ "0-1%",
             per_max < 0.25 ~ "1-25%",
             per_max < 0.5 ~ "25-50%",
             TRUE ~ "50-100%"),
           per_bin = factor(per_bin,
                           levels = c("0-1%", "1-25%",
                                       "25-50%", "50-100%"))),
    by = c("numid", "ID"))

min_med <- nbred %>%
  mutate(minnn = min(MED, na.rm = TRUE)) %>%
  distinct(minnn) %>%
  pull(minnn)
```

```

max_med <- nbred %>%
  mutate(maxxxx =max(MED, na.rm = TRUE)) %>%
  distinct(maxxxx) %>%
  pull(maxxxx)

min_label <- paste0("Min: ",formatC(min_med, format = "f", digits = 3))
max_label <- paste0("Max: ",formatC(max_med, format = "f", digits = 2))

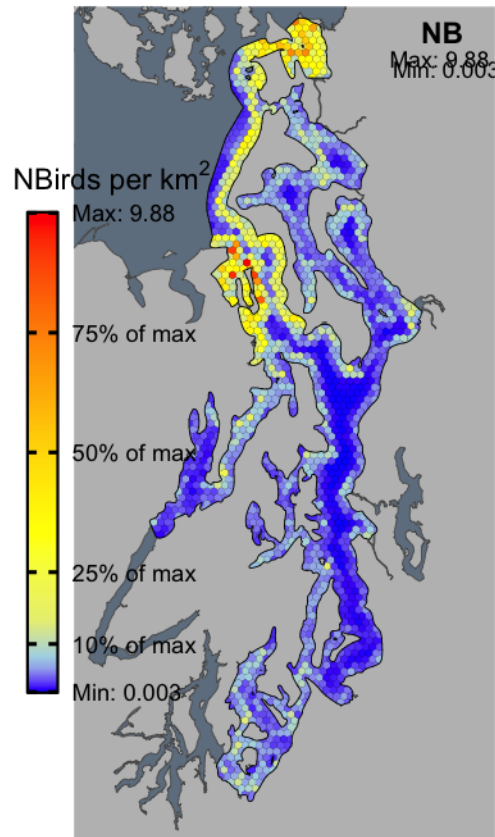
(NB_Plot <- ggplot() +
  geom_sf(data = WA_State, fill = "gray") +
  geom_sf(data = puget, fill = "slategray") +
  geom_sf(data = nbred,
    fill =NA, color = "black", linewidth=0.4) +
  geom_sf(data = nbred,
    aes(fill = per_max), color = NA) +
  scale_fill_gradientn(
    colours = c("blue", "lightblue", "yellow", "orange", "red"),
    values = scales::rescale(c(0, 0.05, 0.15, 0.5, 0.75, 1)),
    limits = c(0, 1),
    name = expression("NBirds per km"^2),
    breaks = c(0,0.1,0.25, 0.5, 0.75, 1),
    labels = c(min_label,"10% of max", "25% of max",
      "50% of max", "75% of max", max_label),
    guide = guide_colourbar(
      direction = "vertical",
      title.position = "top",
      title.hjust = 0.5,
      label.position = "right",
      label.hjust = 0,
      ticks = TRUE,
      ticks.colour = "black",
      ticks.linewidth = 0.7,
      barwidth = unit(4, "mm"),
      barheight = unit(2.5, "in"),
      frame.colour = "black",
      frame.linewidth = 0.5,
      draw.ulim = TRUE,
      draw.llim = TRUE)) +
  coord_sf(
    xlim = c(-123.2, -122),
    ylim = c(47, 48.6),
    crs = st_crs(4326),
    expand = FALSE) +
  scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
  scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
  theme_minimal() +
  theme(
    legend.margin = margin(0, 0, 0, 0),
    # legend.position = "right",
    legend.position = c(0.09, 0.5),
    axis.text = element_blank(),
    axis.title = element_blank(),
    axis.ticks = element_blank())+

```

```

annotate("text", x = -122.3, y = 48.5,
  label = max_label, size = 3, hjust = 0) +
annotate("text", x = -122.29, y = 48.5 - 0.02,
  label = min_label, size = 3, hjust = 0)+
annotate("text", x = -122.15, y = 48.55,
  label = "NB", fontface = "bold"))

```



NB Main

```

# Histogram data
nbred_vals <- nbred %>%
  filter(!is.na(per_max)) %>%
  pull(per_max)

hist_obj <- hist(nbred_vals, breaks = seq(0, 1, length.out = 100), plot = FALSE)

hist_data <- data.frame(
  bin_mid = hist_obj$mids,
  count = hist_obj$counts)

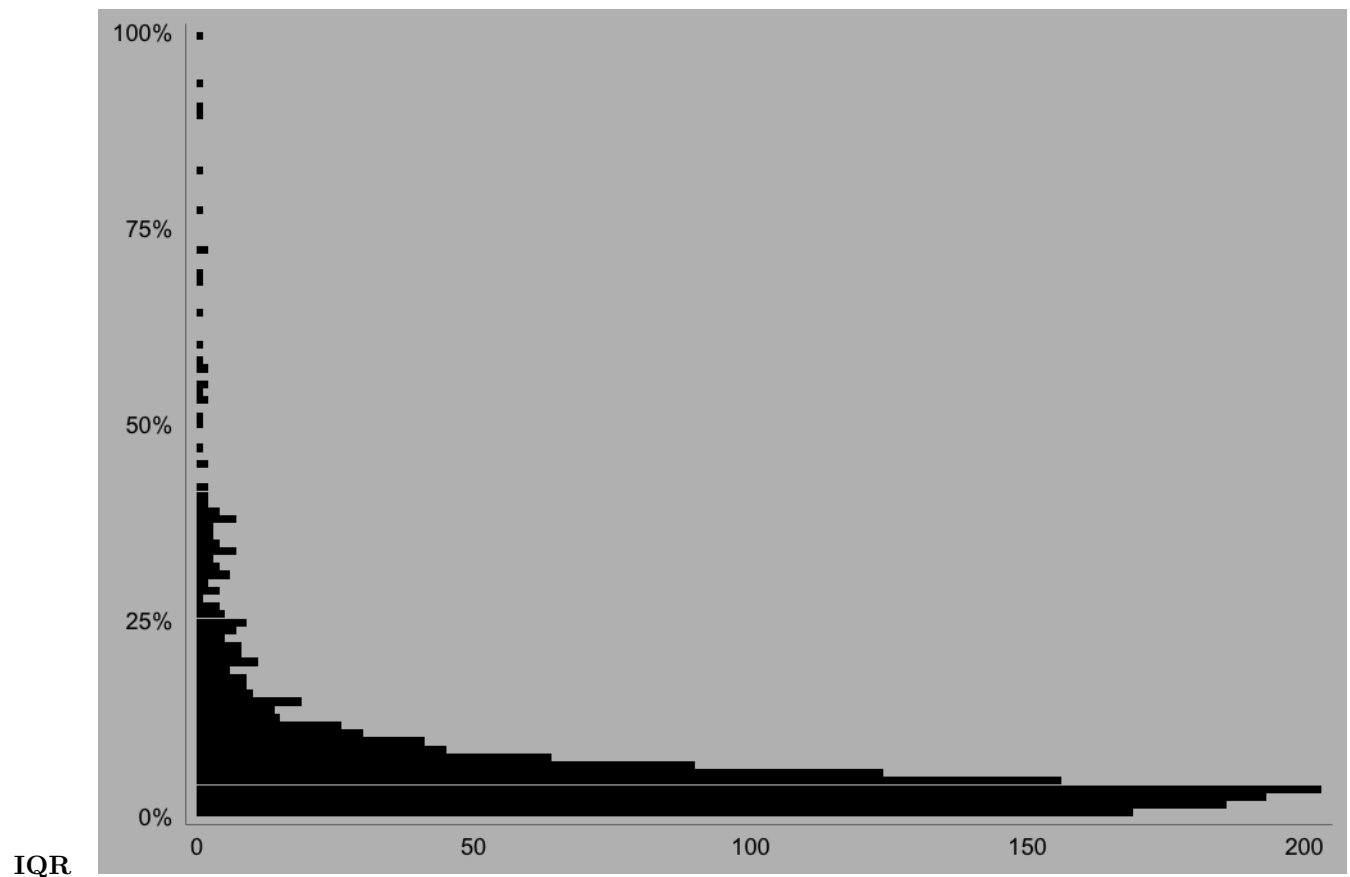
(hist_plot <- ggplot(hist_data, aes(x = bin_mid, y = count)) +
  geom_col(fill = "black", width = 0.01) +
  theme_minimal() +
  coord_flip()+

```

```

theme(
  #plot.margin = margin(2, 2, 2, 2), # top, right, bottom, left in pts
  # axis.text = element_blank(),
  axis.line.x = element_line(linewidth = 0.1, linetype = "solid",
                              colour = "black"),
  axis.line.y = element_line(linewidth = 0.1, linetype = "solid",
                              colour = "black"),
  panel.grid = element_blank(),
  panel.background = element_rect(fill = "gray", color = "gray"),
  panel.spacing = unit(0, "pt"),
  axis.text = element_text(color = "black"),
  axis.title = element_blank(),
  plot.background = element_rect(fill = "gray"),
  axis.ticks = element_blank() +
scale_x_continuous(expand = c(0.01, 0),
                    labels = scales::percent_format(accuracy = 1)) +
scale_y_continuous(expand = c(0.01, 0)))

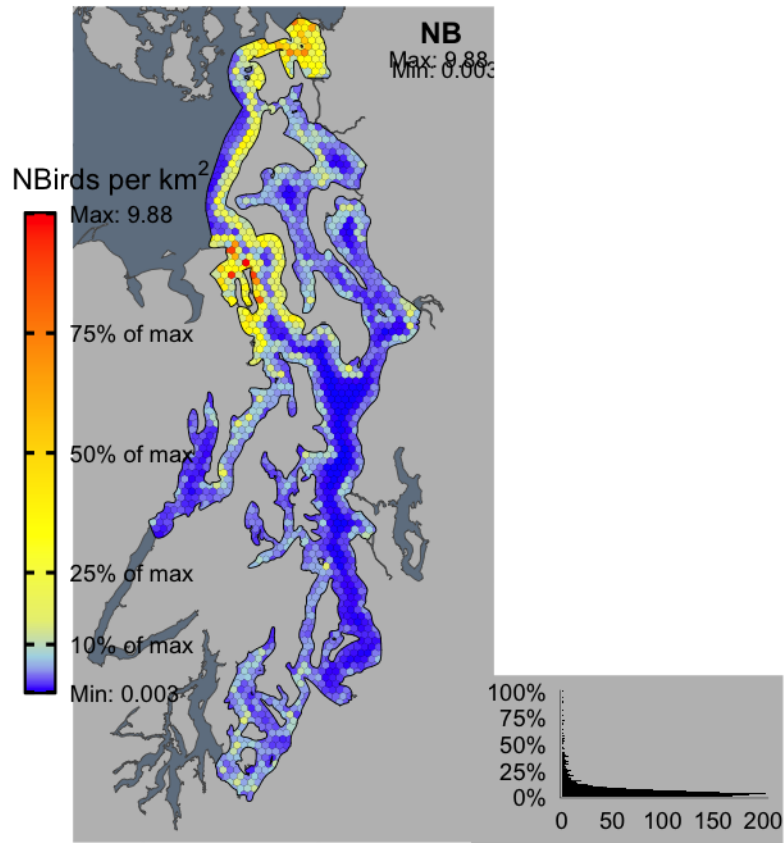
```



```

(nbred_plot <- ggdraw(NB_Plot) +
  draw_plot(hist_plot,
    x = 0.65, y = 0.01,
    width = 0.25, height = 0.2))

```



```
max(nbred$IQR)
```

```
## [1] 4.765269
```

```
mean(nbred$IQR)
```

```
## [1] 0.3567314
```

```
median(nbred$IQR)
```

```
## [1] 0.1932985
```

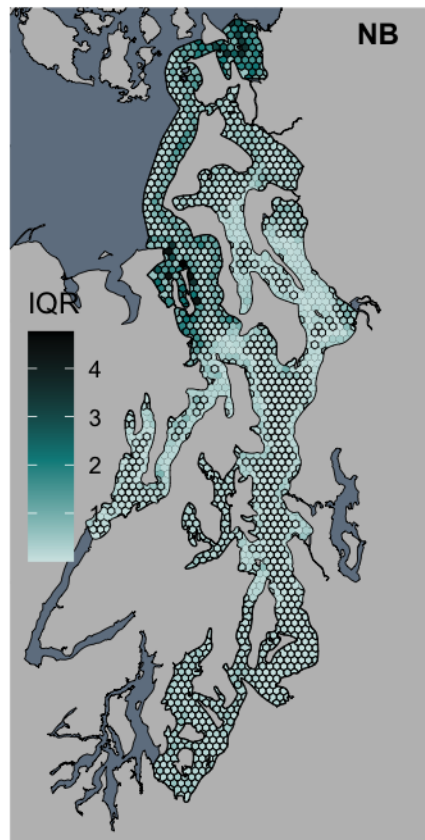
```
flag_nbred <- nbred %>%
  filter(flag == "CAP")

(nbred_iqr <- ggplot() +
  geom_sf(data = WA_State, fill = "gray", color = "black") +
  geom_sf(data = puget, fill = "slategray", color = "black") +
  geom_sf(data = nbred, fill = NA, color = "black", linewidth=0.4) +
  geom_sf(data = nbred, aes(fill = IQR), color = NA) +
  geom_sf(data = flag_nbred, fill = NA, color = "black") +
  coord_sf(
    xlim = c(-123.2, -122),
```

```

ylim = c(47, 48.6),
crs = st_crs(4326),
expand = FALSE) +
scale_fill_gradient2(low = "white", mid = "cyan4", high = "black",
                     midpoint = 2.1)+
scale_x_continuous(breaks = c(-123.5, -123, -122.5, -122)) +
scale_y_continuous(breaks = c(47, 47.5, 48, 48.5)) +
theme_minimal() +
theme(
  legend.margin = margin(0, 0, 0, 0),
  # legend.position = "right",
  legend.position = c(0.13, 0.5),
  axis.text = element_blank(),
  axis.title = element_blank(),
  axis.ticks = element_blank())+
annotate("text", x = -122.15, y = 48.55,
        label = "NB", fontface = "bold"))

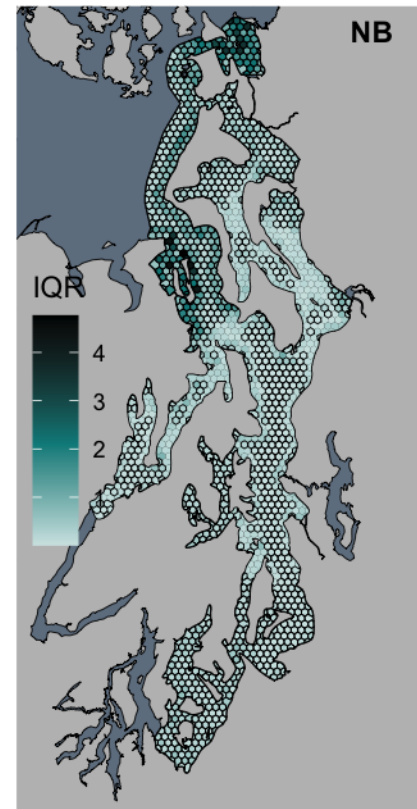
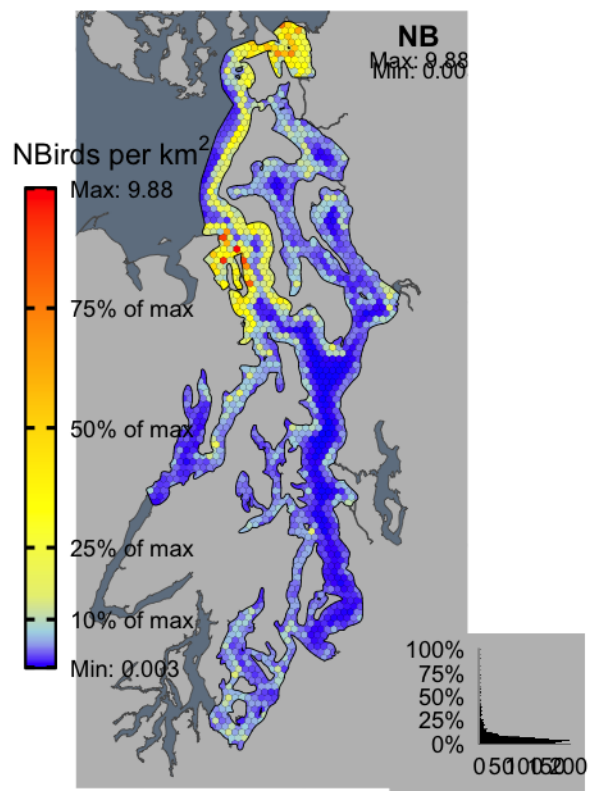
```



```

(nbred_plot2 <- nbred_plot | nbred_iqr)

```



```
#ggsave("RESULTS/MAPS/nbred_plot.pdf", plot = nbred_plot2, width = 12, height = 11, units = "in")
#ggsave("RESULTS/MAPS/nbred_plot.png", plot = nbred_plot2, width = 12, height = 11, units = "in", dpi = 300)
```